Chapter 2

FOUNDATIONS

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This chapter includes standard coverage of mean-variance optimal (MVO) portfolios and their maximum quadratic utility equivalent, along with the optimal active portfolios variant, and the Capital Asset Pricing Model (CAPM), and related expected utility maximization. Because of its practical compuatational amd performance evalationa relevance, we discuss early in the chapter the computation of global minimum variance (GMV) portfolios, and their out-of-sample performance evaluation that includes" (1) Plots of cumulative gross returns of the portfolio and the market: (2) Leverage, turnover, and diversification values; (3) Standard and downside risk measures, amd risk adjusted performance measures.

Additional topics not currently found in M.S. degree level quantitative finance books include: (1) Computing and reporting standard errors for GMV portfolo risk and risk adjusted performance estimates; (2) The zero beta CAPM and the Beta factor; (3) The mathematics of the arbitrage pricing theory (APT); (4) Robust M-estimators of a returns mean and standard deviation that are not much influenced by outliers, along with their computation and a striking application example. These estimator are stepping stones to robust regression estimators introduced and used in later chapters; (5) A brief section that introduces minimum expected shortfall (MES) portfolios and their efficient frontiers in comparison with MVO portfolio efficient frontiers.

The computations used for this chapter make considerable use of the `stocksCRSPmonthly` data set in the `PCRA` package, as well as other R packages, discussed in the Preface. Most of the tables and figures in this chapter are reproducible with the `Ch 2 Foundations Demo.R` code.

2.1 Asset Returns Types

There are two basic definitions of the investment return of an asset, such as a single stock or a portfolio of stocks, over a single investment period, namely its *arithmetic return* and its *logarithmic (or log) return*. In this chapter we focus on arithmetic returns since, as we will see in Section 2.2.2, arithmetic returns of assets in a portfolio result in convenient arithmetic aggregation of returns across the assets in a portfolio. However, we also briefly introduce the log return and point out its main uses.

Arithmetic Return

Let $p_t,\ t = 1, 2, \cdots, T$ be the time series of prices of a traded asset, or of a portfolio of assets, e.g., a stock, a bond, an exchange traded fund (ETF), etc., over some interval of interest, where the integer subscript t may refer to month, weeks or days in the cases of returns with monthly, weekly or daily frequencies. The owner of the asset will experience changes in the asset's price over time, and may, in addition, receive periodic cash flows from the asset, which we denote by c_t. Dividends are the primary form of positive cash flows c_t for stocks, and coupon and principal payments are the primary forms of c_t for bonds. It is customary to assume that such cash flows occur at the end of each period. The *arithmetic total return* r_t, from end of time period $t-1$ to the end of time period t is defined as

$$r_t = \frac{p_t + c_t - p_{t-1}}{p_{t-1}} = \frac{p_t + c_t}{p_{t-1}} - 1 \tag{2.1}$$

where the time period may be in units of months, weeks or days, with which we are mostly concerned in this book.

If we take $c_t \equiv 0$ in (2.1), as we often do for analytic discussion convenience, the arithmetic return reduces to the relative price change from the previous time period to the current period

$$r_t = \frac{p_t - p_{t-1}}{p_{t-1}} = \frac{p_t}{p_{t-1}} - 1. \tag{2.2}$$

and it is customary to refer to this simply as a *arithmetic return*. Note that since prices are non-negative and in principle could be equal to zero, an arithmetic return r_t is bounded below by -1 and can be arbitrarily large, i.e., $-1 \leqslant r_t < \infty$. It is customary to state a return as a percentage by multiplying r_t by 100, and so when a return is expressed as a percent it is bounded below by -100%.

The single period *gross return* R_t is defined to be the ratio of the price in the current time period, plus any end of period cash flow, to the price in the previous period

$$R_t = \frac{p_t + c_t}{p_{t-1}} \tag{2.3}$$

and we immediately see that the single period gross return is expressed in terms of arithmetic total return (2.1) as

$$R_t = r_t + 1 \tag{2.4}$$

and conversely the arithmetic return is expressed in terms of gross return as:

$$r_t = R_t - 1. \tag{2.5}$$

It will often be useful to denote the value (or wealth holding) of an asset by v_t, and to focus on its return over a single period, taken for convenience to start at time $t = 0$ and to end at time $t = 1$. In that case we have $R_1 = \mathrm{v}_1/\mathrm{v}_0$ and so:

$$\mathrm{v}_1 = \mathrm{v}_0 \cdot (r_1 + 1)\,. \tag{2.6}$$

The profit or loss (PL) of the investment is

$$PL = \mathrm{v}_1 - \mathrm{v}_0 = \mathrm{v}_0 \cdot r_1 \tag{2.7}$$

which is positive for a profit and is negative for a loss. For an investment with $\mathrm{v}_0 = \$1$, the PL is simply r_1.

In risk management discussions it is common to focus on loss as a positive quantity, in which case one takes $L = -PL$ and we have

$$L = -\mathrm{v}_0 \cdot r_1. \tag{2.8}$$

Log Return

The single period *log return* is defined as

$$\tilde{r}_t = \ln\left(\frac{p_t + c_t}{p_{t-1}}\right) \tag{2.9}$$

and taking end of period cash flows $c_t \equiv 0$ for simplicity, log return has the following two alternative forms

$$\begin{aligned} \tilde{r}_t &= \ln(p_t) - \ln(p_{t-1}) \\ &= \ln\left(\frac{p_t}{p_{t-1}}\right) \\ &= \ln(1 + r_t) \end{aligned} \tag{2.10}$$

where r_t is the single period arithmetic return. Note that whereas arithmetic returns are bounded below by -1, a log return can take on any positive or negative value: $-\infty < \tilde{r}_t < \infty$.

Logarithmic returns have the great attractiveness of aggregating over time by summation. The log return $\tilde{r}_{1,T}$ from period 1 through period T is defined in the first line below, and subsequent lines follow immediately:

$$\begin{aligned} \tilde{r}_{1,T} &= \ln\left(\frac{p_T}{p_0}\right) \\ &= \ln\left(\frac{p_T}{p_{T-1}} \cdot \frac{p_{T-1}}{p_{T-2}} \cdot \ldots \cdot \frac{p_2}{p_1} \cdot \frac{p_1}{p_0}\right) \\ &= \sum_{t=1}^{T} \tilde{r}_t. \end{aligned} \tag{2.11}$$

A great convenience of the above result is that if the log returns $\tilde{r}_1, \tilde{r}_2, \cdots, \tilde{r}_T$ are uncorrelated with common mean values μ and common variances σ^2, then the mean and variance of $\tilde{r}_{1,T}$ are $T\mu$ and $T\sigma^2$, respectively. This will prove useful in many contexts later on.

2.2 Portfolio Mean and Variance

This section explores the relationship between the mean and variance of a portfolio of risky assets with random returns that are characterized by the their means, variances and covariances, illustrated mainly by simple portfolios of 2 or 3 assets, and ending by deriving a general formula for a vector of portfolio weights that minimize a portfolio's variance. In this section we discuss:

1. Notation for the true mean vector, variance vector, and covariance matrix of a vector of arithmetic returns, and standard estimators of these quantities.
2. Portfolio weights, mean return, variance, and decompositions of portfolio variance and volatility
3. Examples of portfolio mean and variance relationships for portfolios with 2 and 3 assets
4. Derivation of a formula for the weight vector of a portfolio with the smallest possible variance

2.2.1 Returns Mean Vector and Covariance Matrix

Throughout this section, and indeed throughout the book, we use the following basic quantities:

- Asset returns, mean vectors, and covariance matrices
- Portfolio mean return and variance

We now discuss the above two topics.

Asset Returns, Mean Vectors, Covariance Matrices and Correlation Matrices

The basic context of mean–variance portfolio theory is that we are given a portfolio of N assets with returns $r_{t,i}$, $i = 1, 2, \cdots, N$ at times $t = 1, 2, \cdots, T$ of interest, where each time intervals represents a myopic investment horizon. They may, for example, be measured in units of years, months, weeks or days. The column vector of asset returns at time t is:

$$\mathbf{r}_t = \left(r_{t,1}, ..., r_{t,N}\right)', \, t = 1, ..., T \tag{2.12}$$

and when reference to a particular time period is not needed, we drop the time subscript and simply write $\mathbf{r} = (r_1, ..., r_N)'$.

For simplicity, and unless noted otherwise, we make the idealized assumption that the asset returns $r_{t,i}$, $i = 1, 2, \cdots, N$ have constant mean values $\mu_i = \mathrm{E}[r_i]$, $i = 1, 2, \cdots, N$ for $t = 1, 2, \cdots, T$. The column vector of mean returns is

$$\begin{aligned} \boldsymbol{\mu} &= \mathrm{E}[\mathbf{r}] \\ &= (\mathrm{E}[r_1], \mathrm{E}[r_2], \cdots, \mathrm{E}[r_N])' \\ &= (\mu_1, \mu_2, \cdots, \mu_N)'. \end{aligned} \tag{2.13}$$

The returns are assumed to have pairwise covariances

$$\begin{aligned} C_{ij} &= \mathrm{cov}(r_i, r_j) \\ &= \mathrm{E}\left[(r_i - \mu_i)(r_j - \mu_j)\right] \\ &= \mathrm{E}\left[r_i r_j\right] - \mu_i \mu_j \;\; i, j = 1, \cdots, N. \end{aligned} \tag{2.14}$$

where $C_{ii} = \mathrm{E}\left[(r_i - \mu_i)^2\right] = \mathrm{var}(r_i) = \sigma_i^2$.[1]

Note: It is common practice in the literature to use the notation $\sigma_{i,j} = \mathrm{cov}(r_i, r_j)$, $i, j = 1, \cdots, N$ with $\sigma_i^2 = \sigma_{ii}$, and we will sometimes use this notation throughout the remainder of the book.

The $N \times N$ covariance matrix $\mathbf{C}$ of the returns has elements C_{ij}, and has the form:

$$\begin{aligned} \mathbf{C} &= \mathrm{E}[(\mathbf{r} - \boldsymbol{\mu})(\mathbf{r} - \boldsymbol{\mu})'] \\ &= \mathrm{E}[\mathbf{r}\mathbf{r}'] - E[\mathbf{r}\boldsymbol{\mu}'] - E[\boldsymbol{\mu}\mathbf{r}'] + \boldsymbol{\mu}\boldsymbol{\mu}' \\ &= \mathrm{E}\left[\mathbf{r}_t \mathbf{r}_t'\right] - \boldsymbol{\mu}\boldsymbol{\mu}'. \end{aligned} \tag{2.15}$$

It is often of interest to focus on the pairwise correlations

$$\rho_{ij} = \frac{\mathrm{cov}(r_i, r_j)}{\sigma_i \cdot \sigma_j} = \frac{\sigma_{ij}}{\sigma_i \cdot \sigma_j} \;\; i, j = 1, \cdots, N \tag{2.16}$$

where

$$\begin{aligned} \sigma_{ij} &= C_{ij} \\ \sigma_i^2 &= C_{ii} \end{aligned}. \tag{2.17}$$

The corresponding $N \times N$ correlation matrix $\mathbf{R}$ has elements $R_{ij} = \rho_{ij}$. Note that the diagonal elements of $\mathbf{R}$ are all equal to one.

The ij^{th} element of the covariance matrix $\mathbf{C}$ can be obtained from the ij^{th} element of $\mathbf{R}$ by multiplying R_{ij} by $\sigma_i \cdot \sigma_j$. This leads to the following convenient matrix representation of $\mathbf{C}$ in terms of $\mathbf{R}$. Let $\mathbf{D}_\sigma$ be the $N \times N$ diagonal matrix with elements σ_i, $i = 1, 2, \cdots, N$:

[1] The elements C_{ij} of $\mathbf{C}$ are sometimes represented in the literature with the notation σ_{ij}, where the diagonal elements $\sigma_{ii} = \sigma_i^2$ are the variances of the r_i.

$$\mathbf{D}_\sigma = \begin{pmatrix} \sigma_1 & 0 & \cdots & \cdots & 0 \\ 0 & \sigma_2 & 0 & \cdots & 0 \\ \vdots & \ddots & \ddots & \ddots & 0 \\ \vdots & \ddots & \ddots & \ddots & 0 \\ 0 & \cdots & \cdots & 0 & \sigma_N \end{pmatrix}. \tag{2.18}$$

Then simple matrix algebra shows that

$$\mathbf{C} = \mathbf{D}_\sigma \mathbf{R} \mathbf{D}_\sigma. \tag{2.19}$$

Note also that the correlation matrix $\mathbf{R}$ can be obtained from a covariance matrix $\mathbf{C}$ and a volatility matrix $\mathbf{D}_\sigma$ through matrix multiplication:

$$\mathbf{R} = \mathbf{D}_\sigma^{-1} \mathbf{C} \mathbf{D}_\sigma^{-1}. \tag{2.20}$$

Example 2.1. Suppose $N = 2$ and the returns r_1 and r_2 have volatilities $\sigma_1 = 0.1$ and $\sigma_2 = 0.2$, and correlation coefficient $\rho = 0.5$. Then

$$\mathbf{D}_\sigma = \begin{pmatrix} .1 & 0 \\ 0 & .2 \end{pmatrix} \qquad \mathbf{R} = \begin{pmatrix} 1 & .5 \\ .5 & 1 \end{pmatrix}$$

and we have:

$$\mathbf{C} = \begin{pmatrix} .1 & 0 \\ 0 & .2 \end{pmatrix} \begin{pmatrix} 1 & .5 \\ .5 & 1 \end{pmatrix} \begin{pmatrix} .1 & 0 \\ 0 & .2 \end{pmatrix} = \begin{pmatrix} .01 & .01 \\ .01 & .04 \end{pmatrix}. \tag{2.21}$$

Furthermore, one recovers $\mathbf{R}$ from $\mathbf{C}$ by:

$$\mathbf{R} = \begin{pmatrix} .1 & 0 \\ 0 & .2 \end{pmatrix}^{-1} \begin{pmatrix} .01 & .01 \\ .01 & .04 \end{pmatrix} \begin{pmatrix} .1 & 0 \\ 0 & .2 \end{pmatrix}^{-1} = \begin{pmatrix} 10 & 0 \\ 0 & 5 \end{pmatrix} \begin{pmatrix} .01 & .01 \\ .01 & .04 \end{pmatrix} \begin{pmatrix} 10 & 0 \\ 0 & 5 \end{pmatrix} = \begin{pmatrix} 1 & .5 \\ .5 & 1 \end{pmatrix}.$$

Mean–variance portfolio theory relies heavily on the existence of an inverse covariance matrix $\mathbf{C}^{-1}$, which satisfies the basic property that $\mathbf{C}^{-1}\mathbf{C} = \mathbf{C}\mathbf{C}^{-1} = \mathbf{I}$ where $\mathbf{I}$ is an $N \times N$ identity matrix. In the case of the above example, the inverse of $\mathbf{C}$ is:

$$\mathbf{C}^{-1} = \begin{pmatrix} 133.33333 & -33.33333 \\ -33.33333 & 33.33333 \end{pmatrix}. \tag{2.22}$$

We shall assume throughout this book that, if not stated otherwise, the covariance matrix $\mathbf{C}$ of an asset return vector $\mathbf{r}$ is *positive definite*, and so has an inverse $\mathbf{C}^{-1}$.[2] Discussion of returns data requirements such that a covariance matrix estimator $\hat{\mathbf{C}}$ is positive definite is provided in Chapter 4.

[2] An $N \times N$ symmetric matrix $\mathbf{B}$ is said to by *positive definite* if $\mathbf{x}'\mathbf{B}\mathbf{x} > 0$ for all vectors $\mathbf{x} \neq \mathbf{0}$. If the weaker condition $\mathbf{x}'\mathbf{B}\mathbf{x} \geqslant 0$ for all vectors $\mathbf{x} \neq \mathbf{0}$, then $\mathbf{B}$ is said to be *positive semi–definite*, or alternatively *non–negative definite*. Positive definiteness of $\mathbf{B}$ is a sufficient condition for the inverse $\mathbf{B}^{-1}$ to exist.

Sample Mean Vector, Covariance Matrix and Correlation Estimators

The minimum variance optimal (MVO) portfolio theory presented in this section assumes that the returns mean vector $\boldsymbol{\mu}$ and covariance matrix $\mathbf{C}$ are known. But in practice these are unknown and must be estimated from observed historical data $\mathbf{r}_t = (r_{t1}, ..., r_{tN})'$, t = 1,...,T. It is common practice to make the incorrect but practically convenient assumption that the $\mathbf{r}_t$ have a constant mean $\boldsymbol{\mu}$ and a constant covariance matrix $\mathbf{C}$ during this time interval. In that case the standard estimator of $\boldsymbol{\mu}$ is the $N \times 1$ vector of sample means

$$\hat{\boldsymbol{\mu}} = \bar{\mathbf{r}} = \frac{1}{T}\sum_{t=1}^{T}\mathbf{r}_t = (\bar{r}_1, \bar{r}_2, ..., \bar{r}_N)', \quad \text{where } \bar{r}_i = \frac{1}{T}\sum_{t=1}^{T} r_{ti},\ i = 1, 2, \cdots, N \tag{2.23}$$

and the standard estimator of the covariance matrix $\mathbf{C}$ is the $N \times N$ sample covariance matrix estimator $\hat{\mathbf{C}}$, consisting of the pairwise sample covariance estimators $\hat{C}_{ij} = \hat{\text{cov}}(r_i, r_j)$ given by

$$\hat{C}_{ij} = \frac{1}{T}\sum_{t=1}^{T}(r_{ti} - \bar{r}_i)(r_{tj} - \bar{r}_j),\ i, j = 1, 2, \cdots, N \tag{2.24}$$

where $\hat{C}_{ii} = \hat{\sigma}_i^2, i = 1, 2, \cdots, N$ are the sample variance estimators. The corresponding pairwise sample correlation estimates are:

$$\hat{\rho}_{ij} = \frac{\widehat{\text{cov}}(r_i, r_j)}{\hat{\sigma}_i \cdot \hat{\sigma}_j} = \frac{\hat{C}_{ij}}{\sqrt{\hat{C}_{ii}}\sqrt{\hat{C}_{jj}}}\ i, j = 1, \cdots, N \tag{2.25}$$

where $\hat{\rho}_{ii} = 1,\ i = 1, 2, \cdots, N$.

In matrix and vector form, the sample covariance matrix may be written in the form:

$$\hat{\mathbf{C}} = \frac{1}{T}\sum_{t=1}^{T}(\mathbf{r}_t - \hat{\boldsymbol{\mu}})(\mathbf{r}_t - \hat{\boldsymbol{\mu}})'. \tag{2.26}$$

and sample pairwise correlations are extracted from elements of $\hat{\mathbf{C}}$. A sample correlation matrix $\hat{\mathbf{R}}$ may be expressed in terms of the sample correlation matrix in manner similar to the representation (2.20) for a correlation matrix $\mathbf{R}$.

It may be shown that the above sample covariance matrix is the maximum–likelihood estimator (MLE) of the true but unknown covariance matrix $\mathbf{C}$ in the idealized case that the returns have a multivariate normal distribution $MVN(\boldsymbol{\mu}, \mathbf{C})$. [3] Many of the examples in this section are based on formulas that assume a true mean vector and true covariance matrix is known, and we often construct such "true" values by using the above estimates based on actual returns.

We remark that a number of variations on the above standard estimators are of use in portfolio construction, including: exponentially–weighted moving average versions that down–weight observations fur-

[3] See for example Section 4.3 of Mardia et al. (1979).

ther in the past, robust estimators that avoid outliers distortion by down–weighting multi–dimensional returns outliers, shrinkage estimators to reduce estimation error, and factor–model based return and covariance matrix estimators. We discuss these at length in subsequent chapters.

2.2.2 Portfolio Weights, Mean Return and Variance

As mentioned earlier, in this chapter we consider only myopic investors whose investment horizon is a single time period, e.g., a month, a week, or a day, where the investment is made at time $t-1$ and the portfolio return is evaluated at time t. Assume we invest v_{t-1} dollars in a portfolio of assets $A_1, A_2, \cdots, A_N$ at time $t-1$ with $\mathrm{v}_{n,t-1}$ dollars in asset n for $n = 1, 2, \cdots, N$, in which case

$$\mathrm{v}_{t-1} = \sum_{n=1}^{N} \mathrm{v}_{n,t-1}.$$

At time t the value of the assets will change to $\mathrm{v}_{n,t}$, $n = 1, 2, \cdots, N$ and correspondingly

$$\mathrm{v}_t = \sum_{n=1}^{N} \mathrm{v}_{n,t}.$$

The portfolio arithmetic return $r_{P,t}$ from time $t-1$ to time t is:

$$\begin{aligned} r_{P,t} &= \frac{\mathrm{v}_t - \mathrm{v}_{t-1}}{\mathrm{v}_{t-1}} \\ &= \sum_{n=1}^{N} \frac{\mathrm{v}_{n,t-1}}{\mathrm{v}_{t-1}} \cdot \frac{\mathrm{v}_{n,t} - \mathrm{v}_{n,t-1}}{\mathrm{v}_{n,t-1}} \\ &= \sum_{n=1}^{N} w_{n,t-1} \frac{\mathrm{v}_{n,t} - \mathrm{v}_{n,t-1}}{\mathrm{v}_{n,t-1}} \end{aligned}$$

where $w_{n,t-1} = \dfrac{\mathrm{v}_{n,t-1}}{\mathrm{v}_{t-1}}$ are the portfolio *weights* at time $t-1$.

With k_n being the number of shares held of asset n held at both time $t-1$ and time t, and $p_{n,t-1}$and $p_{n,t}$ the price per share of asset n at times $t-1$ and t, respectively, we have

$$\begin{aligned} \frac{\mathrm{v}_{n,t} - \mathrm{v}_{n,t-1}}{\mathrm{v}_{n,t-1}} &= \frac{k_n \cdot p_{n,t} - k_n \cdot p_{n,t-1}}{k_n \cdot p_{n,t-1}} \\ &= \frac{p_{n,t} - p_{n,t-1}}{p_{n,t-1}} \\ &= r_{n,t} \end{aligned}$$

where $r_{n,t}$ is the arithmetic return of the investment in asset n at time t. Thus the portfolio return is

$$r_{P,t} = \sum_{n=1}^{N} w_{n,t-1} r_{n,t} = \mathbf{w}'_{t-1}\mathbf{r}_t \tag{2.27}$$

where the portfolio weights $w_{n,t-1}$ are

$$w_{n,t-1} = \frac{\mathrm{V}_{n,t-1}}{\mathrm{V}_{t-1}} \; n = 1, 2, \cdots, N. \tag{2.28}$$

It is convenient for much of our discussion to drop the time subscript above and write the portfolio return as:

$$r_P = \mathbf{w}'\mathbf{r}. \tag{2.29}$$

It just needs to be remembered that the weights are set at the beginning of the time interval over which returns are computed at the end of the time interval.

Portfolio Mean and Variance

The portfolio mean return $\mu_P = \mathrm{E}[r_P]$ can be expressed in terms of the asset mean return vector and portfolio weight vector as:

$$\begin{aligned} \mu_P &= \mathrm{E}[r_P] \\ &= \mathrm{E}[\mathbf{w}'\mathbf{r}] \\ &= \mathbf{w}' E[\mathbf{r}] \\ &= \mathbf{w}'\boldsymbol{\mu}. \end{aligned} \tag{2.30}$$

The portfolio variance $\sigma_P^2 = \mathrm{var}(r_P)$ may be expressed in terms of a quadratic form in the returns covariance matrix $\mathbf{C}$ as follows:

$$\begin{aligned} \sigma_P^2 &= \mathrm{var}(r_P) \\ &= \mathrm{var}(\mathbf{w}'\mathbf{r}) \\ &= \mathrm{E}\left[(w'r - E(\mathbf{w}'\mathbf{r}))^2\right] \\ &= \mathrm{E}\left[(\mathbf{w}'(\mathbf{r}-\boldsymbol{\mu}))^2\right] \\ &= \mathrm{E}\left[\mathbf{w}'(\mathbf{r}-\boldsymbol{\mu})(\mathbf{r}-\boldsymbol{\mu})'\mathbf{w}\right] \\ &= \mathbf{w}'\mathbf{C}\mathbf{w}. \end{aligned} \tag{2.31}$$

Since portfolio variance is a quadratic form, the assumption of a positive definite covariance matrix means that the portfolio variance is positive for any non–zero weight vector $\mathbf{w}$, and we can take the square root of σ_P^2. Thus, the portfolio volatility is:

$$\sigma_P = (\mathbf{w}'\mathbf{C}\mathbf{w})^{1/2}. \tag{2.32}$$

Portfolio Variance Decomposition

Using the covariance linearity property $\sum_{i=1}^{n} a_i \operatorname{cov}(x, y_i) = \operatorname{cov}(x, \sum_{i=1}^{n} a_i \cdot y_i)$, the portfolio variance may be written in the summation forms:

$$\begin{aligned}
\sigma_P^2 &= \sum_{i=1}^{N} \sum_{j=1}^{N} w_i \cdot w_j \cdot \operatorname{cov}(r_i, r_j) \\
&= \sum_{i=1}^{N} w_i \operatorname{cov}(r_i, \sum_{j=1}^{N} w_j r_j) \\
&= \sum_{i=1}^{N} w_i \operatorname{cov}(r_i, r_P).
\end{aligned} \tag{2.33}$$

The above expression decomposes portfolio variance into a weighted sum of the covariances of the returns r_i with the portfolio return r_P. As such, the covariance $\operatorname{cov}(r_i, r_P)$ represents a contribution to the portfolio variance, and hence a contribution to the portfolio volatility σ_P. This is a very important result that we refer to in the Section 2.5.1 discussion of the *Capital Asset Pricing Model*.

Portfolio Volatility Decomposition

We now show that portfolio volatility σ_P has a linear decomposition analogous to the variance decomposition (2.33). The effect on σ_P of a small perturbation in an individual portfolio weight w_i is measured by the derivative of σ_P with respect to w_i, and such derivatives are referred to as the *marginal contributions of risk* $MCTR_i,\ i = 1, 2, \cdots, N$. A formula for the $MCTR_i$ is easily obtained by observing that the vector of partial derivatives of σ_P with respect to the weights may be expressed as:

$$\begin{aligned}
\frac{\partial \sigma_{\mathbf{P}}}{\partial \mathbf{w}} &= \frac{\partial \left(\mathbf{w}'\mathbf{C}\mathbf{w}\right)^{1/2}}{\partial \mathbf{w}} \\
&= \frac{1}{2} \left(\mathbf{w}'\mathbf{C}\mathbf{w}\right)^{-1/2} 2\,\mathbf{C}\,\mathbf{w} \\
&= \frac{\mathbf{C}\mathbf{w}}{\sigma_P}.
\end{aligned}$$

Thus the marginal contribution to risk of the i-th asset can be written in the following form:

$$\begin{aligned}
MCTR_i &= \frac{(\mathbf{C}\mathbf{w})_i}{\sigma_P} \\
&= \frac{\sum_{j=1}^{N} \operatorname{cov}(r_i, r_j)\, w_j}{\sigma_P} \\
&= \frac{\operatorname{cov}(r_i, r_P)}{\sigma_P}.
\end{aligned} \tag{2.34}$$

It follows that

$$\sum_{i=1}^{N} w_i MCTR_i = \frac{\sum_{i=1}^{N}\sum_{j=1}^{N} w_i \operatorname{cov}(r_i, r_j)\, w_j}{\sigma_P}$$
$$= \sigma_P$$

where we have used the first line of (2.33) to replace the above right-hand-side numerator by σ_P^2. Summarizing, we have the portfolio volatility decomposition

$$\sigma_P = \sum_{i=1}^{N} w_i MCTR_i. \tag{2.35}$$

In the special case of an equally weighted portfolio whose weights are $1/N, i = 1, 2, \cdots, N$, the contribution of the i-th asset to the portfolio volatility is just $MCTR_i/N$. Note that the summand $w_i MCTR_i$ above is the contribution of the i-th asset to the portfolio volatility, and these summands are commonly referred to in the literature as the portfolio's *risk components*.

In a later chapter on Risk Analysis, we discuss that standard deviation, equivalently volatility, is not a proper risk measure. Nevertheless, portfolio standard deviation is commonly referred to as "portfolio risk" in much of the MVO literature. However, we shall refer to asset and portfolio returns standard deviation simply as volatility except in certain contexts where the historical convention is to refer to standard deviation and variance as risk indicators, for example just above where $MCTR_i$ stands for marginal contribution to risk (and with hindsight, it would have been preferable to use $MCTV_i$ as marginal contribution to volatility).

2.2.3 Fully Invested Portfolios with Two and Three Assets

The condition that a portfolio is fully invested in risky assets is represented by the condition that the weight vector satisfies the *full–investment* constraint:

$$\mathbf{w}'\mathbf{1} = \sum_{n=1}^{N} w_n = 1 \tag{2.36}$$

where $\mathbf{1}$ is an N–vector of ones. Note that there is no assumption here that the weights have to be non–negative, so negative weights that correspond to borrowing through *short–selling* are generally possible.

We now investigate the consequences of the full–investment constraint for the special cases of $N = 2$ and $N = 3$ assets.

Two Risky Assets

Consider the simple case where the investor has just two risky assets, with return r_1 and r_2, respectively, to invest in, and the investor knows that these two assets have means $\mu_1 > \mu_2$, volatilities $\sigma_1 > \sigma_2$, and correlation coefficient ρ. In this case the covariance matrix is the two–by–two matrix:

$$\mathbf{C} = \begin{pmatrix} \sigma_1^2 & \sigma_1\sigma_2\rho \\ \sigma_1\sigma_2\rho & \sigma_2^2 \end{pmatrix}. \tag{2.37}$$

The first asset has a higher mean return than the second asset, but also a higher volatility, and that risk gives the investor pause. The second asset has a lower risk, but its mean return is unattractive. The investor wonders if a better return and risk can be obtained by diversifying the investment between the two assets. Thus, the investor forms a portfolio with a non–negative fraction w invested in the first asset and a non–negative amount $1-w$ invested in the second asset, and the portfolio has the random return

$$r_p = w\, r_1 + (1-w)\, r_2, \quad 0 \leqslant w \leqslant 1. \tag{2.38}$$

Here we initially restrict w and $1-w$ to be non–negative, but will subsequently see what happens when we drop this restriction.

The portfolio mean return is

$$\mu_P = w\,\mu_1 + (1-w)\,\mu_2 \tag{2.39}$$

and the portfolio variance $\sigma_P^2 =$ is

$$\begin{aligned} \sigma_P^2 &= w^2\sigma_1^2 + 2w\cdot(1-w)\operatorname{cov}(r_1, r_2) + (1-w)^2\sigma_2^2 \\ &= w^2\sigma_1^2 + 2w\cdot(1-w)\cdot\rho\cdot\sigma_1\sigma_2 + (1-w)^2\sigma_2^2 \end{aligned}. \tag{2.40}$$

The portfolio's standard deviation σ_P is given by the square root of the right hand side of the above equation, and one can easily make a plot of $\mu_P = \mu_P(w)$ versus $\sigma_P = \sigma_P(w)$ for various values of w by evaluating the pairs $(\sigma_P(w), \mu_P(w))$ on a fine grid of values of w between 0 and 1.

Example 2.2. Suppose the first asset has $\sigma_1 = 20\%$, $\mu_1 = 10\%$, the second asset has $\sigma_2 = 15\%$, $\mu_2 = 4\%$, and the correlation between the returns of the two assets is $\rho = 0$. The resulting μ_P versus σ_P plot is shown in Figure 2.1.

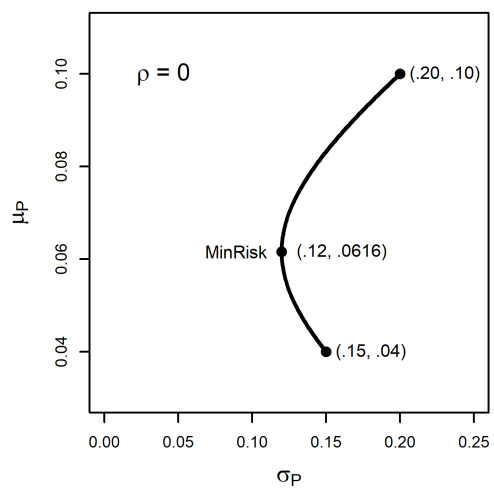

Fig. 2.1 Mean Versus Standard Deviation Plot of Two–Asset Portfolio with Correlation $\rho = 0$.

The portfolio labeled MinRisk occurs when $w = w_{\text{MinRisk}} = 0.36$. and the corresponding volatility and mean return are 0.12 and 0.0616, respectively. If the investor wants a portfolio with a return that has the minimum attainable risk for an investment of \$1,000, then she will invest \$360 (36%) in the first asset, and \$640 (64%) in the second asset. By taking such a diversified position in the two assets, the investor will have achieved a risk of about 80% of that of the second asset, about 60% that of the first asset, with a return reduction of $(0.20 - 0.12)/0.20 = 0.40$, or 40% of the first asset. To some investors this may represent an attractive risk reduction through diversification, at the price of loss of mean return.

Note that an investor would never hold a position on the lower portion of the curve in 2.1 below the MinRisk portfolio. The reason is that for any portfolio on the lower portion of the curve, there is a portfolio on the upper part of the curve with the same risk but higher return.

With the above result for $\rho = 0$ in mind, it is of interest to see how the mean return versus volatility plots look for other values of correlation. One can get a sense of this by considering the two extreme correlations of $\rho = +1$ and $\rho = -1$, i.e., when the two assets are perfectly positively correlated and when they are perfectly negatively correlated. The result for these two cases, along with the previous case of $\rho = 0$, are shown in 2.2.

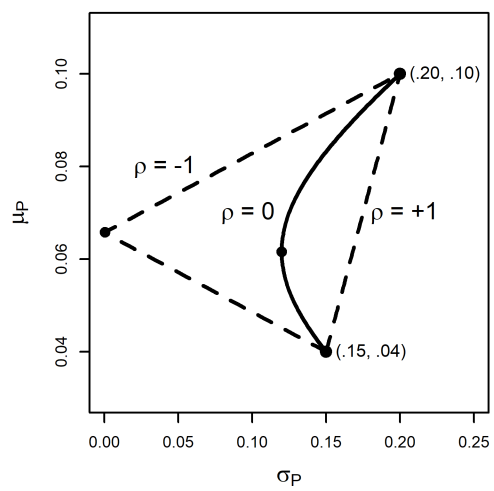

Fig. 2.2 Mean Versus Standard Deviation Plot of Two–Asset Portfolio for Three Values of the Correlation ρ

For $\rho = +1$ the curve is the straight line connecting the positions of the two assets in return versus risk space, and the curves for values $0 < \rho < 1$ lie between this straight line and the curve for $\rho = 0$. For $\rho = -1$ the curve consists of the two straight–line segments that intersect at $\sigma_P = 0$ when $w = 0.43$,for which the portfolio has a mean return 0.0658 with zero risk. An investor would love to find two assets that are perfectly negatively correlated, but this never happens in practice. The curves for values $-1 < \rho < 0$ lie between this piecewise linear curve and the curve for $\rho = 0$.

Diversification Benefit of Small Correlations

The two-assets portfolio variance equation (2.40) and Figure 2.2 reveal the benefit of small asset return correlations, with smaller correlations resulting in lower portfolio variance (portfolio risk) for any fixed pairs of returns means and variances. Typically one is not able to select assets with zero correlation, but use of assets with positive correlations that are as small as possible is generally beneficial in reducing portfolio risk.

Short–Selling

In the above discussion of a two asset portfolio we assumed that both w and $1 - w$ are non–negative, which means that if we invest \$100, then $100 \cdot w$ dollars is invested in the first asset and $100 \cdot (1 - w)$ dollars is invested in the second asset. What is your confidence investing in the first asset is so high that you want to invest more than \$100 in it, but \$100 is all you have available to invest? Then you could borrow money by *short–selling* the second asset, for which the basic mechanism is that your broker

borrows stock to give you a loan under strict contract conditions. Let's say you short–sell \$25 of the second asset, and invest \$125 in the first asset. Then the weight on the first asset is $w = 1.25$ and the weight on the second asset is $1 - w = -.25$. So a short sale in an asset is reflected as a negative weight for the asset.

Let's see what the range of the mean versus volatility curve is for the Example 2.2 if we allow short–selling of the second asset in an amount not to exceed 25%. The result is shown in 2.3.

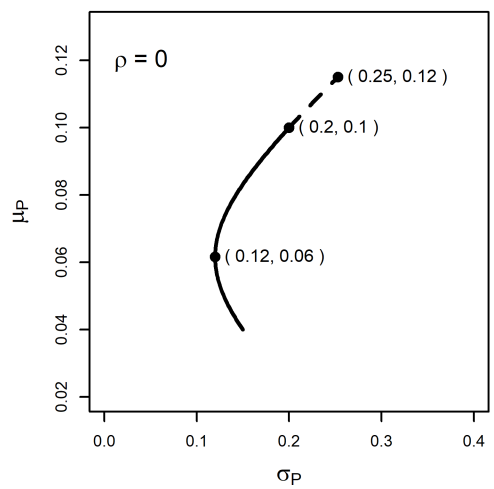

Fig. 2.3 Mean Versus Volatility Curve with Up to 25% Short Selling of Second Asset Allowed

The dashed line extension of the original mean versus volatility solid curve represents that additional portfolio opportunities as a result of short–selling the second asset. For a maximum of 25% short selling of the second asset, the portfolio mean return is 0.25 and the volatility is 0.12, which compares with a maximum of 0.2 and 0.1 without short selling.

Note that by allowing arbitrarily large short selling of the second asset, one can get a portfolio with arbitrarily high mean return, which is of course accompanied by an arbitrarily large risk. In addition, short sellers face the risk of a *short squeeze*, in which it becomes virtually impossible to buy back the shares they have borrowed and shorted to close out their trade, driving prices higher and inducing large (and sometimes career–destroying) losses.

Three Risky Assets

In the case of two risky assets, the portfolio mean versus volatility curve represents the totality of achievable return and risk. This situation changes dramatically with three or more assets. Consider the case of three assets that include the previous two assets with volatility–mean pairs (0.20, 0.10) and (0.15, 0.04),

respectively, plus a third asset with volatility–mean pair (0.10, 0 .02). In this case there are three unique two–asset portfolios whose mean versus volatility curves are shown in 2.4.

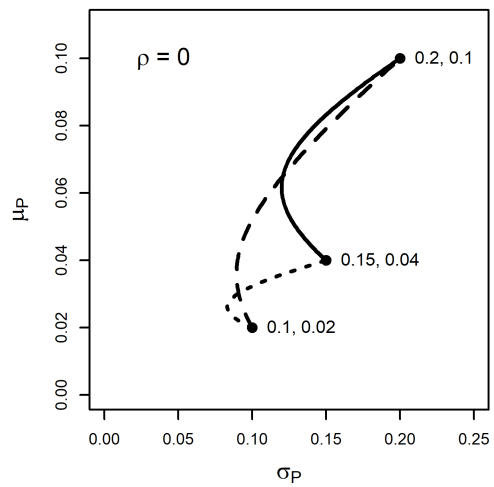

Fig. 2.4 Mean Versus Volatility Curves for the Three Pairwise Choices of Two Asset Portfolios

One may intuitively think that if one could in some way construct a portfolio by combining three two–assset portfolios, one might obtain a portfolio whose mean versus volatility curve is superior to all three of the portfolio curves in Figure 2.4 by virtue of lying to the left of those three portfolio curves. In fact this is possible, not by combining the three portfolios based on two assets at a time, but by forming portfolios of all three assets. Thus we consider the three–asset portfolio return:

$$r_P = \mathbf{w}'\mathbf{r} = (w_1, w_2, w_3)\,(r_1, r_2, r_3)'$$

where the three asset returns are all uncorrelated, with mean vector $\boldsymbol{\mu} = (0.10, 0.04, 0.02)'$ and volatilities vector $\boldsymbol{\sigma} = (0.20, 0.15, 0.10)'$. Since the returns are assumed to be uncorrelated, the covariance matrix is the diagonal matrix $\mathbf{diag}(0.20, 0.15, 0.10)$. Thus the portfolio mean and variance are:

$$\mu_P = \mathbf{w}'\begin{pmatrix} 0.10 \\ 0.04 \\ 0.02 \end{pmatrix} \qquad \sigma_P^2 = \mathbf{w}'\mathbf{C}\mathbf{w} = \mathbf{w}'\begin{pmatrix} (0.20)^2 & 0 & 0 \\ 0 & (0.15)^2 & 0 \\ 0 & 0 & (0.10)^2 \end{pmatrix}\mathbf{w}.$$

To get a sense of what portfolio means versus volatilities can be achieved with the above asset mean vector and covariance matrix, we generated 500 randomly chosen weight vectors $\mathbf{w}_{3,m}$, $m = 1, \cdots, 500$, whose three components are non–negative and satisfy the full–investment constraint that they add to

one. Using the above expressions for μ_P and $\sigma_P = \sqrt{\mathbf{w}_3'\mathbf{C}\mathbf{w}_3}$ results in 500 random pairs $\left(\sigma_{P,m},\ \mu_{P,m},\ m = 1, \cdots, 500\right)$ that are shown in Figure 2.5.[4]

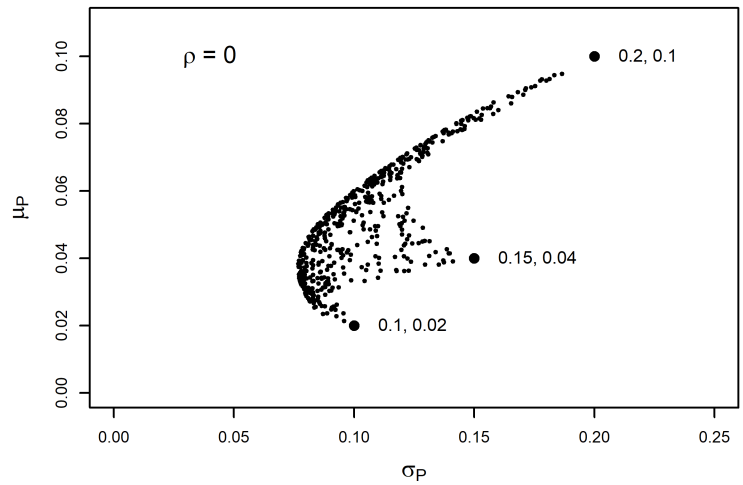

Fig. 2.5 500 Random long–only Portfolios of Three Assets

Note that the solid dots in Figure 2.5 represent the three distinct single asset portfolios obtained when one of the three weights is equal to one, and the other two weights are zero. The distinct feature of the figure is that the points in the plot, which represent all possible portfolio mean–volatility combinations, appear to have a smooth curved left boundary that is similar in shape to the curve in Figure 2.1. This curved boundary appears to represent the minimum variance that a portfolio can achieve for each value of portfolio mean return. Indeed in the following section we show that there is a mathematically well defined curve, called the *portfolio frontier*, that has the minimum possible variance for each possible portfolio mean return. This curve represents the mean and standard deviation of the set of all MVO portfolios.

Figure 2.5 also reveals the MinVar portfolios diversification benefit of having considerably smaller variance than that of any one of the three assets, and also smaller than the variance of each of the three two asset portfolios shown in Figure 2.2.

2.2.4 Global Minimum Variance Portfolios

Formulas exists for the portfolio weights $\mathbf{w}_{\text{gmv}}$, mean return μ_{gmv} and variance σ^2_{gmv} of a portfolio that achieve the minimum possible variance that can be achieved by any *fully-invested* portfolio, where the

[4] The 500 values of $\mathbf{w}_{3,m}$ were generated as uniformly distributed on the simplex $w_1 + w_2 + w_3 = 1$,with each $w_i \geqslant 0$, using the function `simplex.sample(3,500)` in the R package `hitandrun`, and the 500 samples are return in the 3×500 matrix `simplex.sample(3, 500)$samples`.

term fully-invested means that the portfolio weights are constrained to sum to 1. Such a portfolio is referred to as a *global minimum variance* (GMV) portfolio. In this section we will derive those formulas using constraint that the weights sum to 1. It is to be emphasized that with the only constraint being a full-investment constraint, there is no restriction that the weights be positive – it will indeed sometimes happen that some of the weights are negative, and corresponding the sum of the positive weights will be greater than 1 so that the sum of all the weights sum to 1. Before deriving the formulas for the GMV portfolio weights, mean and variance, we provide the following motivational example.

Derivation of GMV Porfolio Weight Vector, Mean Return and Variance

A GMV portfolio weight vector $\mathbf{w}_{\text{gmv}}$ based on a positive definite covariance matrix $\mathbf{C}$ is the solution to the minimization problem

$$\mathbf{w}_{\text{gmv}} = \underset{\mathbf{w}}{\operatorname{argmin}}\, \sigma^2_{\text{gmv}}(\mathbf{w}) = \underset{\mathbf{w}}{\operatorname{argmin}}\, \mathbf{w}'\mathbf{C}\mathbf{w} \tag{2.41}$$

subject to the full investment constraint

$$\mathbf{w}'\mathbf{1} = 1. \tag{2.42}$$

Minimization problems subject to linear equality constraints, such as the one above, are solved using the method of Lagrange multipliers, as follows. A *Lagrangian* is formed by augmenting the objective function of unknown parameters with a zero–equality form of the constraint, multiplied by an unknown Lagrange multiplier. The problem is then solved by setting the derivative of the Lagrangian equal to zero and solving for the unknown parameters, and finally, solving for the Lagrange multiplier by using the equality constraint. We proceed with this method for the above GMV optimization problem.

For the GMV problem, the Lagrangian is

$$L(\mathbf{w}) = \frac{1}{2}\mathbf{w}'\mathbf{C}\mathbf{w} + \gamma\,(1 - \mathbf{w}'\mathbf{1}) \tag{2.43}$$

and setting the derivative of the Lagrangian equal to zero, and solving gives

$$\mathbf{w}_{\text{gmv}} = \lambda\mathbf{C}^{-1}\mathbf{1}$$

where $\mathbf{C}^{-1}$ is positive definite. Solving the constraint equation (2.42) for γ gives $\gamma = (\mathbf{1}'\mathbf{C}^{-1}\mathbf{1})^{-1}$ and so

$$\mathbf{w}_{\text{gmv}} = \frac{\mathbf{C}^{-1}\mathbf{1}}{\mathbf{1}'\mathbf{C}^{-1}\mathbf{1}}. \tag{2.44}$$

Using the general expression (2.30) for portfolio mean return gives

$$\mu_{\text{gmv}} = \mathbf{w}'_{\text{gmv}}\boldsymbol{\mu} = \frac{\mathbf{1}'\mathbf{C}^{-1}\boldsymbol{\mu}}{\mathbf{1}'\mathbf{C}^{-1}\mathbf{1}}. \tag{2.45}$$

We get the expression for the variance of the global minimum variance portfolio by using the general expression ((2.31)) for portfolio variance.

$$\sigma^2_{\text{gmv}} = \mathbf{w}'_{\text{gmv}}\mathbf{C}\mathbf{w}_{\text{gmv}} = (\mathbf{1}'\mathbf{C}^{-1}\mathbf{1})^{-1} \tag{2.46}$$

and correspondingly the volatility of the GMV portfolio is

$$\sigma_{\text{gmv}} = (\mathbf{1}'\mathbf{C}^{-1}\mathbf{1})^{-1/2}. \tag{2.47}$$

Example 2.3. GMV Portfolio for Three Asset Portfolio

For the three–stock portfolio example used for Figures 2.4 and 2.5, where $\boldsymbol{\mu}_3 = (0.10, 0.04, 0.02)'$, $\boldsymbol{\sigma}_3 = (0.20, 0.15, 0.10)'$, and the stock returns are uncorrelated, it turns out that the GMV portfolio weight vector is $\mathbf{w}_{\text{gmv}} = (0.148, 0.262, 0.590)'$, its volatility is $\sigma_{\text{gmv}} = 0.077$, and its mean return is $\mu_{\text{gmv}} = 0.037$. It is left as an exercise to confirm this.

Example 2.4. GMV Portfolio for N Uncorrelated Assets

For the case of uncorrelated assets, the covariance matrix is diagonal with diagonal elements σ^2_n, $i = 1, 2, \cdots, N$, and in this case the GMV portfolio weight vector is of the form

$$\mathbf{w}'_{\text{gmv}} = \frac{\mathbf{C}^{-1}\mathbf{1}}{\mathbf{1}'\mathbf{C}^{-1}\mathbf{1}} = c \cdot \left(\frac{1}{\sigma^2_1}, \frac{1}{\sigma^2_2}, \cdots, \frac{1}{\sigma^2_N}\right) \tag{2.48}$$

where

$$c = \left(\sum_{i=1}^{N} \frac{1}{\sigma^2_n},\right)^{-1}. \tag{2.49}$$

The GMV portfolio mean return is

$$\mu_{\text{gmv}} = \frac{\mathbf{1}'\mathbf{C}^{-1}\boldsymbol{\mu}}{\mathbf{1}'\mathbf{C}^{-1}\mathbf{1}} = c \cdot \sum_{n=1}^{N} \frac{\mu_n}{\sigma^2_n} \tag{2.50}$$

where μ_n is the mean return of the n–th asset. Note that two assets with the same mean return contribute to the GMV portfolio mean return inverse proportion to their variances. Consequently, an asset with a higher mean return but also a higher variance than another asset can contribute less to the portfolio mean return than than the other asset! The GMV portfolio volatility is:

$$\sigma_{\text{gmv}} = (\mathbf{1}'\mathbf{C}^{-1}\mathbf{1})^{-1/2} = \left(\sum_{n=1}^{N} \frac{1}{\sigma^2_n}\right)^{-1/2}. \tag{2.51}$$

Note that the denominator of the expression (2.45) is a quadratic form based on the positive definite matrix $\mathbf{C}^{-1}$ and so it is always positive. However, this is not the case for the numerator which is a linear

form that can be positive or negative. Thus the mean return of the global minimum variance portfolio will be positive if and only if $\mathbf{1}'\mathbf{C}^{-1}\boldsymbol{\mu} > 0$.

Example 2.5. Example of a GMV Portfolio with a Negative Mean Return

Although the mean return of a GMV portfolio is usually positive, there can sometimes be economic conditions that lead to a GMV portfolio with a negative mean return. For example, this happens for a portfolio of IBM, GE and EXON stocks based on weekly returns from November 7, 1980 to October 30, 1981, whose returns are shown in Figure 2.6.

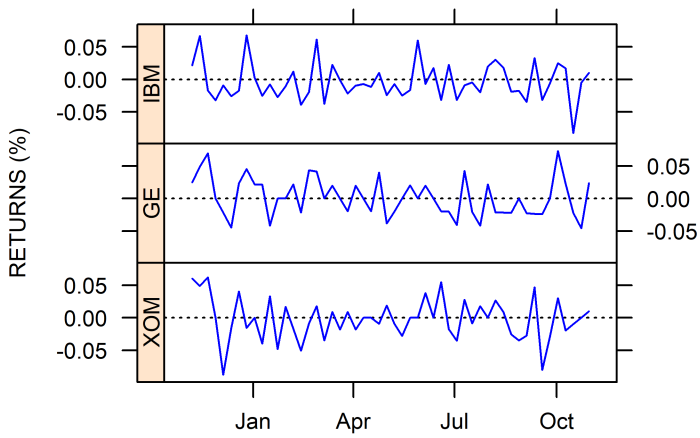

Fig. 2.6 Weekly Returns for IBM, GE and Exon from Late 1980 to Late 1981. Data is from the Center for Research in Security Prices, LLC (CRSP®)

The GMV portfolio has roughly equal weights $\mathbf{w}_{\text{gmv}} = (.349, .350, .301)'$, with a modest annualized volatility $\sigma_{\text{gmv}} = 16.4\%$, and a quite unpleasant negative annualized mean return $\mu_{\text{gmv}} = -6.3\%$.

The Equal Marginal Contribution to Risk Property

A global minimum variance portfolio has the special property that each asset in the portfolio has the same marginal contribution to risk (MCTR). This is easily shown by plugging the GMV weight vector formula (2.44) into the top line of the MCTR definition (2.34):

$$\text{MCTR}_{\text{gmv},i} = \frac{1}{\sigma_{\text{gmv}}} \frac{\left(\mathbf{C}\mathbf{C}^{-1}\mathbf{1}\right)_i}{\mathbf{1}'\mathbf{C}^{-1}\mathbf{1}} = \sigma_{\text{gmv}},\ i = 1, 2, \cdots, N.$$

Conversely, it can be shown that the only portfolio that has equal $\text{MCTR}_{\text{gmv},i}$ values is the GMV portfolio, and a proof of this is left as an exercise.

Global minimum variance portfolios can often be of use in practice, in part due to the fact that GMV portfolio weights depend only on the unknown covariance matrix, which can be estimated reasonably well from the data provided the number of observations are fairly large compared with the number of assets.[5] Interesting studies in this regard were provided by Scherer (2011) and Clarke et al. (2011). Here we illustrate the applications value of GMV portfolios with the following example of a portfolio of 20 stocks, whose covariance matrix is estimated using a moving window of 5 years of monthly returns, so 60 observations of the returns of 20 stocks.

2.3 Portfolio Performance Analysis

GMV portfolios provide a convenient setting for becoming acquainted with *back-testing* based portfolio performance analysis methods that are commonly used in practice for a wide variety of portfolio types, including those treated in this chapter and elsewhere in the book. Back-testing consists of computing the weights of a portfolio according to some criteria for each time position of a *rolling window* of length T, and analyzing the portfolio's performance based on its arithmetic return and cumulative gross return in the subsequent periods. The portfolio is *rebalanced* by computing new weights regularly at fixed rebalancing frequency, such as "weekly", "monthly" or "quarterly". For the GMV portfolio performance analysis in this section, the stock returns are monthly from January 1997 to December 2010, the moving window length is 60 months, and the rebalancing frequency is monthly. Correspondingly, portfolio arithmetic returns and cumulative gross returns are available from February 2002 until December 2010.

The performance analysis methods discussed in the remainder of this section are:

- Cumulative Gross Return, and Drawdown
- Leverage, Turnover, and Diversification
- Returns Based Risk Measures
- Returns Based Risk Adjusted Performance Measures

In what follows, we use the symbol r_t for portfolio arithmetic returns, keeping in mind that for a portfolio P with weight vector $\mathbf{w}_t = (w_{1t}, w_{2t}, \cdots, w_{Kt})'$ and portfolio returns vector $\mathbf{r}_t = (r_{t1}, r_{t2}, \cdots, r_{tK})'$, the portfolios return $r_{P,t}$ is

$$r_{P,t} = \mathbf{w}_t' \mathbf{r}_t \,. \tag{2.52}$$

and the portfolio return $\mathbf{r}_t$at time $t+1$ based on portfolio weight vection

[5] Most importantly, GMV portfolios do not depend on the unknown asset mean returns that are required for the efficient frontier optimal MV portfolios discussed in the remainder of this Section.

2.3.1 Cumulative Gross Return and Drawdown

The *cumulative gross return* $R_{1,T}$ of an individual stock or a portfolio of stocks on a set of T time periods from period 1 to period T is defined as

$$R_{1,T} = \frac{p_T}{p_0} \tag{2.53}$$

where p_0 is the price of the asset at an initial investment time period 0 and p_T is the price of the asset at time period T. Using arithmetic returns

$$\begin{aligned} r_t &= \frac{p_t - p_{t-1}}{p_{t-1}} \\ &= p_t / p_{t-1} - 1 \end{aligned} \tag{2.54}$$

it follows that the cumulative gross return (2.53) can be written in the form:

$$R_{1,T} = (1 + r_1)(1 + r_2) \cdot \cdots \cdot (1 + r_T) \tag{2.55}$$

An important feature of cumulative gross returns is that it accounts for the compounding effect of investing the value of the portfolio at end of each time period to generate a return at the end of the next time period. The *cumulative arithmetic return* is $r_{1,T} = R_{1,T} - 1$, and we susequently refer to it as simply the *cumulative return*. Further discussion of cumulative gross returns and other definitions of return are provided in Chapter 3.

The *drawdown* of a portfolio is defined as follows. Let $p_t, t = 1, \ldots, T$ be the time series values of a portfolio, and define "look-back" maximum values relative to each time u between 1 and t as:

$$p_t^{\max} = \max\{p_u : 1 \leqslant u \leqslant t\}, t = 1, \ldots, T. \tag{2.56}$$

The drawdown d_t at time t is defined as:

$$d_t = p_t^{\max} - p_t, \quad t = 1, 2, \ldots, T. \tag{2.57}$$

and since $d_1 = 0$, we only need to consider d_t for $t \geqslant 2$.

The above definition reveals that drawdowns d_t are either zero or positive. Since large drawdowns represent large losses in some time periods, investors are extremely concerned about the drawdowns of their portfolios, and want to invest in portfolios that have infrequent small drawdowns.

Drawdowns are often stated in *relative drawdown* terms using the formula

$$rd_t = \frac{d_t}{p_t^{\max}}, \; t = 2, 3, \cdots, T$$

which may appear in portfolio performance reports stated in decimal or percentage form. Portfolio manager reports often show time series plots of the portfolio drawdown or relative drawdown. A report

will often also state the maximum value, and perhaps the average value, of the drawdown or the relative drawdown, which are defined, for example, in the case of drawdown, as:

$$
\begin{aligned}
d_{\max} &= \max_t \{d_t : t = 2, \ldots, T\} \\
\bar{d} &= \frac{1}{T-1} \sum_{t=2}^{T} d_t \,.
\end{aligned}
\tag{2.58}
$$

Since drawdowns and relative drawdowns represent losses, they are sometimes reported as negative quantities, and that is the case for the following example.

Example 2.6. Cumulative Gross Return and Drawdown of a GMV Long-Short Portfolio of 10 Smallcap Stocks

Figure 2.7 displays the monthly arithmetic returns of the 10 stocks and the CRSP broad market index (the Market) from January 1997 to December 2010 that are used in this example.

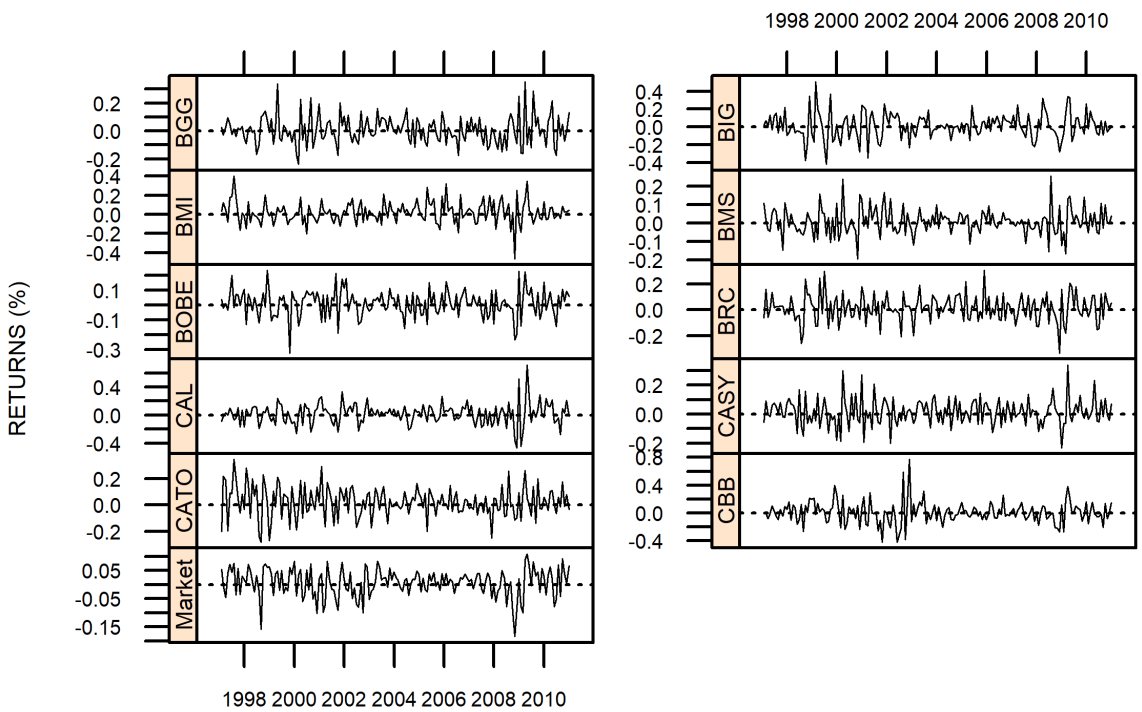

Fig. 2.7 Returns of 10 CRSP Smallcap Stocks and the Market Return from 1997 to 2010

We use the returns of the 10 stocks in Figure 2.7 to compute the cumulative gross returns and drawdown of both a fully-invested GMV long-short portfolio (the GmvLS portfolio), and the Market, from January 2002 to December 2010, where the weights of the GmvLS portfolio are computed each month using a 5-year (60 month) moving window of the stock returns. The reason that the cumulative gross returns and drawdown start in January of 2002 is that the first 5 years of returns from 1997 to 2001 are used to compute the first set of portfolio weights, which are used at the beginning of January 2002 to compute the return of portfolio till the end of January 2002. The GmvLS portfolio weights can be computed each month using (2.44), but *portfolio rebalancing* requires the computation of new portfolio weights for each month, as well as the resulting monthly returns, based on a moving window. These computations are conveniently done using the R package `PortfolioAnalytics` which is designed for such purposes, and we use that package here and elsewhere.

The time series of the 10 GvmLS portfolio weights are displayed in Figure 2.8. The figure provides a good overall visual assessment of the main features of the time evolution of the weights, namely: (a) the stocks with tickers BGG, BIG, CATO and CBB have uniformly small weights during the entire time period 2002 - 2010, and correspondingly the portfolio is concentrated in some of the other 4 stocks; (b) during the early years there are only very small short positions, and the portfolio investment is concentrated in the stocks BMI, BMS, BOBE and CASY; (c) during the height of the financial crisis in 2006–2009 there are substantial short positions in BRC and CAL, and a substantial long position in BMS, whose weight reaches 0.8 in late 2007 and early 2008. Although this GmvLS portfolio is not as diversified as an investor might like, we will see that these behaviors reflect the capability of the GmvLS optimization to maximize cumulative gross return over time relative to the Market by dynamically rebalancing investment in the various stocks in the portfolio.

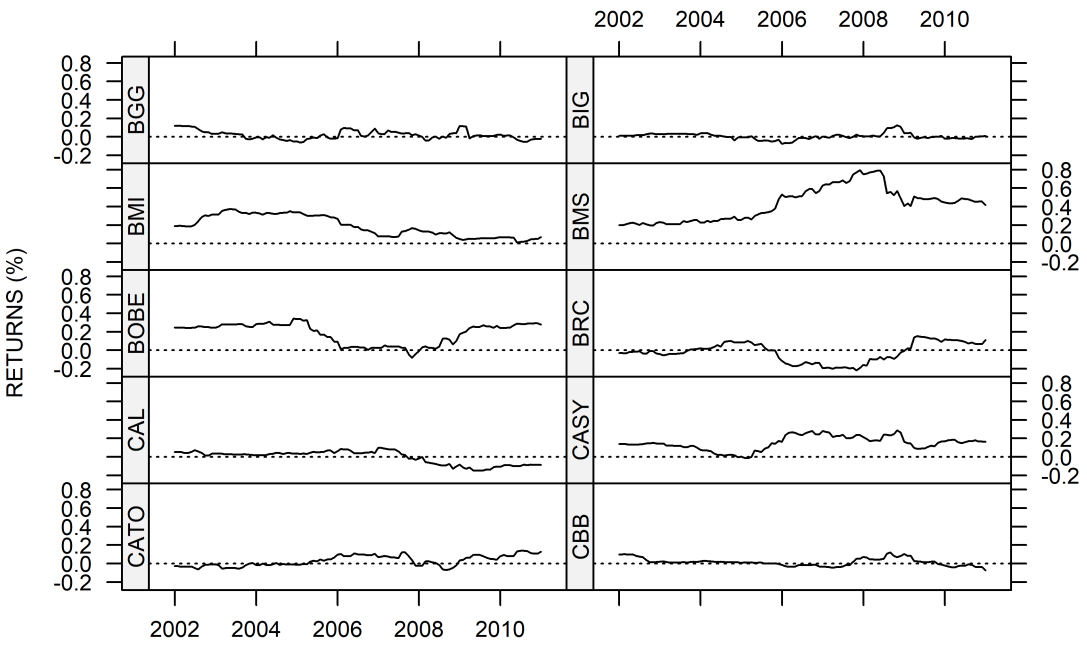

Fig. 2.8 Time Series of Weights of Smallcap10 GmvLO Portfolio

The arithmetic returns of the GmvLS portfolio and the Market displayed in Figure 2.9 reveal that while the GmvLS portfolio is more volatile than the Market, its average returns appear to be somewhat higher than those of the Market. Thus one may think that investments in the GmvLS portfolio is more attractive than an investment in the Market. Unfortunately, plots of arithmetic returns of various portfolios and the Market are not highly effective in assessing their relative return differences. A much better way to do so is to plot the cumulative gross returns, which capture reinvestment benefits, and Figure 2.10 illustrates the benefits of doing so.

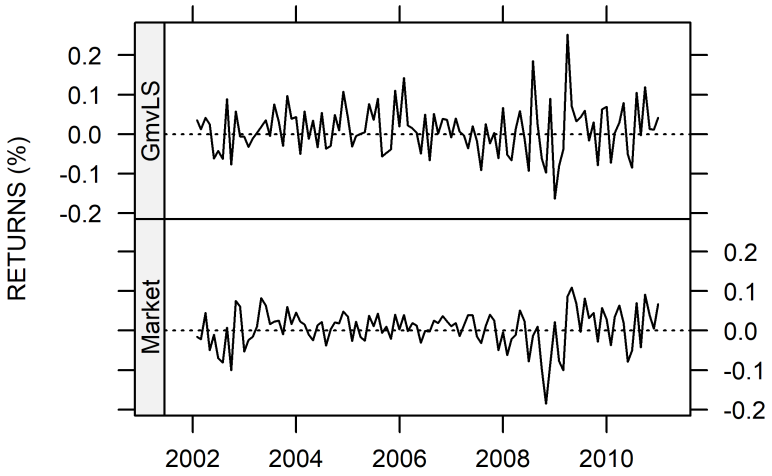

Fig. 2.9 Monthly Arithmetic Returns of the GmvLS Portfolio and the Market from 2002 to 2010

Figure 2.10 uses a solid line for the GmvLS portfolio's cumulative gross return values and a dashed line for those of the Market, assuming initial investments of $1 in each. What is immediately striking about the figure is that the GmvLS portfolio considerably outperforms the Market in terms of cumulative gross return for the entire time period shown, including throughout the financial crisis, which is indicated by the portion of the figure shaded in gray. Furthermore, the GmvLS portfolio has a smaller drawdown than the Market during the crisis period, and it has a shorter recovery time to zero than the Market. This is a payoff for using a theoretically supported GMV portfolio as a foundation for portfolio construction.

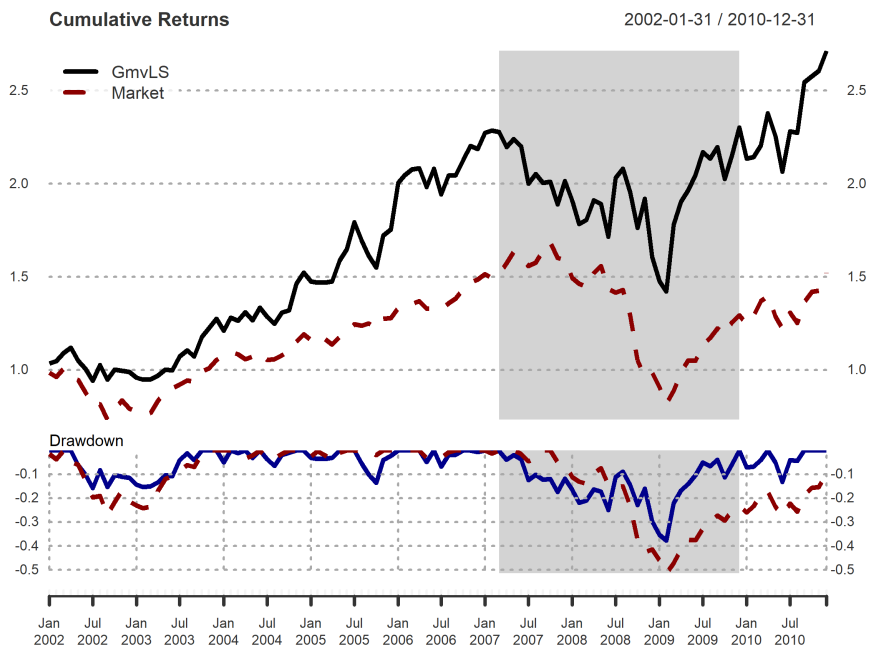

Fig. 2.10 Cumulative Gross Returns of Minimum Variance Long-Short Portfolio Versus Market

There are of course other drawdowns in Figure 2.10, which are shorter and less severe than the one during the crisis. A *drawdown time period* is the shortest positive time interval for which drawdown is equal to zero at the beginning and ending of the period. Sometimes portfolio performance reports will contain listings of the time intervals that contain the several largest drawdowns, e.g., the two or three such intervals. The reader will have no problem in identifying the second most serious drawdown in Figure 2.10 other than the period of the 2007 - 2009 financial crisis. It turns out that the `PerformanceAnalytics` R package has a nice function `table.Drawdowns` that can identify a user specified set of top K drawdowns. The result of using the function for $K = 2$ is Table 2.1, whose dates you should compare with the Drawdown panel in Figure 2.10.

Table 2.1 GmvLO Portfolio's Largest Two Drawdowns Timing and Depths

Begin	Minimum	End	Months	Depth
2007-03-31	2009-02-28	2009-12-31	34	-0.38
2002-05-31	2002-07-31	2003-10-31	18	-0.16

2.3.2 Leverage, Turnover and Diversification

The existence of the shorting of some stocks in the GmvLS portfolio is a form of leverage that allows greater investment in stocks that payoff well than is possible with a long-only constraint. Correspond-

ingly, the sum of the absolute weights can be larger than 1, while the weights sum to 1 due to the full-investment constraint. A common way of characterizing leverage in a long-short portfolio is to define *leverage* as the sum of the absolute values of the weights of the stocks, s, divided by the sum of the weights of the long positions.[6] Figure 2.11 displays the time series of the monthly leverage of the GmvLS portfolio. It is interesting that the GmvLS portfolio has a relatively small leverage with local average ranging from about 1.15 to 1.2 from 2002 to early 2008, but during the latter half of 2008 and the first half of 2009, the leverage grew rapidly to a peak of about 1.45, declining somewhat thereafter.

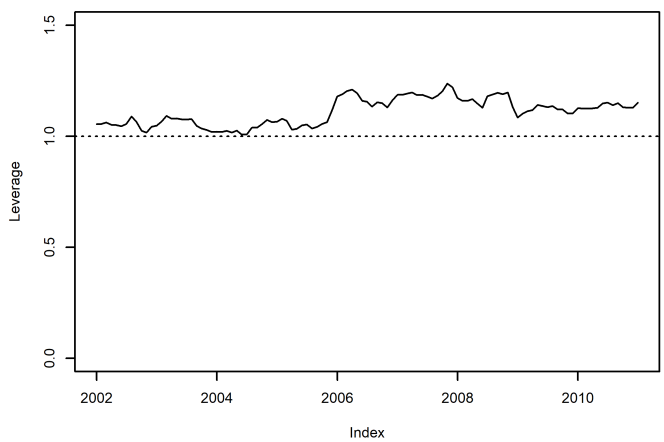

Fig. 2.11 GmvLS Portfolio Leverage Time Series

When a portfolio is rebalanced there is a turnover of assets such as stocks, due to the fact some assets are sold and others are purchased. A commonly used measure of portfolio turnover due to rebalancing at each time t is given by the two-way *turnover* formula

$$\mathrm{TO}_t = \sum_{j=1}^{N} \left| w_{t,j} - w_{t-1,j} \right| \tag{2.59}$$

where the $w_{t-1,j}$, $j = 1, 2, 3, \cdots, N$ are the portfolio weights at the previous rebalancing time $t-1$, and the $w_{t,j}$ are the weights at time t. Turnover is frequently reported in percentage terms. For a long-only portfolio whose weights are between 0 and 1, the triangle inequality shows that the maximum numeric turnover is 2, or 200%. Furthermore, it is easily seen that a *complete turnover value* of 200% is achieved when the sets of non-zero weights at time $t-1$ and time t are disjoint, e.g., a special case of which is $w_{t-1,1} = 1$ and $w_{t,1} = 0$. Portfolio turnover is sometimes reported as a *one-way turnover* given by dividing the above formula by 2, which has that property that $\mathrm{TO}_t = 100\%$ for complete turnover. Reports might also include the average turnover:

[6] The sum of the absolute values of the weights is equal to the sum of the positive weights minus the sum of the negative weights associated with short position.

$$\overline{\mathrm{TO}}_t = \frac{1}{T-1}\sum_{t=2}^{T}\left(\sum_{j=1}^{N}\left|w_{t,j} - w_{t-1,j}\right|\right). \tag{2.60}$$

The sum of the squared portfolio weights, known in economics as the Herfindahl-Hirschman Index (HHI), written as

$$\mathrm{HHI}_t(\mathbf{w}) = \sum_{i=1}^{N} w_{t,i}^2 \tag{2.61}$$

serves very well as a portfolio weights *concentration penalty*.[7]. It is straightforward to show that for fully-invested long-only portfolios $\mathrm{HHI}(\mathbf{w})$ is maximized by choosing one of the weights equal to one and all the other weights equal to zero, and (b) is minimized by an equal weights portfolio. Since many investors are interested in highly diversified portfolios, it is natural to compute and plot the time-varying *diversification index*

$$\begin{aligned}\mathrm{DIV}_t &= 1 - \mathrm{HHI}_t(\mathbf{w}) \\ &= 1 - \sum_{i=1}^{N} w_{t,i}^2\end{aligned} \tag{2.62}$$

to visualize the how diversified the portfolio is over time, and use its time series average

$$\overline{\mathrm{DIV}}_t = \frac{1}{T}\sum_{t=1}^{T}\left(1 - \sum_{i=1}^{N} w_{t,i}^2\right) \tag{2.63}$$

as a single numeric indication of a portfolio's diversification quality over a given time period.[8]

Example 2.7. Turnover and Diversification of the GmvLS Portfolio

Figure 2.12 displays the time series of the GmvLS TO and DIV time series of the GmvLO portfolio, with their sample means. It is notable that a simple long-only constrained global minimum variance This is a basic result from probability theory, which we illustrate by example in Exercise YY. portfolio can give such improved performance relative to the market.

[7] In economics the sum of squared weights is a measure of concentration of companies in a given industry, and its use in mean-variance portfolio optimization was introduced by King (2007).

[8] There is another related concept, the effective number of names, which is defined by $\mathrm{EFN} = 1/\sum_{i=1}^{n} w_i^2$. It is easy to show that $1 \leqslant \mathrm{EFN} \leqslant n$, making it easy to interpret.

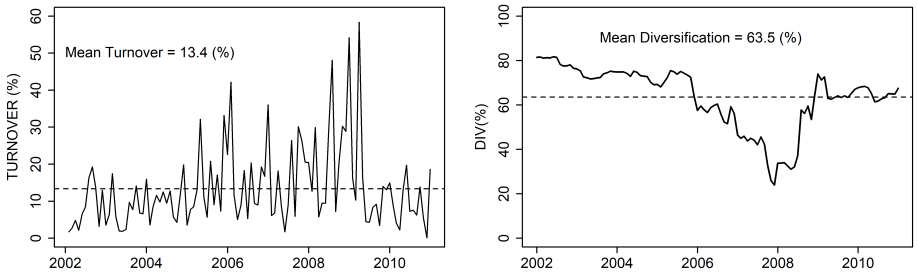

Fig. 2.12 Turnover (left) and Diversification (right) of the 10 Smallcap Stocks GmvLS Portfolio Backtest

2.3.3 Downside Risk Measures

A GMV portfolio minimizes variance, which therefore minimizes standard deviation (SD). The SD measure, also referred to as volatility, was for a long time taken to be "the" measure of portfolio risk. However, starting around the year 2000, quantitative finance research led to a rigorous foundation for risk measures based on a set of *coherent risk measure axioms*, according to which SD is not a coherent risk measure. The coherent risk axioms and their application to downside risk measures, which includes showing that SD is not a coherent risk measure are discussed in detail in the Portfolio Risk Analysis chapter. Meanwhile, we point out here one very intuitive reason why SD fails to be an adequate risk measure, namely: SD is symmetric in the sense that a return of a given value larger than the mean returns contributes the same amount to risk as a return that is below the mean by the same amount. But the latter return represents a loss relative to the mean while the former is gain relative to the mean.

Increasingly, portfolio managers prefer to measure a portfolio's performance using a downside *risk measure* of one kind or another, and the four of these that are most commonly used are: *semi-standard deviation* (SSD), *semi-standard deviation relative to minimum acceptable return MAR* (SSDM), *value-at-risk* (VaR), and *expected shortfall* (ES). A risk measure is defined by a mathematical formula which depends on the distribution function $F = F(r)$ of the returns r. For example, the formulas for the mean and standard deviation are:

$$\begin{aligned} \mu(F) &= \int r dF(r) \\ \sigma(F) &= \left[\int (r - \mu(F))^2 dF(r) \right]^{1/2}. \end{aligned} \tag{2.64}$$

Quantities such as $\mu(F)$ and $\sigma(F)$ are called `functionals` to indicate that their arguments F are in general infinite-dimensional objects, i.e., F can be any one of an infinite number of distributions functions, e.g., all distribution functions for which the above integrals exsist and are finite.

There is a simple general method, called the *plug-in method*, for obtaining a sample based *risk estimator* from the mathematical risk measure functional formula. This plug-in rule does not depend on an assumed parametric distribution form for the returns distribution function $F(r)$, and so therefore the plug-in method results in a *non-parametric* risk estimator.

The *plug-in method* consists of replacing the unknown distribution function F with the *empirical distribution function* $F_n(r)$, which is a "step function" with jumps of height $1/n$ at each of the data values $r_1, r_2, \cdots, r_n$, and has constant values in between. So in the case of the mean functional $\mu(F)$, this results in $\mu(F_n) = \int r dF_n(r)$, where the integral is unusual in that the substitution $F_n(r) = f_n(r)\, dr$ does not work because the derivative $f_n(r)$ of the step function $F_n(r)$ does not exist for the values of r where the steps occur. However, $dF_n(r)$ represents the probabilities of a discrete random variable which takes on the values $r_1, r_2, \cdots, r_n$ with equal probabilities $1/n$.[9] For example, the integral $\mu(F_n) = \int r dF_n(r)$ is

[9] In case of tied values of the r_i, the jumps of $F_n(r)$ would be multiples of $1/n$.

$$\begin{aligned}\mu(F_n) &= \sum_{i=1}^{n} r_i P(r_i) \\ &= (1/n) \sum_{i=1}^{n} r_i\end{aligned} \tag{2.65}$$

which is the sample mean estimate $\bar{r}$.

Sample Standard Deviation

Applying the plug-in rule to the standard deviation functional in (2.64)

$$\hat{\sigma} = \sqrt{\frac{1}{n} \sum_{t=1}^{n} (\bar{r} - r_t)^2}. \tag{2.66}$$

Semi–Standard Deviation

The *Semi-Standard Deviation* downside risk measure is

$$\mathrm{SSD}(F) = \left(\int_{-\infty}^{\mu} (r - \mu)^2 \, dF(r) \right)^{1/2} \tag{2.67}$$

and the plug-in method results in the estimator:

$$\widehat{\mathrm{SSD}} = \sqrt{\frac{1}{T} \sum_{r_t \leqslant \bar{r}} (\bar{r} - r_t)^2}. \tag{2.68}$$

Note that for returns whose distribution is symmetric about the mean, the SSD will have a value $1/\sqrt{2}$ times that of the standard deviation (SD). Thus, some portfolio analysts may prefer to use and report the modified semi-standard deviation $\mathrm{SSD}^* = \sqrt{2} \cdot \mathrm{SSD}$, and doing so results in the equality

$$\mathrm{SSD}^*(F) = \mathrm{SD}(F) \tag{2.69}$$

for symmetric distributions F. Thus one may wish to use the estimator

$$\widehat{\mathrm{SSD}^*} = \sqrt{2} \cdot \widehat{\mathrm{SSD}} \tag{2.70}$$

which has the property that $\widehat{\mathrm{SSD}^*} \approx \widehat{\mathrm{SD}}$ when returns have an approximately symmetric distribution, and the extent to which this approximation fails to hold is a reflection of the extent of skewness of the returns distribution.

Semi-Standard Deviation Relative to MAR

This downside risk measure is motivated by investors who view returns as risky only if they fall below some personal *minimum acceptable return* (MAR). Investors with different MAR's will hold different views of a given portfolio's risk. The Semi–Standard Deviation Relative to MAR is defined as:

$$\mathrm{SSDM}(F,\ \mathrm{MAR}) = \left(\int_{-\infty}^{\mathrm{MAR}} (r - \mathrm{MAR})^2 \, dF(r) \right)^{1/2}. \tag{2.71}$$

and the plug-in method results in the estimator:

$$\widehat{\mathrm{SSDM}}(\mathrm{MAR}) = \sqrt{\frac{1}{T} \sum_{r_i < \mathrm{MAR}} (r_i - \mathrm{MAR})^2}. \tag{2.72}$$

When MAR is the mean of the distribution, the resulting measure $\mathrm{SSDM}(F,\ \mathrm{MAR})$ becomes the semi-standard deviation measure $\mathrm{SSD}(F)$.[10]

Value-at-Risk

The *value-at-Risk* (VaR) of a random return R with distribution function $F(r) = P(R \le r)$ is defined for a "small" positive probability α in terms of an *α-quantile functional* $q_\alpha(F)$ that satisfies the equality

$$P(R \le q_\alpha(F)) = \alpha\,. \tag{2.73}$$

In risk management practice, the value of α is typically 0.01, 0.025, or 0.05. For distribution functions F that are continuous and strictly increasing $q_\alpha(F) = F^{-1}(\alpha)$.

The quantile function $q_\alpha(F)$ is negative for those typical small value of α used in risk management by banks. Since the risk management literature and risk management practitioners treat loss as a positive quantity, the *value-at-risk functional* is defined as the negative of the quantile functional:

$$\mathrm{VaR}_\alpha(F) = -q_\alpha(F). \tag{2.74}$$

The plug-in rule of obtaining an estimate of the above VaR functional by substituting the empirical distribution function $F_n(r)$ for the unknown distribution function $F(r)$ does not work. This because the empirical quantile $q_\alpha(F_n(r))$ is not well defined, and one must resort to a generalized inverse, as discussed in Appendix B.15. None-the-less, it works well in practice to compute the intuitive *plug-in* VaR estimator

$$\widehat{\mathrm{VaR}}_\alpha = -r_{(\lceil \alpha T \rceil)} \tag{2.75}$$

where $\lceil x \rceil$ is the smallest positive integer greater or equal to x, and $r_{(\lceil \alpha T \rceil)}$ is obtained from the ordered returns values that in practice can be assumed to satisfy the strict inequalities $r_{(1)} < r_{(2)} < \cdots < r_{(T)}$. Thus for $\alpha = 0.05$, we have $\lceil \alpha T \rceil = 1$ for $T = 10$ and $\lceil \alpha T \rceil = 5$ for $T = 100$.

The VaR risk measure in terms of an initial portfolio investments of v_0 dollars is

$$\mathrm{VaR}_\alpha(F) = -\mathrm{v}_0 \cdot q_\alpha(F) \tag{2.76}$$

[10] More generally, we can replace the exponents of 2 and 1/2 in (2.71) by k and $1/k$, $k > 0$, in which case the bracketed term is known as a *Lower Partial Moment of Order k*. The properties of lower partial moments, as well as their use in portfolio construction and in the derivation of a mean–downside risk CAPM, were first explored in Bawa (1976), Fishburn (1977) and Bawa and Lindenberg (1977).

and its sample based estimator is:

$$\widehat{\mathrm{VaR}}_\alpha = -\mathrm{v}_0 \cdot r_{(\lceil \alpha T \rceil)}. \tag{2.77}$$

There are two main weakness of the VaR risk measure: The first and most important one is that $\mathrm{VaR}_\alpha(F)$ tells portfolio risk analysts and managers nothing about the size of the losses larger than the value of $\mathrm{VaR}_\alpha(F)$. The second weakness, discussed in the Chapter 8, is that $\mathrm{VaR}_\alpha(F)$ is not a coherent risk measure. The main reason for introducing $\mathrm{VaR}_\alpha(F)$ here is that it plays an important role in the Expected Shortfall risk measure discussed next.

Expected Shortfall

Expected shortfall (ES) is essentially the average of the losses that exceed the VaR. So it is not surprising that, with losses as a positive quantity, ES has the integral form

$$\mathrm{ES}_\alpha(F) = -\frac{1}{\alpha} \cdot \int_{-\infty}^{q_\alpha(F)} r \cdot dF(r). \tag{2.78}$$

where $q_\alpha(F) = -\mathrm{VaR}_\alpha(F)$. The plug-in estimator of ES is

$$\begin{aligned} \widehat{\mathrm{ES}}_\alpha &= -\frac{1}{\lceil \alpha T \rceil} \sum_{r_t \leqslant r_{(\lceil \alpha T \rceil)}} r_t \\ &= -\frac{1}{\lceil \alpha T \rceil} \sum_{i=1}^{\lceil \alpha T \rceil} r_{(i)} \end{aligned} \tag{2.79}$$

where $r_{(\lceil \alpha T \rceil)} = -\widehat{\mathrm{VaR}}_\alpha$. Thus $\widehat{\mathrm{ES}}_\alpha$ is the sample average of the losses beyond $\widehat{\mathrm{VaR}}_\alpha$. In terms of an initial portfolio investments of v_0 dollars, the ES estimator is:

$$\widehat{\mathrm{ES}}_\alpha = -\mathrm{v}_0 \cdot \frac{1}{\lceil \alpha T \rceil} \sum_{r_t \leqslant r_{(\lceil \alpha T \rceil)}} r_t. \tag{2.80}$$

Generally speaking, the values of $\widehat{\mathrm{ES}}_\alpha$ are quite different than those of a standard deviation estimate that investors are quite familiar with for portfolio risk comparisons, but they will not be familiar with the typical values of $\widehat{\mathrm{ES}}_\alpha$. One might hope for some transformation that would result in $\mathrm{ES}_\alpha(F)$ being equal to the standard deviation for some restricted set of distributions F. A common way of achieving this is to assume that F is a normal distribution $N(\mu, \sigma^2)$ with mean μ and standard deviation σ. In that case Exercise 2.13 shows that with $\mathrm{NES}_\alpha = \mathrm{ES}_\alpha(N(\mu, \sigma^2))$ the general formula (2.78) becomes:

$$\mathrm{NES}_\alpha = -\mu + \sigma \frac{\varphi(z_\alpha)}{\alpha} \tag{2.81}$$

where φ is the standard normal density function, and z_α is the α-quantile of the standard normal distribution function Φ, i.e., $z_\alpha = \Phi^{-1}(\alpha)$. The above expression leads to

$$\sigma_{\mathrm{NES},\alpha} = (\mu + \mathrm{NES}_\alpha) \cdot \frac{\alpha}{\varphi(z_\alpha)} \tag{2.82}$$

where the second factor on the right-hand-side is a known constant after the risk manager sets the value of α, but μ and NES_α are unknown. Given a value for $\widehat{\text{ES}}_\alpha$ computed according to (2.79), and the sample mean estimate $\hat{\mu} = \bar{r}$, one can estimate σ_{NES} with:

$$\hat{\sigma}_{\text{NES},\alpha} = \left(\bar{r} + \widehat{\text{ES}}_\alpha\right) \cdot \frac{\alpha}{\varphi(z_\alpha)}. \tag{2.83}$$

The $\bar{r}$ component of $\hat{\sigma}_{\text{NES}}$ will be close to μ for reasonably large sample sizes, say T equal to at least 100 or 200, and if in addition the returns are nearly normally distributed then $\widehat{\text{ES}}_\alpha$ will be close to NES_α, and $\hat{\sigma}_{\text{NES}}$ will be close to the standard deviation estimate $\widehat{\text{SD}}$. But otherwise $\hat{\sigma}_{\text{NES}}$ and $\widehat{\text{SD}}$ will differ, with the extent of the difference reflecting how far the distribution of the returns $r_1, r_2, \cdots, r_T$ is from being a normal distribution.

Uses of Expected Shortfall

It should be noted that there are three distinct uses of ES. The first is as a routine risk performance indicator, as in the the following Example. The second use is for the construction of *minimum downside risk* portfolios, which are briefly introduced in Section 2.6, and fully treated in the Chapter 9. The third use is for bank regulatory purposes as described for example in Bank of International Settlements documents BIS (2005) and Bank-International-Settlements (2019).

Example 2.8. Downside Risk Estimator Analysis of the GmvLS Portfolio and the Market

We applied the SD, SSD, VaR and ES estimators to the GmvLS portfolio and the Market monthly arithmetic returns from 2002 through 2010, and the results are displayed in the "Estimate (%)" column of Table 2.2. Note the SD and SSD risk estimates are both larger for the GmvLS portfolio than for the Market, and one may think of this ordering as a "risk price" paid for the uniformly higher cumulative gross returns of the GmvLS portfolio relative to the Market in Figure 2.10. It is interesting to note that the ES and VaR risks are almost identical for the GmvLS portfolio and the Market, which evidently is because these are tail risk measures, whereas SSD is only a downside risk estimator that depends on all the returns below the mean return, and of course the SD depends on all the returns data.

The StdError column of Table 2.2 provides standard errors (SE's) of the estimates. These SE's are computed based on the assumption that the returns are independent and identically distributed (i.i.d.), but taking into account the non-normality of the returns. The formulas for computing these standard errors are described in Section 2.3.5, and the theoretical foundations for these SE formulas is discussed in the Chapter 15.

Table 2.2 Standard Deviation (SD), Semi-Standard Deviation (SSD), Expected Shortfall (ES) and Value-at-Risk (VaR) of GmvLS Portfolio and Market.

	Estimate (%)	StdError (%)
SD		
GmvLS SD	6.21	0.56
Market SD	4.78	0.42
SSD		
GmvLS SSD	4.18	0.33
Market SSD	3.69	0.40
ES		
GmvLS ES	10.18	1.61
Market ES	10.81	2.00
VaR		
GmvLS VaR	7.96	0.81
Market VaR	8.05	1.10

In closing this discussion of downside risk measures, we point out that according to the risk measures coherence axioms discussed in the Chapter 8, $\mathrm{ES}_\alpha(F)$ is a coherent risk measure, but $\mathrm{SD}(F)$, $\mathrm{SSD}(F)$ and $\mathrm{SSDM}(F)$ fail to satisfy all the coherence axioms.

2.3.4 Risk Adjusted Performance Ratios

It often proves useful to combine a measure of return with a measure of risk to create a *risk–adjusted measure of performance* that characterizes the two most important aspects of a portfolio's performance with just one number, and a survey of the literature brings to light a vast array of such measures, each claiming for itself some special advantage over the others. In practice, (see for example Cogneau and Hubner (2009)) however, their claimed advantages often prove illusory, or are confined to a few special cases that are not of general interest.

We focus, therefore, on four measures that we believe to be of broad utility to performance analysts: the *Sharpe Ratio* (SR), the *Downside Sharpe Ratio* (DSR), the *Sortino Ratio* (SOR), and the *Expected Shortfall Ratio* (ESR). The last three of these ratios are obtained from the Sharpe Ratio by replacing the SD symmetric risk measure in the denominator of the Sharpe Ratio with the SSD, SSDM and ES downside risk measures, respectively. These downside risk of a portfolio cannot be computed from the downside risk of its constituents using a quadratic form as is the case for portfolio volatility risk, so these risk measures play a larger role in performance assessment of mean-variance optimal portfolios than in their construction. However, the path breaking work of Rockafellar and Uryasev (2000) in constructing optimal Expected Shortfall portfolios, is likely to change this.

We start our exploration of the above four risk adjusted performance ratios by examining the Sharpe Ratio. This is not because the Sharpe Ratio is an absolute best measure of risk–adjusted performance, but because of its long history, its wide use, its key role in portfolio efficient frontiers that we discuss in Section 2.4.3, its simple time scaling property, and because the majority of performance measures that have been proposed are variants of it.

The Sharpe Ratio

The Sharpe Ratio[11] of a portfolio, or individual asset such as a stock, is defined as

$$\begin{aligned} \mathrm{SR} &\triangleq \frac{\mu_e}{\sigma_e} \\ &= \frac{\mu - r_f}{\sigma} \end{aligned} \tag{2.84}$$

where μ_e and σ_e are the one–period expected return and standard deviation of its excess returns (i.e. its returns in excess of the risk–free rate). The second line follows from the observation that the return of a riskless asset is certain, so that the volatility of the portfolio's excess returns is equal to that of its total returns.

Use of the plug-in rule with time-varying excess returns

$$r_{e,t} = r_t - r_{f,t},\ t = 1, 2, \cdots, T \tag{2.85}$$

results in the Sharpe Ratio estimator

$$\widehat{\mathrm{SR}} = \frac{\bar{r}_{e,T}}{\hat{\sigma}_{e,T}} \tag{2.86}$$

where[12]

$$\hat{\sigma}_{e,T} = \sqrt{\frac{1}{T}\sum_{t=1}^{T}\left(\left(r_t - r_{f,t}\right) - \left(\bar{r}_T - \bar{r}_{f,T}\right)\right)^2}. \tag{2.87}$$

N.B. When we use a subscript T on a quantity, e.g., as in $\hat{\sigma}_{e,T}$, $\bar{r}_T$ and $\bar{r}_{f,T}$, it means that the quantity has been computed based on returns from time $t = 1$ to $t = T$. Furthermore, in order to adequately reflect the time varying risk-free rate $r_{f,t}$ we strongly recommend to use it as in the time-varying excess returns $r_{e,t}$ as above.

The Importance of Using Excess Returns

The Sharpe Ratio definition (2.84) shows that when risk-free rates are positive, as they almost always are, omission of the risk-free rate in the numerator inflates the Sharpe Ratio, and makes a portfolio's performance appear better than it actually is. Thus we cannot stress too strongly the importance of using excess returns when computing a Sharpe Ratio, as well as when computing the downside risk adjusted performance ratios discussed shortly.

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[11] Sharpe (1966) originally characterized (2.84) as the Reward–to–Variability Ratio.

[12] Equation 2.87 is a slightly biased estimator of the standard deviation of the excess return. For normally distributed data, the bias can be almost completely eliminated by replacing T with $T - 1.5$ in the denominator of 2.87.

Example 2.9. Inflation of the Sharpe Ratio Due to Not Using Excess Returns.

Figure 2.13 shows that while the monthly risk-free rates in the U.S. have been negligible between 2009 and 2015, they are quite non-negligible to various degrees during the earlier years, with a largest average level of about 0.003 (0.3%) from the beginning of 1999 to the end of 2000.

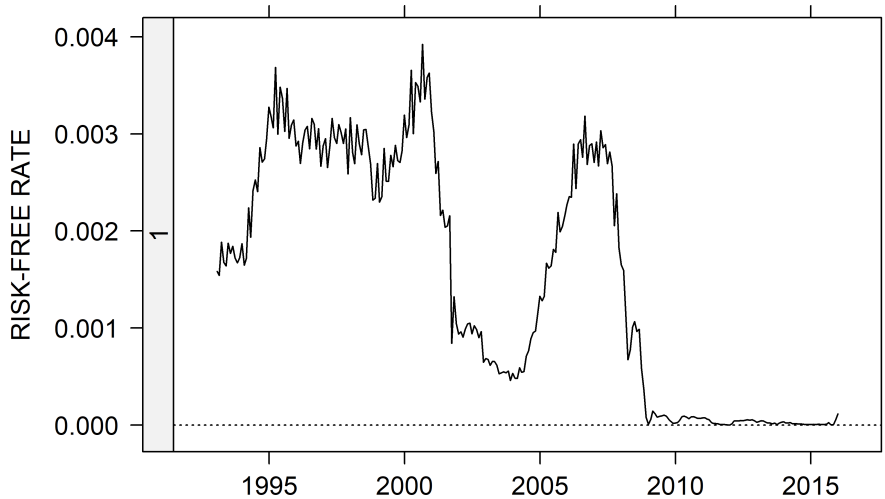


Fig. 2.13 Risk-Free Rate From 1993 - 2015

Focusing on the time period 1999 to 2000, we considered all the stocks in the `stocksCRSPmonthly` data set whose mean monthly return for that period fell between 0.005 and 0.007 (0.5% and 0.7%), i.e., larger than the risk-free rate during that time period, but not so large as to completely dominate the the risk-free rate. From the 21 stocks so found, we chose the one with ticker FMC, and Figure 2.14 displays the returns of FMC and the risk-free rate for 1999 to 2000.

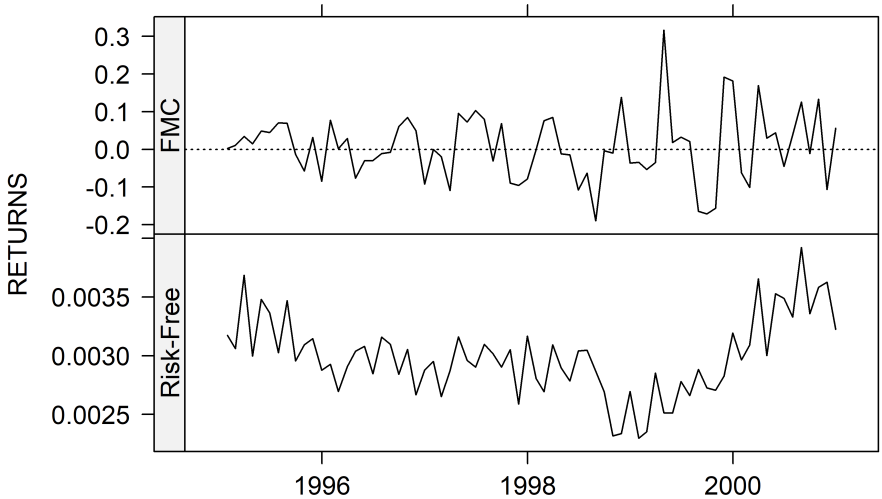


Fig. 2.14 FMC Returns and Risk-Free Rate 1999 - 2000

For the monthly returns of the stock FMC during 1999-2000, the annualized mean excess return is 4.72% and the annualized volatiity is 31.13%, resulting in the correctly computed Sharpe Ratio value:

$$\begin{aligned}\widehat{\mathrm{SR}}_{\mathrm{FMC}} &= \frac{4.72\%}{31.14\%} \\ &= 0.15.\end{aligned}$$

When the raw returns are used the annualized mean return return is 18.31% and the annualized volatility is 31.16%, resulting the incorrect Sharpe Ratio value

$$\begin{aligned}\widehat{\mathrm{SR}}_{\mathrm{FMC}}^{\mathrm{Incorrect}} &= \frac{18.31\%}{31.16\%} \\ &= 0.27,\end{aligned}$$

which is 75% larger than the correctly computed Sharpe Ratio! Note that since the denominators in the above two ratios are essentially equal, this inflation is almost totally due to the numerator difference between the larger mean of the raw returns and the mean return of the excess returns.

The above results suggests the Exercise 2.14 to compute the percent of the stocks in the `stocksCRSPmonthly` data set for which the incorrect Share ratio is at least 25%, 50%, and 75% larger than the correctly computed Sharpe Ratio.

Time Scaling Property of Sharpe Ratios

Thus far we have derived results assuming that returns and performance measures are computed at the same frequency, e.g. we use monthly returns to compute a monthly Sharpe Ratio and use annual returns to compute an annual Sharpe Ratio. We now ask if we can compute a long–horizon performance

measure (e.g. an annual Sharpe Ratio) using higher frequency (e.g. monthly or weekly) observations. This solution that we now discuss for the Sharpe Ratio also works in the case of the Information Ratio discussed in 2.4.4, but are not immediately applicable to other reward–to–risk measures such as those discussed below.

In the special case when the portfolio's excess returns are i.i.d., the T–period Sharpe Ratio based on r_t and $r_{f,t}$ multi-period cumulative returns is given by

$$\mathrm{SR}_T = \frac{\mu_T - r_{f,T}}{\sigma_T} \tag{2.88}$$

where $r_{f,T}$ is the cumulative risk-free returns, and μ_T and σ_T are the cumulative r_t mean and volatility, whose somewhat complicated formulas are defined precisely in Chapter 3. However, practitioners usually use the approximation

$$\begin{aligned} \mathrm{SR}_T &\approx \frac{T \cdot \left(\mu - r_f\right)}{\sqrt{T}\,\sigma} \\ &= \sqrt{T} \cdot SR, \end{aligned} \tag{2.89}$$

where μ, σ, and SR are the single period quantities, and this leads to the T period Sharpe Ratio estimate:

$$\begin{aligned} \widehat{\mathrm{SR}}_T &= \frac{T \cdot \left(\bar{r} - \bar{r}_f\right)}{\sqrt{T} \cdot \hat{\sigma}} \\ &= \sqrt{T} \cdot \widehat{SR}. \end{aligned} \tag{2.90}$$

It is the norm in investment management to measure returns at a monthly frequency and to quote Sharpe Ratios on an annualized basis, so that in the i.i.d. case

$$\widehat{\mathrm{SR}}_{\text{Annual}} = \sqrt{12} \cdot \widehat{\mathrm{SR}}_{\text{Monthly}}. \tag{2.91}$$

Just as (2.84) reflects the influence of mean–variance optimal portfolios, as we discuss in Section 2.4.3, so do the next three downside risk–adjusted performance measures reflect the influence of mean–downside risk portfolio optimization, which we explore in Chapter 9.

An important role of Sharpe Ratios in the context of minimum-variance optimal portfolios with a risk-free rate ("cash") and risky asset is discussed in Section 2.4.3.

The Downside Sharpe Ratio

For the same reason that investors may prefer the downside semi-standard deviation SSD risk measure to the symmetric standard deviation measure SD, they may prefer to use a *Downside Sharpe Ratio*, denoted DSR, for which the SD in the denominator of the Sharpe Ratios if replaced with the SSD. The resulting DSR can be made to have values similar to the SR for returns that are approximately

normally distributed by using $\sqrt{2} \times \text{SSD}$ in place of SSD in the denominator.[13]. This scaling results in the DSR being exactly equal to SR for returns that are symmetrically located around the risk–free rate, and approximately so for many financial time series on account of their modest mean. It is particularly effective for time series that are measured monthly or more frequently, as, in these cases, the mean is not far removed from the risk free rate, and the factor of $\sqrt{2}$ is a reasonable approximation to the true scaling factor, which depends on the entire distribution, and not just its left tail. The resulting estimator for the Downside Sharpe Ratio is

$$\widehat{\text{DSR}} = \frac{\bar{r} - \bar{r}_f}{\sqrt{2} \cdot \widehat{\text{SSD}}} \tag{2.92}$$

where the denominator is equal to the equation (2.70) estimator $\widehat{\text{SSD}^*}$, which is close to $\widehat{\text{SD}}$ for returns that have an approximately symmetric empirical distribution function F_n.

Ziemba (2005) provides a rich set of examples of the use of DSR with returns measured at multiple frequencies, and also links it to log–optimal investment,[14]mutual fund performance, horse racing and gambling.

The Sortino Ratio

The *Sortino Ratio*, denoted SOR (MAR), is the best known downside–risk based performance measure. This risk-adjusted return measures is motivated by investors who view returns as risky only if they fall below some personal *minimum acceptable return*, or MAR, and investors with different MAR's will report different Sortino ratio values. The Sortino ratio was defined in Sortino and Price (1994) to be the ratio of excess return to semi–deviation with both measured relative to MAR. The Sortino ratio estimator is:

$$\widehat{\text{SOR}}\,(\text{MAR}) = \frac{\bar{r} - \text{MAR}}{\widehat{\text{SSDM}}\,(\text{MAR})}. \tag{2.93}$$

The ES Ratio

The *ES Ratio Estimator* is defined as

$$\widehat{\text{ESR}}_\alpha = \frac{\bar{r} - \bar{r}_f}{\widehat{\text{ES}}_\alpha} \tag{2.94}$$

where α is a more or less small left-tail probability, and the denominator is the expected shortfall risk estimator (2.79).

The tail probability $\alpha = 0.025$ was recommended by BIS Bank-International-Settlements (2019) for banking regulatory risk management. However, this choice results in a large variance of the estimator $\widehat{\text{ES}}_\alpha$ unless the sample size T is very large. We suggest using larger tail probabilities for general portfolio tail risk adjusted performance evaluation, e.g. $\alpha = 0.25$. Alternatively, for downside risk adjusted performance measurement, as opposed to tail risk adjusted performance, we recommend use of $\alpha = 0.5$, in which case the ES estimator is the average of returns below the median of the returns, and the result-

[13] See Ziemba (2005) for details. Ziemba originally set $MAR = 0$, but allowed that this was an ad–hoc decision, and considered other values as well. We set $MAR = r_f$ to emphasize the similarity between the traditional Sharpe Ratio and its downside derivative.

[14] See Chapter 14 and Appendix C for a discussion of logarithmic utility and log–optimal investing, and their application to liability driven investing.

ing ES will be very close the the square-root of the *semi-variance* measure to which Markowitz (1959) devoted a whole chapter.

As we discussed previously the estimate $\widehat{\text{ES}}_\alpha$ generally has a substantially different range of values than a residual returns standard deviation, such as the $\hat{\sigma}^P_{e,T}$ in the Sharpe Ratio estimator (2.86), and correspondingly the $\widehat{\text{ESR}}_\alpha$ values will typically be substantially different than the SR estimate values. This problem can be alleviated by replacing $\widehat{\text{ES}}_\alpha$ in the denominator of the above ESR estimator with the estimator $\hat{\sigma}_{\text{NES},\alpha}$ given by (2.83), and this results in the *normal distribution calibrated* ESR estimator

$$\widehat{\text{NESR}}_\alpha = \frac{\overline{r} - \overline{r}_f}{\hat{\sigma}_{\text{NES},\alpha}}. \tag{2.95}$$

When the returns distribution is approximately normally distributed, one expects that $\widehat{\text{NESR}}_\alpha$ and $\widehat{\text{ESR}}_\alpha$ to have similar values, and otherwise the extent to which they differ will reflect the extent of the non-normality of the returns data.

Example 2.10. Risk Adjusted Performance Ratios Analysis of the GmvLS Portfolio and the Market

We applied the SR, DSR, SOR and ESR estimators to the GmvLS portfolio and the Market arithmetic returns from 2002 through 2010, and the results are displayed in Table 2.3. Note that each of these performance ratios is larger for the GmvLS portfolio than for the Market, by amounts of 34%, 22%, 24% and 45% respectively. However, recall that Table 2.2 shows that denominator risk estimates SD, SSD and ES are all larger for the GmvLS portfolio than for the market, so the larger values of the performance ratios in Table 2.3 for the GmvLS portfolio relative to the market are due to the even larger returns of the GmvLS portfolio.

Table 2.3 Sharpe Ratio (SR), Downside Sharpe Ratio (DSR), Sortino Ratio (SOR), and Expected Shortfall Ratio (ESR) of GmvLS Portfolio and Market.

	Estimate	StdError
SR		
GmvLS SR	0.18	0.09
Market SR	0.11	0.10
DSR		
GmvLS DSR	0.19	0.10
Market DSR	0.10	0.09
SOR		
GmvLS SOR	0.19	0.19
Market SOR	0.10	0.15
ESR		
GmvLS ESR	0.13	0.08
Market ESR	0.05	0.06

The StdError column of Table 2.3 provides standard errors (SE's) of the estimates. These SE's are computed based on the assumption that the returns are independent and identically distributed (i.i.d.), but taking into account the non-normality of the returns. We next describe the formulae used to compute these standard errors, and discuss their theoretical foundations in Chapter 15.

2.3.5 Standard Error Computation

We include standard error (SE) columns in Tables 2.2 and 2.3 because risk and performance estimates are of limited value for comparing alternative securities and portfolios without some measure of the uncertainty of the estimates. Our goal is to strongly encourage the practice of computing SE's for all formally reported security and portfolio risk and performance estimates, using a relatively new general analytic method for computing SE's based on *influence functions*. Influence functions are discussed in Chapter 15, and their use in particular for deriving estimator asymptotic formulas and computing estimator standard errors are discussed in Section 15.3.[15]

Quantitative finance practitioners only infrequently report portfolio risk and performance estimate SE's. The reason for this is in large part because analytic methods of computing such SE's are based on estimator *asymptotic variance* formulas, which are scattered in the statistics and quantitative finance literature, and not well known.[16] Thus we briefly discuss such formulas.

The *asymptotic variance* of an estimator $\hat{\theta}_T = \hat{\theta}(r_1, r_2, \cdots, r_T)$ based on reutrns $r_1, r_2, \cdots, r_T$ is the variance $\mathrm{V}(\theta)$ of the limiting zero mean normal distribution of the random quantity

$$\sqrt{T}\left(\hat{\theta}_T - \theta\right) \tag{2.96}$$

as the sample size T goes to infinity. For purposes of the discussion here, and for the results in Tables 2.2 and 2.3, it is assumed that the returns are independent and identically distributed (i.i.d.). The *asymptotic variance* $\mathrm{V}(\hat{\theta}_\infty, \theta)$ of the infinite sequence $\hat{\theta}_\infty$ of estimators $\hat{\theta}_T$ of the unknown parameter value θ is the variance of the limiting normal distribution of (2.96). We note that $\mathrm{V}(\hat{\theta}_\infty, \theta)$ depends on the returns distribution function $F_\theta = F_\theta(r)$, which is typically a more or less fat-tailed and skewed, but for notational simplicity we omit the dependence of $\mathrm{V}(\hat{\theta}_\infty, \theta)$ on F_θ.

Given an asymptotic variance formula $\mathrm{V}(\hat{\theta}_\infty, \theta)$ for an estimator $\hat{\theta}_T$, the estimator's approximate sample size T variance is defined to be

$$\mathrm{var}(\hat{\theta}_T) \approx \frac{1}{T}\mathrm{V}(\hat{\theta}_\infty, \hat{\theta}_T) \tag{2.97}$$

[15] This computational method is implemented in the open source *Risk and Performance Estimator Standard Errors* (`RPESE`) package, which was used to compute the SE's in Tables 2.2 and 2.3. The `RPESE` package, along with its introductory Vignette "Risk and Performance Estimators Standard Errors" is available at `https://cran.r-project.org/web/packages/RPESE`. See also Zhang et al. (2021).

[16] The same is true with regard to bootstrap resampling methods of computing SE's.

The standard error of an estimator is then defined as the square root of the approximation on the right-hand side of (2.97):

$$\mathrm{SE}(\hat{\theta}_T) \triangleq \frac{1}{\sqrt{T}} \left[\mathrm{V}(\hat{\theta}_\infty, \hat{\theta}_T)\right]^{1/2}. \tag{2.98}$$

Given a formula $\mathrm{V}(\hat{\theta}_\infty, \theta)$ for the asymptotic variance of an estimator $\hat{\theta}_T$, the computation of the standard error $\mathrm{SE}(\hat{\theta}_T)$ is quite easy. The problem is that formulas $\mathrm{V}(\hat{\theta}_\infty, \theta)$ for a wide range of risk and performance estimators are not at all well known.[17].

Fortunately, influence functions allow one to easily derive $\mathrm{SE}(\hat{\theta}_T)$ formulas for a very wide range of risk and performance estimators. As discussed in Chapter 15, an influence function

$$\mathrm{IF}(r\,;\hat{\theta}_\infty) = \mathrm{IF}(r\,;\hat{\theta}_\infty, \theta) \tag{2.99}$$

is the limit of the *empirical influence functions*

$$\mathrm{EIF}_{\hat{\theta}_n}(r\,;\mathbf{r}) = \frac{\hat{\theta}_{n+1}(r, \mathbf{r}) - \hat{\theta}_n(\mathbf{r})}{1/(n+1)} \tag{2.100}$$

as the sample size n goes to infinity. Thus the EIF is a numerical approximation to a derivative, and it will not be surprising to find in Chapter 15 that the mathematical formula for $\mathrm{IF}(r\,;\hat{\theta}_\infty)$ is that of a special kind of derivative. That formula allows one to compute the influence function of a wide variety of estimators using the rules of ordinary calculus, including the following examples for the volatility and Sharpe ratio estimators.

Example 2.11. The Volatility Estimator Influence Function.

The influence function of a volatility estimator $\sigma = \mathrm{SD}$ is shown in Section 15.2 to be

$$\mathrm{IF}(r\,;\sigma) = \frac{1}{2\sigma}\left((r-\mu)^2 - \sigma^2\right) \tag{2.101}$$

where σ is the risk parameter of interest and μ is a *nuisance parameter*, both of which need to be estimated.

Example 2.12. The Sharpe Ratio Estimator Influence Function

It is shown in Zhang et al. (2021) that Sharpe Ratio Estimator (SR) has the influence function

[17] Notable exceptions of well-known formulas include: (1) The asymptotic variance of the standard deviation, which is $0.25\left(m_4 - \sigma^4\right)/\sigma^2$ where m_4 is the fourth central moment; (2) The asymptotic variance of the Sharpe Ratio for possibly non-normal returns, which was derived in Mertens, E. (2002). "Comments on the Correct Variance of Estimated Sharpe Ratios in Lo (2002, FAJ) When Returns Are IID", and is available at `http://www.elmarmertens.org/research/workingpapers`.

$$\mathrm{IF}\,(r;\mathrm{SR}) = -\frac{\mathrm{SR}}{2}\left(\frac{r-\mu}{\sigma}\right)^2 + \frac{r-\mu}{\sigma} + \frac{\mathrm{SR}}{2} \tag{2.102}$$

where SRis the risk-adjusted performance parameter of interest, and μ and σ are nuisance parameters, all three of which need to be estimated.

Figure 2.15 displays $\mathrm{IF}(r\,;\hat{\sigma})$ and $\mathrm{IF}(r;\widehat{\mathrm{SR}})$ plots for the edhec data set convertible arbitrage (CA) hedge fund style index of monthly returns from January 1997 to May 2021.[18]

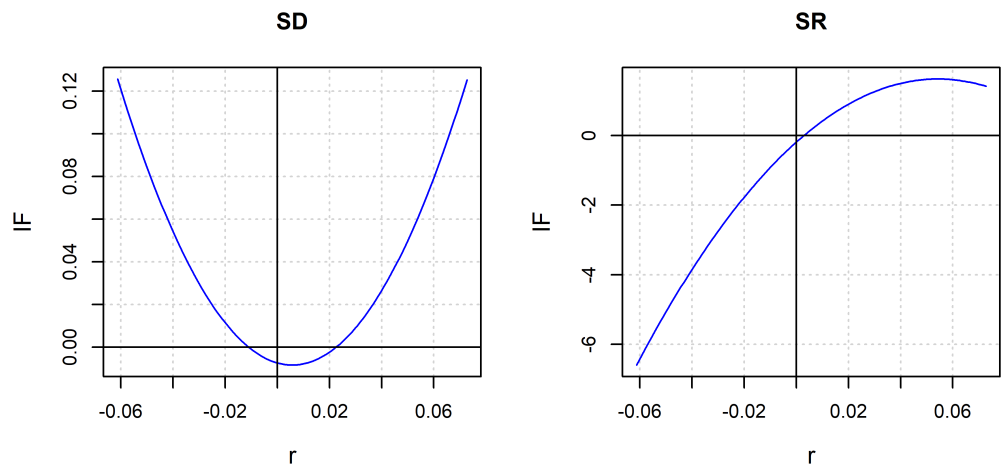


Fig. 2.15 Influence functions of the standard deviation (SD) and semi-standard deviation (SemiSD) with unknown parameters computed using the CA hedge fund style index of monthly returns from 1997 to 2021.

The quadratic character of SD estimator influence function indicates that both positive and negative outlier returns have a large positive influence on the estimate of SD, and that "inlier" returns with values close to the mean have a *negative* influence. By way of contrast the SR estimator influence function exhibits a "negative quadratic" behavior, which is due to the fact that outlies increase the denominator of the Sharpe Ratio in a quadratic manner, and thereby decrease the SR influence functtion value.

The Standard Error Formula

It is discussed in Section 15.3 that the asymptotic variance $\mathrm{V}(\hat{\theta}_\infty,\theta)$ formula of an estimator $\hat{\theta}_T$ is given in terms the estimator's influence function $\mathrm{IF}(r\,;\hat{\theta}_\infty)$ by the rather simple general formula

$$\mathrm{V}(\hat{\theta}_\infty,\theta) = \mathrm{E}\left[\mathrm{IF}^2(r_t\,;\theta)\right]. \tag{2.103}$$

This formula can be used in (2.98), with the above right-hand side replaced by the sample aveage of the $\mathrm{IF}^2(r_t\,;\theta)\,,\, t = 1,2,\cdots,T$, to compute an estimator's SE with the formula:

[18] The plots in Figure (2.15) were made using the RPESE package, which has functions to compute and plot influence functions for a variety of risk and performance estimators, for a range of notional returns values r, based on a set of returns $r_1, r_2, \cdots, r_T$ that are used to compute both the main parameter estimate and nuisance parameter estimates.

$$\mathrm{SE}(\hat{\theta}_T) = \frac{1}{\sqrt{T}} \left[\frac{1}{T} \sum_{t=1}^{T} \mathrm{IF}^2(r_t\,;\hat{\theta}_T) \right]^{1/2} \tag{2.104}$$

This is the formula used for computing estimator standard errors in the `RPESE` package (available at `https://cran.r-project.org/web/packages/RPESE/index.html`) under the default option that the SE computation is based on the assumption of i.i.d. returns.[19]

Example 2.13. Volatility Estimate SE Formula.

In the case of a volatility estimate, the formula (2.104) results in

$$\mathrm{SE}_{\mathrm{IFdirect}}(\hat{\sigma}) = \frac{1}{\sqrt{T}} \left[\frac{1}{T} \sum_{t=1}^{T} \left(\frac{\left(r_t - \bar{r}_T\right)^4 - \hat{\sigma}_T^2}{2\hat{\sigma}_T} \right)^2 \right]^{1/2}. \tag{2.105}$$

where $\hat{\sigma}_T^2$ is the sample variance, and $\bar{r}_T$ is the sample mean estimate of the nuisance parameter μ.

In summary, the main points of this section are: (i) computation of SE's for the risk estimators Table (2.2) and performance estimators in Table (2.3), as well as others discussed in Zhang et al. (2021), can be easily done using the `RPESE` package; (ii) The estimator SE computation method consists of averaging the squared values of the influence function evaluated at returns $r_1, r_2, \cdots, r_T$, where the latter are used to compute unknown parameter values; (iii) Influence function formulas for a wide range of risk and performance estimators, including those in Tables (2.2) and (2.3), are provided in Zhang et al. (2021).

2.4 Mean-Variance Optimal Portfolios

A basic principle of financial investments is that:

> "*Higher returns cannot be obtained without taking on higher risk*".

There will be little debate about the meaning of the word "returns" in this principle, as most investors will take returns to mean arithmetic returns as defined in (2.2), and the principle will not change for those that prefer logarithmic returns since it is a monotonic transformation of arithmetic returns. However, the meaning of the word "risk" is more subjective, and as we have seen in Section 2.3.3 there are several possible choices of a risk measure. That said, Harry Markowitz in his path–breaking research papers Markowitz (1952a), Markowitz (1952b), Markowitz (1956), and subsequent 1959 monograph *Portfolio Selection: Efficient Diversification of Investments* Markowitz (1959), used variance as *the* risk measure for portfolio construction purposes. His subsequent 1987 book *Mean–Variance Analysis in Portfolio Choice and Capital Markets* Markowitz (1987), and subsequent Nobel Prize in Economic Sciences (along

[19] This is referred to as the direct method in Section 15.3.

with Merton Miller and William Sharpe) in 1990, helped to further establish his contributions, whose central method is that of portfolio mean–variance optimization (MVO).

The Portfolio Mean–Variance Optimality (MVO) Principle:

> *"An investor should select a portfolio of assets such that the variance of the portfolio's returns is minimized at any specified level of the portfolio's mean return."*

This principle has manifested itself in the academic world in a variety of course contexts, including: (1) MBA "Investment" courses, taken by thousands of students each year, where it occupies a prominent position; (2) Over 100 MS degree programs in quantitative finance world–wide (see e.g., www.quantnet.com for rankings of the top 30 U.S. programs); (3) graduate degree program courses in engineering, particularly Industrial Engineering. The principle has also manifested itself very strongly in commercial portfolio management software such as that offered by MSCI Barra, Axioma, Bloomberg, Sungard and Northfield.

In this section we provide the basics of MVO portfolio theory for the mathematically tractable cases in which the only constraints are one or two simple *linear equality* constraints, and note that the approach is easily extended to more general equality constraints. For such MVO problems, the method of Lagrange multipliers allows one to derive formulas for the optimal portfolio weights and relationship between the portfolio mean return and standard deviation. Furthermore, we stress that the theory in this section is limited to the case of *single-period portfolio optimization*, where optimal portfolio weights are used for investing at the beginning of a time period, for example 9 months of daily returns, 2 years of weekly returns, and 5 years of monthly returns, among others, and optimal performance is evaluated at the end of the period. The topic of *multi-period portfolio optimization* is more challenging, and is discussed in a later chapter.

We cannot stress too strongly that the mathematical MVO results obtained in this section are quite limited due to the fact MVO portfolio construction in practice minimally involves the use of linear inequality constraints, e.g., for long–only portfolios and weights box constrainted portfolios, market risk constrained portfolio weights, and concentration penalties, for which numerical optimization is required to obtain optimal portfolio weights. Such MVO portfolios are discussed in Chapter 5, and readers who are mostly interested in applications can go directly to Chapter 5. However, such readers may eventually want to return to the other subsections in this chapter to gain an understanding of the underlying theory for the idealized case of only linear equality constraints, and readers who are highly interested in the underlying theory will do well to read this entire section before going to Chapter 5.

The MVO material in this section provides the theoretical foundations for the following three distinct special cases of MVO portfolio construction:

1. Portfolios fully invested only in risky assets, e.g., stocks, bonds, hedge funds
2. Portfolios fully invested in a combination of cash and risky assets
3. Benchmark relative portfolios, commonly known as *actively-managed* portfolios.

In the first case, "full investment" means that the portfolio weights satisfy the simple equality constraint that they sum to one (100% investment in risky assets), In the second case it means that the fraction of the investment in cash plus the sum of the risky assets weights is equal to 1 (the percentage in cash plus the percentage in risky assets sums to 100%). In the third case, it turns out the active portfolio weights sum to zero, and such portfolios are also referred to as *self–financing* or *dollar neutral* portfolios. Understanding the behavior of the MVO solutions for these three distinct cases, which involve only linear equality constraints, can provide guidance on what to expect for MVO optimization for these three cases with long–only, box and other linear inequality constraints.

An important aspect of initial treatment of MVO portfolios is discussion of the two main approaches to obtaining MVO portfolio formulas, namely minimum–variance (MVO) portfolios and maximum quadratic utility (MaxQU) portfolios. These two approaches turn out to result in equivalent sets of portfolios that one would normally use in practice. But since the MaxQU method is typically used in practice, much of our subsequent discussion will focus on MaxQU portfolios. We will also briefly mention MaxMean portfolios.

This Section covers the following main topics:

1. Maximum Qudratic Utility Portfolios of Risky Assets
2. Maximum Quadratic Utility Porfolios with Cash and Risky Assets
3. Maximum Quadratic Utility Benchmark Relative Portfolios
4. Expected Utility Maximization and Mean Variance Optimal Portfolios

For portfolios of assets such that the full investment constraint (2.36) is satisfied, along with other possible linear constraints, there exist three distinct types if nean-variance optimal (MVO) portfolio optimization problems which have an analytic solution for the optimal portfolio weight vector $\mathbf{w}$. These three problems are as follows:

1. *MVO Portfolios*. These portfolios minimize the portfolio variance $\sigma_P^2(\mathbf{w}) = \mathbf{w}'\mathbf{C}\mathbf{w}$ with respect to $\mathbf{w}$, subject to the full–investment constraint $\mathbf{w}'\mathbf{1} = 1$, a portfolio mean return *equality constraint* $\mathbf{w}'\boldsymbol{\mu} = \mu_P$, where μ_P is specified by the portfolio manager.
2. *MaxQU Portfolios*. These portfolios maximize the quadratic utility (QU) function $\mathrm{QU}(\mathbf{w}) = \mu_P(\mathbf{w}) - \frac{1}{2}\lambda\sigma_P^2(\mathbf{w})$ subject to the full–investment constraint, where $\lambda > 0$ is the portfolio manager's *risk aversion parameter*.
3. *MaxMean Portfolios*. These portfolios maximize the portfolio mean return $\mu_P(\mathbf{w}) = \mathbf{w}'\boldsymbol{\mu}$, subject to the full–investment constraint and a portfolio variance *equality constraint* $\mathbf{w}'\mathbf{C}\mathbf{w} = \sigma_P^2$, where σ_P is specified by the portfolio manager.

The MVO problem has a solution for all possible positive and negative values of μ_P, and in particular it has solutions for values of μ_P less than, as well as greater than, the mean return μ_{gmv} of the global minimum variance portfolio. But, as we shall see, for every MVO portfolio with $\mu_P = \mu_{\mathrm{gmv}} - c$ with $c > 0$,

there exists a MVO portfolio with $\mu_P = \mu_{\text{gmv}} + c$. The latter portfolios are called *efficient* MVO portfolios, and the former are *inefficient* MVO portfolios.

The QU portfolio problem has solutions for all values of $\lambda > 0$, and in the limit when the portfolio manager's risk aversion becomes arbitrarily large, i.e., when $\lambda \to \infty$, the portfolio becomes the global minimum variance portfolio. The MaxMean problem has a solution only for values of the portfolio variance that are at least as large as that of the global minimum variance portfolio.[20] We note the following facts:

1. A MaxQU portfolio is a MVO portfolio, as we show in the Theorem below.
2. MaxQU and MaxMean portfolios are equivalent in that the set of all solutions to these two problems are the same.
3. The risk aversion parameter λ is individual to each investor and must be determined separately. Wilcox et al. (2025) describe a remarkably simple and effective method to identify an investor's λ using the Generalized Logarithmic Utility function described in Rubinstein (1976) – it requires an answer to only one question and is presented in Appendix C.

Theorem 2.1. *A MaxQU portfolio is MVO optimal.*

Proof. Assume that MaxQU and MVO portfolios are subject to the same linear equality constraints. Let $\tilde{\mu}_{\text{qu}}$ and $\tilde{\sigma}^2_{\text{qu}}$ be the mean return and variance of a MaxQU portfolio corresponding to optimal weight vector $\tilde{\mathbf{w}}_{\text{qu}}$. Let $\tilde{\mathbf{w}}_{\text{minvar}}$ be the weights of the MVO portfolio that is constrained to have mean return $\tilde{\mu}_{\text{qu}}$, and let $\tilde{\sigma}^2_{\text{minvar}}$ be the variance of this portfolio. By the definition of a MVO portfolio it must be that $\tilde{\sigma}^2_{\text{minvar}} \leqslant \tilde{\sigma}^2_{\text{qu}}$. Suppose that $\tilde{\sigma}^2_{\text{minvar}} < \tilde{\sigma}^2_{\text{qu}}$. But this contradicts that assumption that $\tilde{\mathbf{w}}_{\text{qu}}$is optimal, for then $Q(\mathbf{w})$ can be increased from the value $Q(\tilde{\mathbf{w}}_{\text{qu}})$ by replacing $\tilde{\mathbf{w}}_{\text{qu}}$ with $\tilde{\mathbf{w}}_{\text{minvar}}$, thereby leaving the mean return term of the quadratic utility unchanged, but reducing the variance component. Thus it must be that $\tilde{\sigma}^2_{\text{minvar}} = \tilde{\sigma}^2_{\text{qu}}$.

It turns out that MaxQU portfolios and efficient MVO portfolios are equivalent in the sense that the set of all efficient MVO portfolios is the same as the set of all MaxQU portfolios.

The MVO approach is the one typically used in mathematically oriented graduate level books that cover mean-variance optimal portfolio theory, e.g., as in Chapter 5.2 of Cochrane (2005), Chapter 2.3 of Pennacchi (2008) and Chapter 3 of Huang and Litzenberger (1988). However, since the QU approach has been the primary methodology focus of commercial portfolio optimization software, we concentrate on the MaxQU approach throughout the book. Nonetheless, before proceeding to discuss the MaxQU method, we briefly describethe main MVO portfolio mathematics results. In passing, we note that the

[20] In order to obtain a MaxMean portfolio one must first solve for the global minimum variance portfolio variance in order to set an achievable variance constraint for the MaxMean problem. This is one of the reasons that MaxMean portfolios have received relatively little attention in the literature. The other reason is that due to the inequality variance constraint, this problem is not a quadratic programming problem.

complete set of MVO efficient and inefficient portfolios is used for obtaining *zero-covariance portfolios*, which are also called *zero-beta portfolios*. The latter are discussed in Section 2.5.2.

N.B. Throughout the remainder of the book we use the term MVO and the acronym MVO interchangebly.

2.4.1 MVO Portfolios

Consider a portfolio with return vector $\mathbf{r}$, whose mean and covariance matrix are $\boldsymbol{\mu}$ and $\mathbf{C}$, with $\mathbf{C}$ positive definite with inverse $\mathbf{C}^{-1}$. The weight vector $\mathbf{w}_{\text{mv}}$ of a fully invested *minimum variance* (MVO) portfolio solves the constrained optimization problem

$$\mathbf{w}_{\text{mv}} = \underset{\mathbf{w}}{\operatorname{argmin}}\, \sigma_P^2(\mathbf{w}) = \underset{\mathbf{w}}{\operatorname{argmin}}\, \mathbf{w}'\mathbf{C}\mathbf{w} \tag{2.106}$$

subject to full–investment and mean return constraints:

$$\mathbf{w}'\mathbf{1} = 1, \quad \mathbf{w}'\boldsymbol{\mu} = \mu_P. \tag{2.107}$$

The Lagrangian for this problem is

$$L(\mathbf{w}) = \frac{1}{2}\mathbf{w}'\mathbf{C}\mathbf{w} + \lambda_1(\mu_P - \mathbf{w}'\boldsymbol{\mu}) + \lambda_2(1 - \mathbf{w}'\mathbf{1}). \tag{2.108}$$

Setting the derivative of the Lagrangian equal to zero and solving for the MVO weight vector results in:

$$\mathbf{w}_{\text{mv}} = \lambda_1\mathbf{C}^{-1}\boldsymbol{\mu} + \lambda_2\mathbf{C}^{-1}\mathbf{1} \tag{2.109}$$

where expressions for the Lagrange multipliers λ_1 and λ_2 can be obtained by applying the constraints (2.107).

Straightforward algebra in 2.15 results in the MVO portfolio weight expression

$$\mathbf{w}_{\text{mv}} = \mathbf{w}_1 + \mu_P\mathbf{w}_2 \tag{2.110}$$

and the formulas for $\mathbf{w}_1$ and $\mathbf{w}_2$, which depend on the values of $\boldsymbol{\mu}$ and $\mathbf{C}$. Note that the above formula provides the MVO portfolio weight vector as a function of the portfolio mean return μ_P which may be positive or negative. Exercise 2.16 shows furthermore that the covariance of the returns $r_{P_1} = \mathbf{w}'_{\text{mv},1}\mathbf{r}$ and $r_{P_2} = \mathbf{w}'_{\text{mv},2}\mathbf{r}$ of two MVO portfolios is given by the expression

$$\operatorname{cov}(r_{P_1}, r_{P_2}) = \sigma^2_{\text{gmv}} + \frac{1}{d\sigma^2_{\text{gmv}}}\left(\mu_{P_1} - \mu_{\text{gmv}}\right)\left(\mu_{P_2} - \mu_{\text{gmv}}\right) \tag{2.111}$$

where the unusual constant

$$d = (\boldsymbol{\mu}'\mathbf{C}^{-1}\boldsymbol{\mu})(\mathbf{1}'\mathbf{C}^{-1}\mathbf{1}) - \left(\mathbf{1}'\mathbf{C}^{-1}\boldsymbol{\mu}\right)^2 \tag{2.112}$$

is *positive* except for a very special case that does not arise in practice.[21] The formula (2.111) results in the somewhat surprising fact that the covariance of any MVO portfolio with the global minimum variance (GMV) frontier is equal to the variance of the GMV.

The special case of formula (2.111) with $P_1 = P_2 = P$ results in the MVO portfolio variance expression

$$\sigma^2_{\text{mv}} = \sigma^2_{\text{gmv}} + \frac{1}{d\sigma^2_{\text{gmv}}}\left(\mu_P - \mu_{\text{gmv}}\right)^2. \tag{2.113}$$

Since d is positive, this formula can be rearranged to give the following basic formula for the MVO portfolio mean return μ_{mv} as a function of σ_{mv}:

$$\mu_{\text{mv}} = \mu_{\text{gmv}} \pm \sqrt{d} \cdot \sigma_{\text{gmv}} \cdot \left(\sigma^2_{\text{mv}} - \sigma^2_{\text{gmv}}\right)^{1/2}, \sigma_{\text{mv}} \geqslant \sigma_{\text{gmv}}. \tag{2.114}$$

The the complete μ_{mv} versus σ_{mv} unbounded curve defined by the above equation for both the positive and negative signs is called the portfolio *frontier*.[22] In mathematics this type of curve is called a *hyperbola*. The part of the curve with the positive sign for the second term is called the portfolio *efficient frontier*. Obviously, among all possible frontier portfolios, a portfolio manager will choose portfolios that lie on the efficient frontier. Note that the value of μ_{mv}in creases more rapidly with σ_{gmv} the greater the value of d.

An example of a MVO frontier and efficient frontier for a three asset portfolio is provided in Example 2.14 and Figure (2.16) in the following discussion of MaxQU fully-invested portfolios.

2.4.2 MaxQU Portfolios of Risky Assets

A maximum quadratic utility (MaxQU) fully invested portfolio is obtained by maximizing the quadratic utility objective function

$$\begin{aligned} Q(\mathbf{w}) &= \mu_P(\mathbf{w}) - \tfrac{1}{2}\lambda\sigma^2_P(\mathbf{w}) \\ &= \mathbf{w}'\boldsymbol{\mu} - \tfrac{1}{2}\lambda\mathbf{w}'\mathbf{C}\mathbf{w} \end{aligned} \tag{2.115}$$

where the risk parameter λ is positive, subject to the full–investment constraint

$$\mathbf{w}'\mathbf{1} = 1. \tag{2.116}$$

It is clear that as λ goes to infinity, the quadratic form term dominates the QU objective function, and the optimal portfolio becomes the GMV portfolio. We show subsequently that, as one might intuitively expect, the smaller the value of the risk aversion parameter, the larger the portfolio return and the larger the portfolio volatility risk.

[21] We note that the constant d also appears in the formulas for MaxQU fully-invested portfolios in the next Section (2.4.2), and in the formulas for MaxQU benchmark relative portfolios in Section (2.4.4).

[22] We note in passing that classical presentations of the above formulas are often stated in terms of the abstract quantities a, b, c, and d, whereas here we state them in terms of the main portfolio mean and variance quantities μ_{gmv}, μ_{mv}, σ^2_{gmv}, σ^2_{mv}

The Lagrangian for the MaxQU portfolio optimization problem is

$$L(\mathbf{w}) = \mathbf{w}'\boldsymbol{\mu} - \frac{\lambda}{2}\mathbf{w}'\mathbf{C}\mathbf{w} + \gamma \cdot (1 - \mathbf{w}'\mathbf{1}), \quad \lambda > 0 \tag{2.117}$$

where γ is the Lagrange multiplier. Setting the derivative of the Lagrangian equal to zero gives

$$\boldsymbol{\mu} - \lambda\mathbf{C}\mathbf{w} - \gamma \cdot \mathbf{1} = 0$$

and so the portfolio weight vector has the form

$$\mathbf{w} = \lambda^{-1}\mathbf{C}^{-1}(\boldsymbol{\mu} - \gamma \cdot \mathbf{1}). \tag{2.118}$$

Applying the full–investment constraint gives:

$$\mathbf{1}'\mathbf{w} = \lambda^{-1}\mathbf{1}'\mathbf{C}^{-1}(\boldsymbol{\mu} - \gamma \cdot \mathbf{1}) = \lambda^{-1}\left(\mathbf{1}'\mathbf{C}^{-1}\boldsymbol{\mu} - \gamma\mathbf{1}'\mathbf{C}^{-1}\mathbf{1}\right) = 1 \tag{2.119}$$

and solving for γ gives

$$\gamma_{qu} = \frac{\mathbf{1}'\mathbf{C}^{-1}\boldsymbol{\mu} - \lambda}{\mathbf{1}'\mathbf{C}^{-1}1}. \tag{2.120}$$

Plugging this γ into (2.118) and rearranging terms shows that the MaxQU optimal weight vector $\mathbf{w}_{\text{qu}}$ has the form:

$$\begin{aligned} \mathbf{w}_{qu} &= \frac{\mathbf{C}^{-1}\mathbf{1}}{\mathbf{1}'\mathbf{C}\mathbf{1}} + \lambda^{-1}\mathbf{C}^{-1}\left(\boldsymbol{\mu} - \frac{1'\mathbf{C}^{-1}\boldsymbol{\mu}}{1'\mathbf{C}^{-1}\mathbf{1}} \cdot \mathbf{1}\right) \\ &= \mathbf{w}_{\text{gmv}} + \lambda^{-1}\mathbf{C}^{-1}\left(\boldsymbol{\mu} - \mu_{\text{gmv}} \cdot \mathbf{1}\right). \end{aligned} \tag{2.121}$$

Let's define the second term above, except for the λ^{-1} factor, as a portfolio weight:

$$\mathbf{w}_0 = \mathbf{C}^{-1}\left(\boldsymbol{\mu} - \mu_{\text{gmv}} \cdot \mathbf{1}\right). \tag{2.122}$$

Since $\mathbf{w}_{\text{qu}}$ and $\mathbf{w}_{\text{gmv}}$ are both fully–invested portfolios, whose weights sum to one, it must be that the weights $\mathbf{w}_0$ must sum to zero (this fact can also be shown directly from the above formula).

Thus $\mathbf{w}_0$ represents a *zero–investment* portfolio, which is alternatively referred to as a *dollar neutral* portfolio, and every QU optimal portfolio has the *two–fund representation* of the optimal QU weight:

$$\mathbf{w}_{\text{qu}} = \mathbf{w}_{\text{gmv}} + \lambda^{-1} \cdot \mathbf{w}_0, \quad \lambda > 0. \tag{2.123}$$

It is left as a straightforward exercise to show that a QU optimal portfolio with weight vector $\mathbf{w}_{\text{qu}}$ has a mean return:

$$\begin{aligned} \mu_{\text{qu}} &= \boldsymbol{\mu}'\mathbf{w}_{\text{qu}} \\ &= \mu_{\text{gmv}} + \lambda^{-1}\sigma^2_{\text{gmv}} \cdot d \end{aligned} \tag{2.124}$$

where $d > 0$ is given by (2.112).

Note that the MaxQU portfolio mean return given by (2.124) increases toward infinity with as the positive risk aversion parameter λ decreases toward zero, and decreases toward μ_{gmv} as risk aversion λ tends to infinity.

With some more vector–matrix algebra manipulation that we leave as an Exercise, one can show that a MaxQU optimal portfolio has variance:

$$\begin{aligned}\sigma_{\text{qu}}^2 &= \mathbf{w}_{\text{qu}}'\mathbf{C}\mathbf{w}_{\text{qu}} \\ &= \sigma_{\text{gmv}}^2 \cdot \left(1 + \lambda^{-2} \cdot d\right).\end{aligned} \tag{2.125}$$

Recalling that d is positive we see that the MaxQU optimal portfolio variance has a minimum value equal to that of the global minimum variance portfolio as the risk aversion parameter tends to infinity, and has a variance that increases with decreasing risk aversion.

Rearranging equation (2.124) we have

$$\mu_{\text{qu}} - \mu_{\text{gmv}} = \lambda^{-1}\sigma_{\text{gmv}}^2 \cdot d \tag{2.126}$$

and rearranging ((2.125)) we have

$$\sigma_{\text{qu}}^2 - \sigma_{\text{gmv}}^2 = \lambda^{-2}\sigma_{\text{gmv}}^2 \cdot d. \tag{2.127}$$

Squaring the first of these and using the result to eliminate λ^{-2} in the second equation gives:

$$\sigma_{\text{qu}}^2 - \sigma_{\text{gmv}}^2 = \frac{1}{\sigma_{\text{gmv}}^2 \cdot d}(\mu_{\text{qu}} - \mu_{\text{gmv}})^2 \tag{2.128}$$

which, recalling that $d > 0$, allows one to express the MaxQU portfolio mean return μ_{qu} in terms of its volatility σ_{qu} as:

$$\mu_{\text{qu}} = \mu_{\text{gmv}} + \sqrt{d} \cdot \sigma_{\text{gmv}} \cdot \left(\sigma_{\text{qu}}^2 - \sigma_{\text{gmv}}^2\right)^{1/2}, \sigma_{\text{qu}} \geqslant \sigma_{\text{gmv}} \tag{2.129}$$

It is interesting that for a fixed increment $\sigma_{\text{qu}}^2 - \sigma_{\text{gmv}}^2$, the MaxQU portfolio mean return μ_{qu} increases with both σ_{gmv} and $\sqrt{d}$.

Note that the above equation is identical to the MVO equation (2.114) on the efficient frontier where $\mu_{\text{qu}} \geqslant \mu_{\text{gmv}}$. Thus we now have, in addition to Theorem 2.1, an explicit formula confirmation that the mean versus volatility performance of a fully invested MaxQU optimal performance is identical to that of a fully–invested MVO portfolio on the portfolio efficient frontier.

Example 2.14. Portfolio Frontier of Three Asset Portfolio and Portfolio Weights Along the Efficient Frontier

Consider a portfolio of three stocks whose returns are uncorrelated, with means $\boldsymbol{\mu}_3 = (0.10, 0.04, 0.02)$ and variances $\boldsymbol{\sigma}_3^2 = (0.20^2, 0.15^2, 0.10^2)$. Evaluation of μ_{gmv} and σ_{gmv}^2 with the formulas (2.45) and (2.46),

and evaluation d with the formula (2.112), allows one to use the formula (2.129) to plot the portfolio mean return versus its volatility. The result is the efficient frontier in the upper part of the curve shown in Figure (2.16). The lower part of the curve is the inefficient part of the overall portfolio frontier defined by (2.114) for MV optimal portfolios.

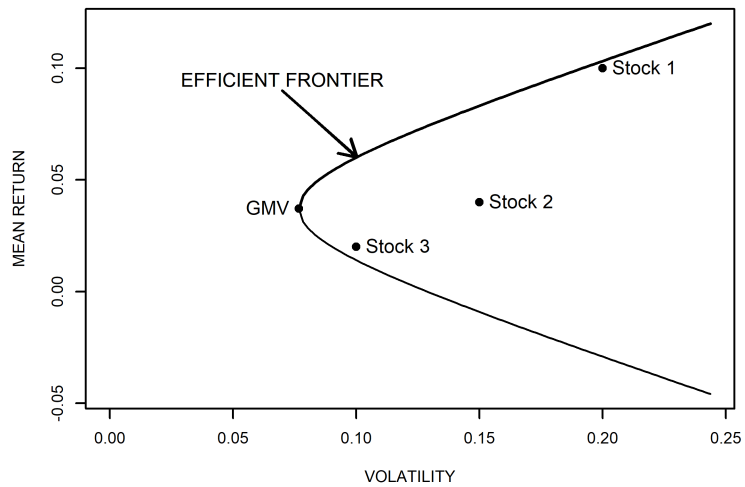


Fig. 2.16 MVO Portfolio Frontier and QU Efficient Frontier for 3 Uncorrelated Returns Means $\boldsymbol{\mu}_3 = (0.10, 0.04, 0.02)'$, and Volatilities $\boldsymbol{\sigma}_3 = (0.20, 0.15, 0.10)'$.

In addition to the efficient frontier curve, one also needs to be able to compute the portfolio weights for points along the efficient frontier. One can do so in terms of either the portfolio mean return μ_{qu} or the portfolio volatility σ_{qu}. To do so in terms of the portfolio mean return, you solve for λ^{-1}in (2.126) and substitute the result in the MaxQU weight formula (2.123) to obtain the following alternative two-fund representation of a MaxQU optimal portfolio:

$$\mathbf{w}_{\text{qu}} = \mathbf{w}_{\text{gmv}} + \frac{\mu_{\text{qu}} - \mu_{\text{gmv}}}{\sigma^2_{\text{gmv}} \cdot d} \cdot \mathbf{w}_0, \quad \mu_{\text{qu}} \geqslant \mu_{\text{gmv}}. \tag{2.130}$$

The above representation specifies the entire family of efficient frontier weights as a function of the portfolio mean return μ_{qu} as independent variable. Furthermore, it follows from the above representation that any combination $\mathbf{w}_{\text{qu},\lambda} = (1-\lambda)\mathbf{w}_{\text{qu},1} + \lambda\mathbf{w}_{\text{qu},2}$ of efficient frontier portfolio weight vectors $\mathbf{w}_{\text{qu},1} \neq \lambda\mathbf{w}_{\text{qu},2}$ with $0 < \lambda < 1$ is an efficient portfolio weight vector given by (2.130) with $\mu_{\text{qu},\lambda} = (1-\lambda)\mu_{\text{qu},1} + \lambda\mu_{\text{qu},2}$.

Likewise, you can express $\mathbf{w}_{\text{qu}}$ in terms of the portfolio volatility by solving for λ^{-1}in (2.127) and substituting the result in (2.123) to obtain:

$$\mathbf{w}_{\text{qu}} = \mathbf{w}_{\text{gmv}} + \left(\frac{\sigma^2_{\text{qu}} - \sigma^2_{\text{gmv}}}{d \cdot \sigma^2_{\text{gmv}}}\right)^{1/2} \cdot \mathbf{w}_0, \quad \sigma_{\text{qu}} \geqslant \sigma_{\text{gmv}}. \tag{2.131}$$

The volatility, mean return and portfolio weights along the efficient frontier in Figure 2.16 are shown in Table 2.4 for five equally spaced portfolio volatilities. The first two rows of the table display the portfolio mean values and volatilities for those five volatilities along the efficient frontier in Figure 2.16, and other three rows provide the corresponding sets of portfolio weights.

Table 2.4 Portfolio volatility, mean return and weights along the efficient frontier in Figure 2.16 for five equally spaced portfolio volatilities.

	P1	P2	P3	P4	P5
MU	0.077	0.119	0.160	0.202	0.244
VOL	0.037	0.069	0.087	0.104	0.120
W1	0.149	0.544	0.765	0.968	1.163
W2	0.262	0.295	0.314	0.331	0.347
W3	0.588	0.161	-0.079	-0.299	-0.510

The values of the portfolio weights at the five efficient frontier points are also shown in the Figure 2.17 barplot.

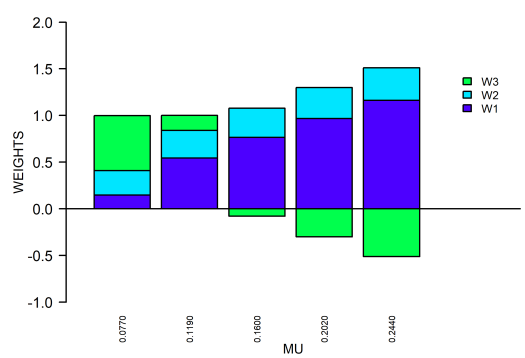


Fig. 2.17 Weights at Five Values Along the Efficient Frontier for the Three Stocks Portfolio

Note that since there is no constraint to prevent portfolio weights from being either negative or greater than one, it is not surprising to see that both of these cases occur along the upper region of the efficient frontier. In particular, the portfolio weight of Stock 3 is increasingly negative when the portfolio mean return is 16% and higher, reaching -0.510 when the mean return is 24.4%, and in that case the weight of Stock 1 has a value 1.163.

The following example is a more realistic than the above toy portfolio of three stocks.

Example 2.15. Efficient Frontier for Portfolio of Ten Smallap Stocks.

This example uses the monthy returns of 10 smallcap stocks from 1997 to 2001, shown in Figure 2.7, ignoring the Market returns of that Figure. We use those returns to estimate the mean vector and co-

variance matrix from the five years of monthly returns of the 10 stocks, and use formula (2.129) with the quantities μ_{gmv}, σ_{gmv}, and d in that formula computed as described in Example 2.14. Figure 2.18 shows the MaxQU efficient frontier for full investment in ten risky stoccks, along with the mean–volatility positions of the ten stocks. The improved performance of the efficient frontier reflects the diversification benefit of MaxQU/MVO mean-variance investing.

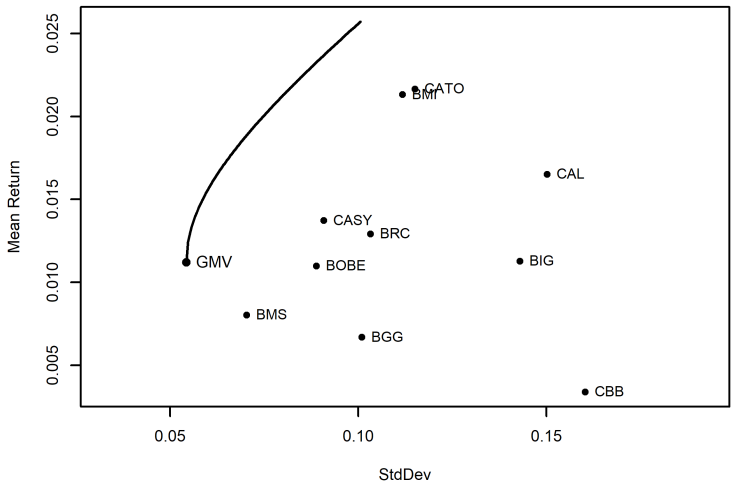


Fig. 2.18 Efficient Frontier for Full Investment in 10 Smallcap Stocks 1997–2001

The weights along the efficient frontier in Figure 2.18 are shown in Figure 2.19. The left–hand bar in the figure shows the weights for the GMV portfolio, and the right–hand bar shows the weights for the upper right–hand end of the efficient frontier. Note that the GMV portfolio involves a very tiny bit of leverage associated with a tiny amount of shorting. However., the amount of leverage increases rapidly along the efficient frontier with increasing portfolio volatility, with CBB, BGG and BIG accounting in roughly equal ammounts for the almost all of the shorting. This is not surprising given that two of those stocks have the lowest mean returns among the 10 stocks. Of course the covariance matrix of the set of all 10 returns has influence beyond that of the means of the returns, which evidently accounts for the amount of shorting of BIG, even though its mean return is considerably larger than that of BMS and nearly equal to that of BOBE.

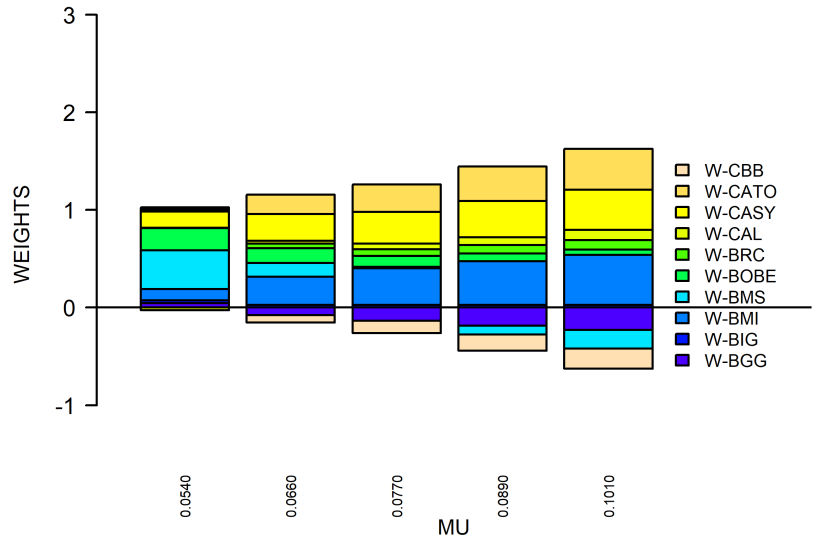


Fig. 2.19 Weights Along Efficient Frontier for Full Investment in 10 Smallcap Stocks 1997–2001

N.B. The shorting that occurs along the efficient frontier in Figure 2.19 is not something a portfolio manager would do in practice. If the manager was going to invest in these ten stocks (which in itself is questionable), then some weights constraints would be imposed, e.g., long–only constraints that allow no shorting, or box constraints that prevent shorting, or allow only a small amount of shorting. Constrained MaxQU porfolio optimization is discussed in Chapter 5.

Sensitivity of Optimal Weights to Changes in Asset Mean Returns

It is important to understand the sensitivity of the optimal weights given by 2.145 to changes in the mean returns of the assets, and it turns out larger correlations between the assets give rise to greater such sensitivity. To illustrate this claim, we consider the special case of two assets with mean vector $\boldsymbol{\mu} = (\mu_1, \mu_2)'$ and covariance matrix:

$$\mathbf{C} = \begin{pmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{pmatrix}.$$

With optimal weights vector $\mathbf{w} = (w_1, w_2)'$ given by 2.145, it is straightforward to shows that

$$\frac{dw_1}{d\mu_1} = \lambda^{-1}\frac{1}{\sigma_1^2(1-\rho^2)}. \tag{2.132}$$

Here we see that larger absolute values of the correlation ρ lead to larger sensitivities of $w_1$to changes in μ_1.

A Special Efficient Frontier Portfolio

Note that if you try to maximize the quadratic utility (2.115) without imposing the full investment constraint (2.116), the optimal weights have the form

$$\mathbf{w} = \lambda^{-1}\mathbf{C}^{-1}\boldsymbol{\mu} \tag{2.133}$$

and an attempt to impose the full–investment constraint (2.116) results in $\lambda = \mathbf{1}'\mathbf{C}^{-1}\boldsymbol{\mu}$. But since it is assumed that risk aversion is positive, i.e., $\lambda > 0$, imposing this constraint can only be done if $\mathbf{1}'\mathbf{C}^{-1}\boldsymbol{\mu} > 0$, which is not necessarily the case. But when the inequality $\mathbf{1}'\mathbf{C}^{-1}\boldsymbol{\mu} > 0$ does hold, the portfolio weight vector is

$$\breve{\mathbf{w}} = \frac{\mathbf{C}^{-1}\boldsymbol{\mu}}{\mathbf{1}'\mathbf{C}^{-1}\boldsymbol{\mu}}. \tag{2.134}$$

It should be noted that in that case, the above weight vector is also obtained from the general formula (2.121) by the choice $\lambda = \mathbf{1}'\mathbf{C}^{-1}\boldsymbol{\mu}$.

The portfolio with the above weight vector has mean and variance

$$\breve{\mu}_{qu} = \frac{\boldsymbol{\mu}'\mathbf{C}^{-1}\boldsymbol{\mu}}{\mathbf{1}'\mathbf{C}^{-1}\boldsymbol{\mu}} \qquad \breve{\sigma}^2_{qu} = \frac{\boldsymbol{\mu}'\mathbf{C}^{-1}\boldsymbol{\mu}}{(\mathbf{1}'\mathbf{C}^{-1}\boldsymbol{\mu})^2} \tag{2.135}$$

Using the positivity of d given by (2.112), one can establish the anticipated inequalities

$$\breve{\mu}_{qu} \geqslant \mu_{\text{gmv}} \qquad \breve{\sigma}^2_{qu} \geqslant \sigma^2_{\text{gmv}} \tag{2.136}$$

where μ_{gmv} and σ^2_{gmv} are the mean and variance of the GMV portfolio, and the inequalities are strict in practice.

2.4.3 MaxQU Portfolios with Cash and Risky Assets

We now turn to the highly important case where the portfolio manager invests in a combination of risky assets and *cash*. Here cash is taken to be a risk–free instrument such as a 90–day Treasury Bill (T–Bill) in the U.S., and the return on the risk–free instrument is assumed to be r_f over the time horizon of the investment. Now the discussion of MaxQU portfolios is in terms of asset excess returns and mean excess returns, both in excess of r_f, and portfolio excess returns and mean excess returns, also in excess of r_f.

In the previous subsection Theorem 2.1 shows that a MaxQU portfolio is a MVO. This result also holds in the case of portfolios with a mix of cash and risky assets, by replacing mean return with mean excess return. Consequently, one can focus on the MaxQU approach for portfolios with cash and risky assets. The main results of this section are:

1. The relationship between the portfolio mean excess return and portfolio volatility is linear.

2. At any level of achievable portfolio mean excess return, the QU portfolio volatility risk is less or equal to that of QU portfolio consisting of risky assets only.

Excess Returns and Mean Excess Returns

The vector of asset excess returns is

$$\mathbf{r}_e = \mathbf{r} - \mathbf{1} r_f \tag{2.137}$$

where $\mathbf{1}$ is a vector of one's, and correspondingly the vector of mean excess returns is

$$\boldsymbol{\mu}_e = \boldsymbol{\mu} - \mathbf{1} r_f. \tag{2.138}$$

We are focused on portfolios where the total investment is split between a fraction w_0 in cash and a fraction $\mathbf{w}'\mathbf{1} = \sum_{i=1}^{N} w_i$ in risky assets, resulting in the constraint

$$w_0 + \mathbf{w}'\mathbf{1} = 1. \tag{2.139}$$

Note that there is no constraint on short–selling the risky assets, and there is no constraint that w_0 be positive, i.e., a portfolio manager could have w_0 negative by borrowing to leverage the portfolio with $\mathbf{w}'1 > 1$. With that flexibility in mind, the portfolio mean return $\mu_P(\mathbf{w})$ is the sum of the return on cash and the return on the portfolio of risky assets, namely

$$\begin{aligned} \mu_P(\mathbf{w}) &= w_0 r_f + \mathbf{w}'\boldsymbol{\mu} \\ &= w_0 r_f + \mathbf{w}'(\boldsymbol{\mu}_e + \mathbf{1} r_f) \\ &= r_f + \mathbf{w}'\boldsymbol{\mu}_e \end{aligned} \tag{2.140}$$

where we have used (2.138) in going from the first line to the second line above, and used (2.139) in going from the second to the third. It follows that the portfolio mean excess return is:

$$\begin{aligned} \mu_{P,e}(\mathbf{w}) &= \mu_P(\mathbf{w}) - r_f \\ &= \mathbf{w}'\boldsymbol{\mu}_e. \end{aligned} \tag{2.141}$$

IMPORTANT POINT: The above expressions for $\mu_P(\mathbf{w})$ and $\mu_{P,e}(\mathbf{w})$ also hold in the special case of (2.139) where $w_0 = 0$, for which the portfolio that is fully invested in risky assets.

The MaxQU Solution

The QU optimal portfolio weights maximize the quadratic utility

$$\begin{aligned} Q(\mathbf{w}) &= \mu_P(\mathbf{w}) - \tfrac{1}{2}\lambda\sigma_P^2(\mathbf{w}) \\ &= w_0 r_f + \mathbf{w}'\boldsymbol{\mu} - \tfrac{1}{2}\lambda\mathbf{w}'\mathbf{C}\mathbf{w} \\ &= r_f + \mathbf{w}'\boldsymbol{\mu}_e - \tfrac{1}{2}\lambda\mathbf{w}'\mathbf{C}\mathbf{w}. \end{aligned} \tag{2.142}$$

where $\lambda > 0$ is a *risk aversion* parameter, and $\mathbf{C}$ is the covariance matrix of the risky assets. Note that the constraint (2.139) was used in going from the second line to the third line above, and so the third line implicitly contains that constraint. Consequently, choosing the weight vector to maximize the third line does not require a Lagrange multiplier method. Instead one just solves for the weight vector $\mathbf{w}$ that maximizes the third line and plug the result into the constraint equation (2.139), thereby obtaining the value for w_0:

$$w_0 = 1 - \mathbf{w}'\mathbf{1}. \tag{2.143}$$

Since r_f is a constant in the third line of (2.142), it will suffice to choose $\mathbf{w}$ to maximize the equivalent objective:

$$Q^*(\mathbf{w}) = \mathbf{w}'\boldsymbol{\mu}_e - \frac{1}{2}\lambda\mathbf{w}'\mathbf{C}\mathbf{w}. \tag{2.144}$$

Setting the derivative of $Q^*(\mathbf{w})$ with respect to $\mathbf{w}$ equal to zero and solving for $\mathbf{w}$ gives

$$\mathbf{w} = \lambda^{-1}\mathbf{C}^{-1}\boldsymbol{\mu}_e \tag{2.145}$$

and w_0 is then obtained from (2.143).

N.B. For the sake of notational simplicity, we do not use any special notation on the weight vector $\mathbf{w}$ to indicate that it is a MaxQU portfolio weight, as this will generally be clear from the context, e.g., from the formula for such a $\mathbf{w}$.

Note that as risk aversion $\lambda \to \infty$ the optimal portfolio consists of an all cash position, and as risk aversion $\lambda \to 0$ the portfolio weights become unbounded in absolute value, and correspondingly the amount of short selling becomes unbounded. The inverse λ^{-1} of the risk aversion parameter is a *risk tolerance* parameter. As risk tolerance goes to zero the portfolio consists of an all cash position, and as risk tolerance goes to infinity the portfolio weights become unbounded. In the latter case, it is an interesting exercise to determine what happens to the cash position.

Using (2.141), the optimal portfolio mean excess return is

$$\mu_e = \mathbf{w}'\boldsymbol{\mu}_e = \lambda^{-1}\boldsymbol{\mu}_e'\mathbf{C}^{-1}\boldsymbol{\mu}_e > 0 \tag{2.146}$$

where the positivity of μ_e follows from the fact that λ is positive and $\mathbf{C}^{-1}$ is positive definite.. Solving the above equation for λ^{-1} and substituting the result into (2.145) gives:

$$\mathbf{w} = \frac{\mathbf{C}^{-1}\boldsymbol{\mu}_e}{\boldsymbol{\mu}_e'\mathbf{C}^{-1}\boldsymbol{\mu}_e} \cdot \mu_e. \tag{2.147}$$

Note that one can think of μ_e as a "target mean return" that specifies the MaxQU weight vector through the above formula. The optimal portfolio variance is

$$\sigma^2 = = \mathbf{w}'\mathbf{C}\mathbf{w} = \frac{1}{\boldsymbol{\mu}_e'\mathbf{C}^{-1}\boldsymbol{\mu}_e} \cdot \mu_e^2 \tag{2.148}$$

Since the above formula contains only positive quantities, one can take square roots on both sides to obtain $\mu_e = \sigma\sqrt{\boldsymbol{\mu}_e'\mathbf{C}^{-1}\boldsymbol{\mu}_e}$, and substituting this into (2.147) gives the alternate optimal weight vector representation:

$$\mathbf{w} = \frac{\mathbf{C}^{-1}\boldsymbol{\mu}_e}{\sqrt{\boldsymbol{\mu}_e'\mathbf{C}^{-1}\boldsymbol{\mu}_e}} \cdot \sigma. \tag{2.149}$$

The above can be thought of as expressing the MaxQU weight in terms of a portfolio's target volatility.

Since $\mu_e > 0$, we can take the square root of both sides to obtain the following linear relationship between the optimal portfolio mean excess return and volatility:

$$\mu_e = \sqrt{\boldsymbol{\mu}_e'\mathbf{C}^{-1}\boldsymbol{\mu}_e} \cdot \sigma. \tag{2.150}$$

We state the above result in slightly different form as the following theorem.

Theorem 2.2. Quadratic utility optimal portfolios that contain cash and risky assets have the efficient frontier straight line equation

$$\mu = r_f + \sqrt{\boldsymbol{\mu}_e'\mathbf{C}^{-1}\boldsymbol{\mu}_e} \cdot \sigma. \tag{2.151}$$

This straight line is referred to as the *Capital Allocation Line* (CAL), and the slope of the CAL is the Sharpe Ratio

$$\text{SR} = \frac{\mu_e}{\sigma} = \sqrt{\boldsymbol{\mu}_e'\mathbf{C}^{-1}\boldsymbol{\mu}_e}. \tag{2.152}$$

Example 2.16. Linear Efficient Frontier for 10 Smallcap Stocks.

Here we consider the returns of the same 10 smallcap stocks as in Figure 2.7, but since we are now focused on portfolios that are a mix of cash and risky stocks, we replace the Market return in that figure with the return of a Risk-Free asset (the CRSP `ret13WkBill`), as shown in Figure 2.20. Note that the RiskFree returns display a slowly varying non-stationary character.

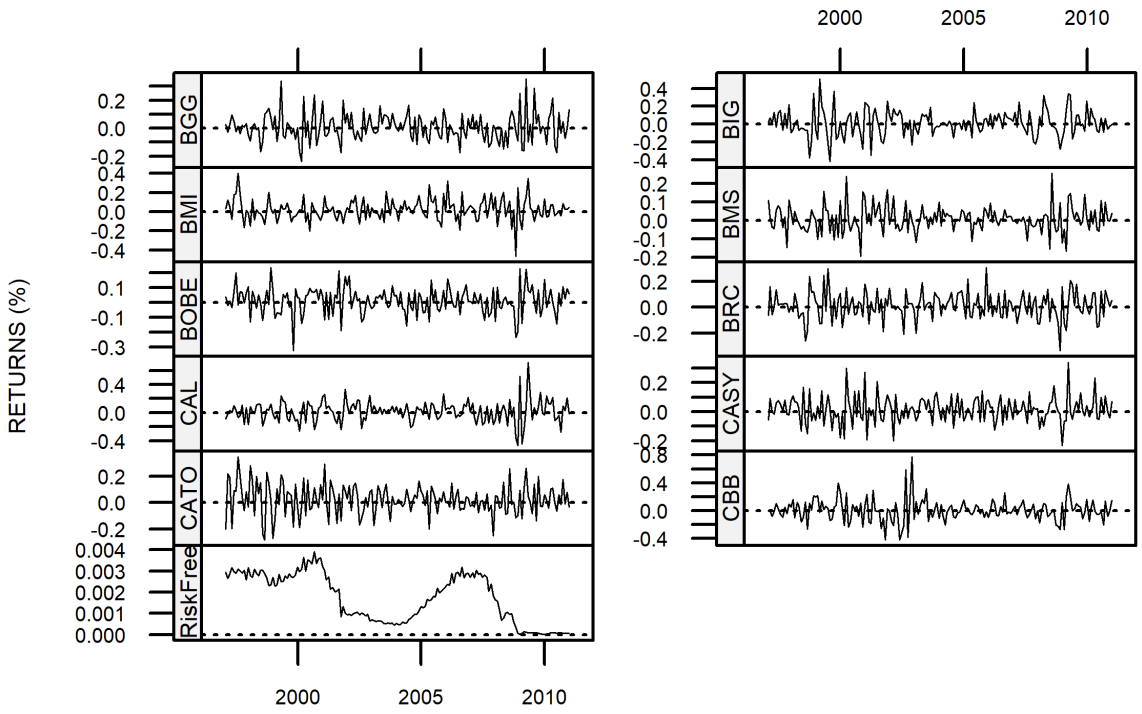


Fig. 2.20 Returns of 10 Smallcap Stocks and Risk-Free 13 Week Bill 1997 - 2001

Figure 2.21 shows the linear efficient frontier and the mean–volatility positions of the ten stocks whose returns are shown in Figure 2.20, along with the mean-volatility position of an equally weighted portfolio of the stocks. The figure legend shows the monthly risk–free rate of 0.5% used for the Sharpe Ratio on the efficient frontier, computed using the formula (2.152).

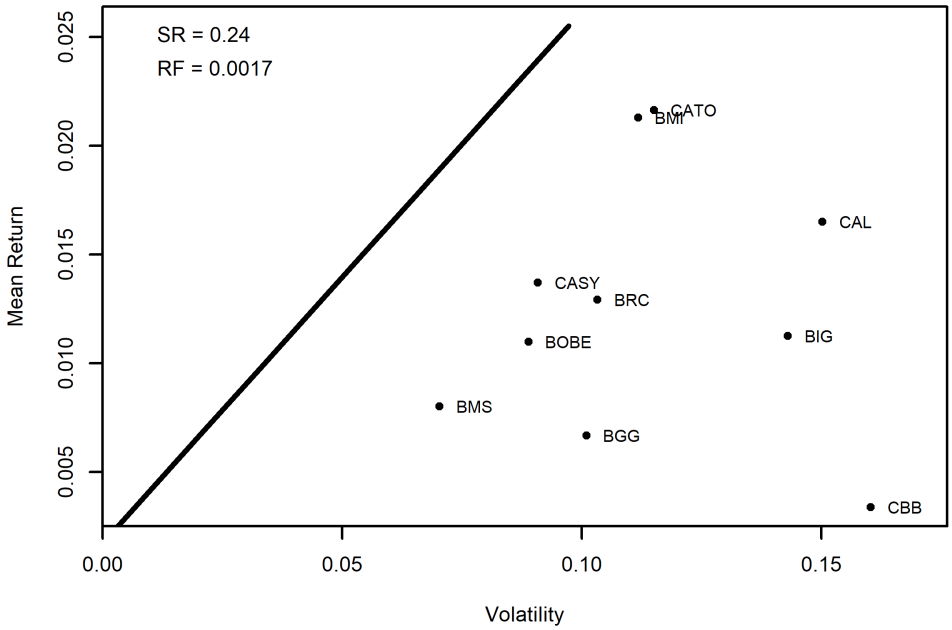


Fig. 2.21 Cash Plus Risky Assets Capital Allocation Line for 10 Smallcap Stocks 1997 - 2001

Figure 2.22 shows the cash position fraction w_0 of the total investment, and the risky stocks fraction $\mathbf{w}'_{\text{qu}}\mathbf{1}$ of the total investment, at 10 equally spaced points along the linear efficient frontier. The first point along the efficient frontier is the zero volatility, all cash, position, in which all wealth is invested in the risk–free asset that returns 0.5% monthly (6% annual). As one moves beyond the half–way point along the efficient frontier, the weight w_0 in the risk–free asset becomes negative, representing borrowing at the risk free, and the fraction $\mathbf{w}'_{\text{qu}}\mathbf{1}$ invested in the risky stocks increases beyond one. Borrowing has provided *leverage* to invest more in risky stocks than would be possible without borrowing.

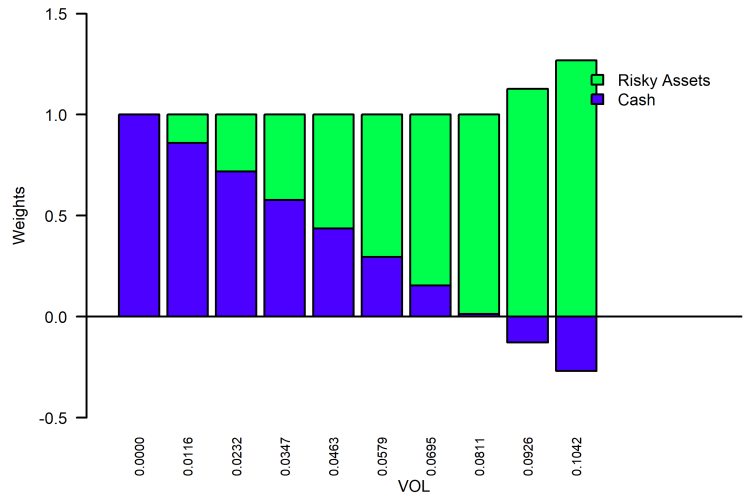


Fig. 2.22 Weights of Cash Versus Risky Assets Allocation Along the Efficient Frontier of Figure 2.21

Risk Aversion Along the Cash plus Risky Assets CAL

Note that equation (2.146) allows one to express the risk aversion parameter in terms of the asset mean excess returns, the asset returns covariance matrix and the portfolio mean excess return

$$\lambda = \frac{\boldsymbol{\mu}_e' \mathbf{C}^{-1} \boldsymbol{\mu}_e}{\mu_e}.$$

Equation (2.148) shows that the numerator in the above expression is the square of the Sharpe Ratio along the efficient frontier

$$\boldsymbol{\mu}_e' \mathbf{C}^{-1} \boldsymbol{\mu}_e = \frac{\mu_e^2}{\sigma^2}$$

and making this substitution gives the following well–known expression for the risk aversion parameter along the efficient frontier as a function of the portfolio volatility, given the Sharpe Ratio (the slope of the linear efficient frontier):

$$\lambda = \frac{\mu_e}{\sigma^2} = \frac{\mathrm{SR}}{\sigma}. \tag{2.153}$$

Existence of a No–Cash Position on the CAL

The CAL linear efficient frontier was derived with the assumption that there is a mix of cash and risky assets, and one might think that as a special case the efficient frontier should contain a position that is totally invested in risky assets. However, in order for that to be the case the condition $\mathbf{1}'\mathbf{w} = 1$ must hold, which implies that

$$\mathbf{1}'\mathbf{w} = \lambda^{-1}\mathbf{1}'\mathbf{C}^{-1}\boldsymbol{\mu}_e = 1. \tag{2.154}$$

Recalling that the risk aversion parameter λ is positive, we see that in order to have a full investment in risky assets position on the efficient frontier it must be that

$$\lambda = \mathbf{1}'\mathbf{C}^{-1}\boldsymbol{\mu}_e > 0. \tag{2.155}$$

The above expression is a linear form (not a quadratic form), and there is no guarantee that it is positive. Usually it is positive, as is easily shown for example in the case of the 10 smallcap stocks returns efficient frontier of Figure 2.21, where $\lambda = 2.10$ and $\lambda^{-1} = 0.48$.

Since we claim that the condition $\mathbf{1}'\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}_e > 0$ does not always hold, one naturally asks whether or not this condition has any special meaning. In fact this condition does have a special meaning, and to see that this is so, recall the expression (2.45) for the mean return of the GMV portfolio, and consider the mean excess return of the GMV portfolio:

$$\begin{aligned}
\mu_{\text{gmv},e} &= \frac{\mathbf{1}'\mathbf{C}^{-1}\boldsymbol{\mu}}{\mathbf{1}'\mathbf{C}^{-1}\mathbf{1}} - r_f \\
&= \frac{\mathbf{1}'\mathbf{C}^{-1}\boldsymbol{\mu}}{\mathbf{1}'\mathbf{C}^{-1}\mathbf{1}} - \frac{\mathbf{1}'\mathbf{C}^{-1}\mathbf{1}\cdot r_f}{\mathbf{1}'\mathbf{C}^{-1}\mathbf{1}} \\
&= \frac{\mathbf{1}'\mathbf{C}^{-1}\boldsymbol{\mu}_e}{\mathbf{1}'\mathbf{C}^{-1}\mathbf{1}}.
\end{aligned} \tag{2.156}$$

We see that the condition for the mean excess return of the GMV portfolio to be positive is identical to the condition (2.155) that there exists a portfolio on the cash-risky efficient frontier, i.e., on the CAL, that is fully invested in risky assets. Thus there exists a portfolio on the CAL that is fully invested in risky assets exists if, and only if, the mean excess return of the GMV portfolio is positive. It is important to know that in practice the mean excess return of a GMV portfolio is not always positive, as the following example illustrates.

Example 2.17. A GMV Portfolio with Mean Return Less Than the risk–free Rate

If one sets about looking for a GMV portfolio with mean return less than the risk–free rate, it is natural to look for time periods with high interest rates. Figure 2.23 shows the history of 90–day T–Bill annualized interest rates from 1934 to 2014. Indeed, as

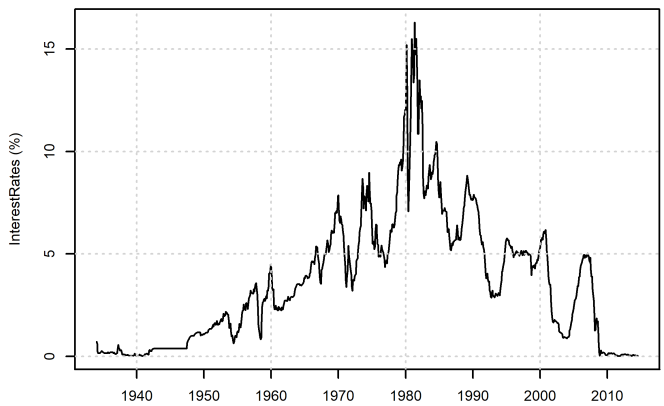


Fig. 2.23 Annualized 90 Day T–Bill Rates from 1934 to 2014. Source: Federal Reserve

an example of such behavior is obtained by computing the mean return of the GMV portfolio for the weekly returns of 10 stocks with tickers IBM, GE, XOM, GT, PG, AA, CVX, AXP, WFC, MMM from November 7, 1980 to October 30, 1981. The risk-free rate for that period is an astonishing 14.6%, and the mean return of the GMV portfolio of the 10 stocks is $\mu_{gmv} = -2.3\%$. So for this portfolio of 10 stocks there is no position along the linear efficient frontier that is fully invested in the stocks.

The Meeting of Two Efficient Frontiers: the Tangency Portfolio

Given that the mean return versus standard deviation shape of the efficient frontier with risky assets only is a hyperbola, and that the efficient frontier with a mix of risky assets and cash is linear, one naturally wonders how the two efficient frontiers are related. Recall that in the discussion involving equations (2.155) and (2.156), we showed that the linear efficient frontier (the CAL) with a mix of risky assets and cash has a position with full investment in risky assets if and only if the mean return of the global minimum variance portfolio is larger than the risk free rate, and in that case the weight vector for full investment in risky assets is obtained by plugging $\lambda^{-1} = \left(\mathbf{1}'\mathbf{C}^{-1}\boldsymbol{\mu}_e\right)^{-1}$ into the optimal linear efficient frontier formula $\mathbf{w} = \lambda^{-1}\mathbf{C}^{-1}\boldsymbol{\mu}_e$ given by (2.145). Furthermore, since the portfolio variance along the CAL involves no constraints on the weight vector $\mathbf{w}$, the variance along the CAL is always less or equal to the variance of the hyperbolic efficient frontier with weights constraint to sum to one, and the point where the two efficient frontiers coincide is a point of tangency. We therefore have the following efficient frontiers tangency result.

It is important to note that the investor's risk aversion determines her portfolio position along the CAL, i.e., the investor's risk aversion determines the ratio of cash holdings to the amount invested in the tangent portfolio T. Specifically, the risk aversion is determined by the portfolio volatility and the Sharpe Ratio according to formula 2.153. Points along the linear efficient frontier with mean return less than

that of T correspond to lending at the risk–free rate, and points along the line with mean return larger than that of T correspond to borrowing at the risk–free rate, i.e., using leverage.

Theorem 2.3. Tangency of Efficient Frontiers and Maximum Sharpe Ratio. If condition ((2.155)) holds, then the hyperbolic efficient frontier based on risky assets only must be tangent to the linear efficient frontier based on cash–plus–risky assets, the tangency portfolio has the maximum Sharpe Ratio among all efficient frontier portfolios,and the tangency portfolio T has weight vector:

$$\mathbf{w}_{\mathrm{T}} = \frac{\mathbf{C}^{-1}\boldsymbol{\mu}_e}{\mathbf{1}'\mathbf{C}^{-1}\boldsymbol{\mu}_e}. \tag{2.157}$$

Example 2.18. The Linear and Hyperbolic Efficient Frontiers for the 10 Smallcap Stocks Portfolios

The sample mean vector and sample covariance matrix of the 10 midcap stocks are used as "true" quantities to compute and display both the linear efficient frontier for cash–plus–risky assets portfolios, and the hyperbolic efficient frontier for only–risky–assets portfolios. The result is show in Figure 2.24.

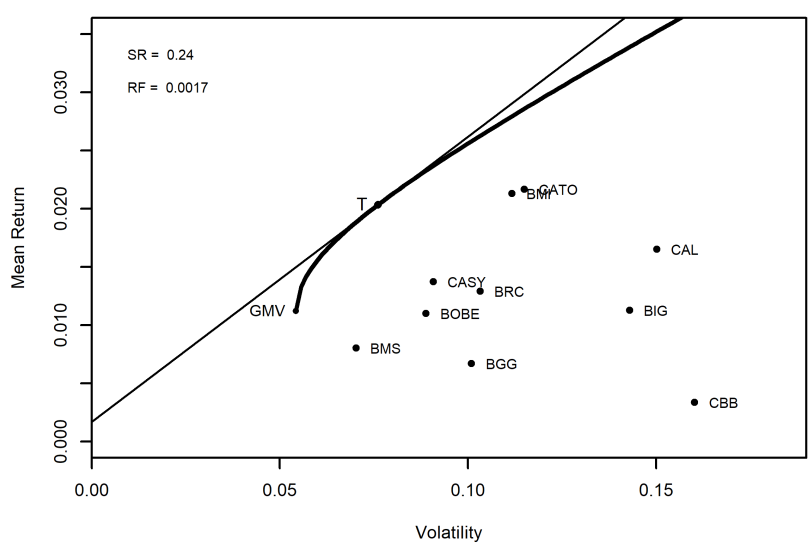


Fig. 2.24 Cash Plus Risky Assets and Risky Assets only Efficient Frontier for 10 Smallcap Stocks

The general relationship between the linear frontier (with a mix of risky assets and cash) and the hyperbolic efficient frontier (for risky assets only), exhibited in Figure 2.24, reflects the following fundamental *Separation Principle*.

The Separation Principle

The QU portfolio based on risky assets and cash along the CAL is a linear combination of cash and a tangent portfolio T, where (2.157) shows that the latter is totally determined by the mean vector and covariance matrix of the portfolio's asset returns and the risk free rate, and hence is independent of the investor's risk preferences. This result is referred to as the *separation principle*, which may be thought of as a two–fund theorem with cash being one fund and the tangent portfolio T being the other fund.

Scaling Portfolio Return and Risk

We can create a new portfolio P' with any desired volatility $\sigma_{P'}$ on the linear efficient frontier defined by (2.151) by investing a fraction

$$w_P = \frac{\sigma_{P'}}{\sigma_P} \tag{2.158}$$

of our assets in P, and the remaining fraction $w_C = 1 - w_P$ in cash.

The one–period expected return and volatility of P' are related via

$$\mu_{P'} = r_f + \mathrm{SR}_P \cdot \sigma_{P'}, \tag{2.159}$$

We can now abstract away from standard deviation and rewrite (2.159) as

$$\mu_{P'} = r_f + L \cdot \left(\mu_P - r_f\right) \tag{2.160a}$$

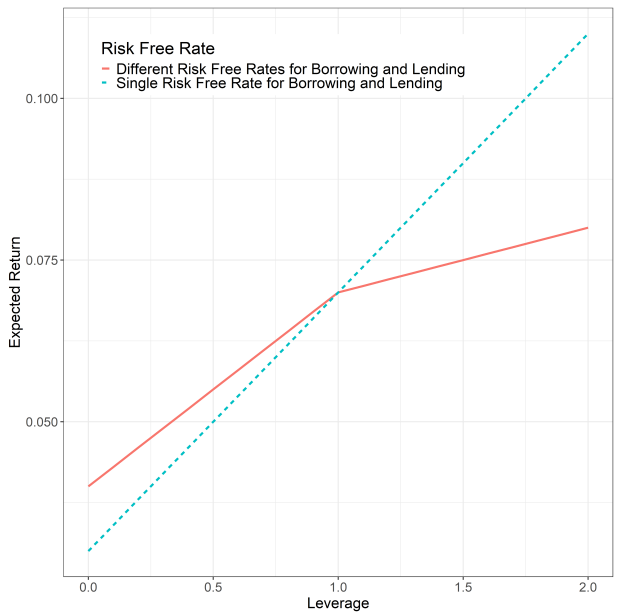
$$\sigma_{P'} = |L| \cdot \sigma_P, \tag{2.160b}$$

where $L \in (-\infty, \infty)$ is the portfolio's *leverage*. If $L < 0$, we sell our entire holding in P, and, in addition, sell short an additional $|L|$ units of P, and invest the entire proceeds in cash. If $0 \leqslant L < 1$, we sell $1 - L$ units of P and invest the proceeds in cash. If $L = 1$, we merely hold P. Finally, if $L > 1$, we borrow $L - 1$ units of cash and use it to purchase additional units of P. Observe that it is only the portfolio's excess return, and not its total return, that can be levered, as leverage comes with a cost – in this case, the risk free rate.

The Sharpe Ratio usefully characterizes P's trade–off between risk and return: in a single–period setting, a portfolio P_1 will always be preferred to another portfolio P_2 with a lower Sharpe Ratio, as for any given level of risk or leverage, portfolios P' constructed using P_1 will have a higher return than those constructed using P_2. Keep in mind that the quantities in (2.84) can exhibit significant time variation, and, in addition, that the denominator is usually larger than the numerator. As a consequence, point estimates of the Sharpe Ratio computed using realized returns are often burdened with a high variance, hampering statistical inference about its mean.

One problem with excess returns though is that the riskless asset is not uniquely determined. For example, investors may hold differing views on the tenor of the risk free rate, and an asset that is riskless

in nominal terms is not riskless in real terms., two investors may compute different Sharpe Ratios for the same portfolio.



Source: Authors' own calculations

Fig. 2.25 Expected Return vs. Leverage

More subtly, the lending and borrowing rates available to investors may differ, with the borrowing rate typically being higher than the lending rate. This induces a kink in (2.160a) at $L = 1$, as can be seen in Figure 2.25. The dotted line assumes that $r_f = 3\%$ for both borrowers and lenders, while the solid line assumes that $r_f = 4\%$ when lending and 6% when borrowing. In both cases, we assume that $\mu_P = 7\%$. Computing a realistic Sharpe Ratio requires one to first determine the intended level of leverage and to then use the relevant risk free rate to compute it.

The prospective Sharpe Ratio of risky assets must necessarily be positive – if not, there would be little reason for a risk averse investor to hold them. As can be seen from Figure 10.2, though, their sample Sharpe Ratio can be negative, and for extended periods: Bessembinder (2018) and Bessembinder et al. (2021) show that the aggregate performance of U.S. and global stock market indices is driven largely by a few big winners, and that more than half the stocks in their global sample under perform one–month U.S. Treasury Bills!

MaxQU Portfolios with General Equality Constraints

In practice portfolio managers will be interested in making use of a number of other linear equality constraints such as constraints on exposures to factor risk for factors such as Market exposure, firm Size (log of market capitalization in $), book-to-market ratio (BM), earnings–to–price ratio (EP), and

industry sector exposures. Such general constraints can be specified by

$$\mathbf{A}\mathbf{w} = \mathbf{b} \tag{2.161}$$

where $\mathbf{A}$ is a $k \times n$ matrix and $\mathbf{b}$ is a $k \times 1$ vector. For this set of linear constraints, the method of Lagrange multipliers can be used with a $k \times 1$ dimensional Lagrange multiplier vector $\boldsymbol{\gamma} = (\gamma_1, \gamma_2, \cdots, \gamma_k)'$, and the Lagrangian:

$$L(\mathbf{w}) = \mathbf{w}'\boldsymbol{\mu} - \frac{\mathbf{1}}{\mathbf{2}}\lambda\mathbf{w}'\mathbf{C}\mathbf{w} + \boldsymbol{\gamma}'(\mathbf{A}\mathbf{w} - \mathbf{b}) \tag{2.162}$$

It is left as an exercise to show that the Lagrange multiplier vector and QU optimal weight vector are given by

$$\begin{aligned} \boldsymbol{\gamma} &= \mathbf{G}^{-1}\mathbf{A}\mathbf{C}^{-1}\boldsymbol{\mu} - \lambda\mathbf{G}^{-1}\mathbf{b} \\ \mathbf{w} &= \mathbf{C}^{-1}\mathbf{A}'\mathbf{G}^{-1}\mathbf{b} + \lambda^{-1}\mathbf{C}^{-1}\left(\mathbf{I}_n - \mathbf{A}'\mathbf{G}^{-1}\mathbf{A}\mathbf{C}^{-1}\right)\boldsymbol{\mu} \end{aligned} \tag{2.163}$$

where $\mathbf{I}_n$ is the $n \times n$ identity matrix and $\mathbf{G} = \mathbf{A}\mathbf{C}^{-1}\mathbf{A}'$.

2.4.4 MaxQU Benchmark Relative Optimization

Many asset managers construct their portfolios to perform well relative to a *benchmark* portfolio such as the SP500 and the Russell 1000 largecap index, among many others. This kind of management is referred to as *benchmark relative management*, or more simply *active management*, and in this context portfolio optimization is done relative to a benchmark. To differentiate *active managers* whose portfolio performance is measured relative to a benchmark from managers who use MVO as in the previous sections, the latter kind of managers are often referred to as *absolute returns managers*. We now briefly describe the approach taken by active managers in optimizing their portfolios.

For time period t, let $r_{B,t}$ be the return of the manager's benchmark portfolio B and let $r_{P,t}$ be the manager's portfolio, both of which are based on the same cross-section of asset returns $\mathbf{r}_t = (r_{t,1}, r_{t,2}, \cdots, r_{t,N})'$ at time t. It is assumed that $\mathbf{r}_t$ is a stationary multivariate time series with constant mean value $E(\mathbf{r}) = \boldsymbol{\mu}$ and constant covariance matrix $\mathbf{C}$. For notational convenience we suppress the subscript t, and let r_B, r_P, and $\mathbf{r}$ represent the values of these quantities at any fixed t. Let $\mathbf{w}_B$ be the benchmark weights and let $\mathbf{w}_P$ be the manager's portfolio weights, both of which are assumed to satisfy the full-investment constraint $\mathbf{1}'\mathbf{w}_B = \mathbf{1}'\mathbf{w}_P = 1$. Then the benchmark and portfolio returns $r_B = \mathbf{w}_B'\mathbf{r}$ and $r_P = \mathbf{w}_P'\mathbf{r}$ have expected values $\mu_B = \mathbf{w}_B'\boldsymbol{\mu}$ and $\mu_P = \mathbf{w}_P'\boldsymbol{\mu}$, and variances σ_B^2 and σ_P^2, respectively.

The manager's *active return* is defined as

$$r_A = r_P - r_B \tag{2.164}$$

and the corresponding mean active return $\mu_A = \mathrm{E}(r_A)$ and active variance $\sigma_A^2 = \mathrm{var}(r_A)$ are given by

$$\mu_A = \mu_P - \mu_B \tag{2.165}$$
$$\sigma_A^2 = \sigma_P^2 + \sigma_B^2 - 2 \cdot cov(r_P, r_B). \tag{2.166}$$

The active variacne σ_A^2 is caled the *Tracking Error Variance* (TEV), and σ_A is called the *Tracking Error* (TE).

The active weights are defined as

$$\mathbf{w}_A = \mathbf{w}_P - \mathbf{w}_B \tag{2.167}$$

and since $\mathbf{1}'\mathbf{w}_B = \mathbf{1}'\mathbf{w}_P = 1$, the active weights satisfy the *self-financing* (a.k.a. *dollar neutral*) condition:

$$\mathbf{1}'\mathbf{w}_A = 0. \tag{2.168}$$

Noting that the active returns can now be expressed as

$$r_A = \mathbf{w}_A'\mathbf{r} \tag{2.169}$$

one has the following formulas for the mean active return and TEV in terms of the active weights are:

$$\mu_A = \mathbf{w}_A'\boldsymbol{\mu} \tag{2.170}$$
$$\sigma_A^2 = \mathbf{w}_A'\mathbf{C}\mathbf{w}_A. \tag{2.171}$$

Whereas the Sharpe Ratio is a standard performance measure of absolute returns portfolio managers, the information ratio (IR) of a portfolio P with respect to a benchmark B is defined as

$$\mathrm{IR}_{P,B} = \frac{\mu_A}{\sigma_A} \tag{2.172}$$

is a standard measure of performance of an active manager.[23] One can think of the IR_P as being obtained from the Sharpe Ratio by replacing the time-varying risk-free rate $r_{f,t}$ with the time varying benchmark $r_{B,t} = \mathbf{w}_B'\mathbf{r}_t$. The IR_P estimator based on a sample of multivariate returns $\mathbf{r}_t$, $t = 1, 2, \cdots, T$ is

$$\widehat{\mathrm{IR}}_{P,B} = \frac{\bar{r}_{A,T}}{\hat{\sigma}_{A,T}} \tag{2.173}$$

where $\bar{r}_{A,T}$ and $\hat{\sigma}_{A,T}$ are the sample mean and sample standard deviation of $r_{A,T} = (r_{A,1}, r_{A,2}, \cdots, r_{A,T})$.

Any of the downside risk estimators of Section (2.3.3) can be used in place of $\hat{\sigma}_{A,T}^{P,B}$ in the above IR estimate, thereby resulting in a *downside risk information ratio*. For example, the *downside information ratio* estimator is defined by analogy to the downside Sharpe Ratio (2.92) as

$$\widehat{\mathrm{DIR}}_{P,B} = \frac{\bar{r}_{A,T}}{\sqrt{2} \cdot \widehat{\mathrm{SSDA}}_T} \tag{2.174}$$

[23] Goodwin (1998) is a useful reference for the Information Ratio.

where SSDA is the semi-standard deviation of the active returns r_A. By analogy to the ES ratio (2.94), an *expected shortfall information ratio estimator* is defined as

$$\widehat{\text{ESIR}}_{\alpha,P,B} = \frac{\bar{r}_{A,T}}{\widehat{\text{ESA}}_{\alpha,T}} \tag{2.175}$$

where $\text{ESA}_{\alpha,T}$ is the expected shortfall of the active return r_A. One could also define a *Sortino information ratio* where the active return is measured relative to an investor specified minimum acceptable active return $MAAR$, i.e., $r_A \to r_A - MAAR$.

MinTEV Optimization

Just as efficient portfolios can be obtained using either a MVO or MaxQU optimization method, optimal portfolios that are actively managed relative to a benchmark can be obtained using either a MinTEV or an equivalent MaxQU method. Here, just as in the discussion of MVO portfolios, we shall focus on the MaxQU method, but we first briefly state the MinTEV optimization problem here. Mimicing the MV optimization problem in Section 2.4.1. we now minimize the TEV subject to a constraint on mean active return and the self-financing constraint:

$$\min_{\mathbf{w}_A} \mathbf{w}_A' \mathbf{C} \mathbf{w}_A \tag{2.176}$$

subject to

$$\mathbf{w}_A' \mathbf{1} = 0 \tag{2.177}$$

$$\mathbf{w}_A' \boldsymbol{\mu} = \mu_A. \tag{2.178}$$

The solution to this problem was originally given by Roll (1992).

MaxQU Benchmark Relative Optimization

The MaxQU approach to benchmark relative optimization is straightforward. One just finds the active weight vector that maximizes the active quadratic utility function

$$\begin{aligned} Q_A(\mathbf{w}) &= \mu_A(\mathbf{w}) - \tfrac{1}{2}\lambda\sigma_A^2(\mathbf{w}) \\ &= \mathbf{w}_A' \boldsymbol{\mu} - \tfrac{1}{2}\lambda \mathbf{w}_A' \mathbf{C} \mathbf{w}_A \end{aligned} \tag{2.179}$$

subject to the self-financing (dollar neutral) constraint

$$\mathbf{w}_A' = 0. \tag{2.180}$$

The Lagrangian is

$$L(\mathbf{w}_A) = \mathbf{w}_A' \boldsymbol{\mu} - \frac{1}{2}\lambda \mathbf{w}_A' \mathbf{C} \mathbf{w}_A - \gamma \cdot \mathbf{w}_A' \mathbf{1}. \tag{2.181}$$

Setting the derivative of $L(\mathbf{w}_A)$ equal to zero and solving for the optimal active weights gives

$$\mathbf{w}_A = \lambda^{-1}\mathbf{C}^{-1}\left(\boldsymbol{\mu} - \gamma \cdot \mathbf{1}\right). \tag{2.182}$$

Applying (2.180) shows that the Lagrange multiplier γ is equal to the mean return μ_{gmv} of the GMV portfolio:

$$\gamma = \frac{\mathbf{1}'\mathbf{C}^{-1}\boldsymbol{\mu}}{\mathbf{1}'\mathbf{C}^{-1}\mathbf{1}} = \mu_{\text{gmv}} \tag{2.183}$$

and so

$$\mathbf{w}_A = \lambda^{-1}\mathbf{C}^{-1}\left(\boldsymbol{\mu} - \mu_{\text{gmv}}\mathbf{1}\right). \tag{2.184}$$

Henceforth unless otherwise noted $\mathbf{w}_A$ is the optimal active weight vector given by the above expression. The active portfolio manager's optimal portfolio weights are the sum of the benchmark weights and optimal active portfolio weights:

$$\mathbf{w}_P = \mathbf{w}_B + \mathbf{w}_A. \tag{2.185}$$

It is important to note that optimal active portfolio weights do not depend on the benchmark weights $\mathbf{w}_B$. Thus the active portfolio weights may be used in connection with any benchmark that the manager decides to use!

N.B. Somewhat surprisingly, the expression (2.184) for the optimal active weights is identical to the second term of (2.121) in Section (2.4.2). This provides an important alternative interpretation of the MaxQU fully invested portfolio weight vector (2.121): it is a portfolio manager's optimal benchmark relative weight vector using the GMV portfolio as the benchmark!

Application of the general formula (2.170) for the mean active return with $\mathbf{w}_A$ the optimal active weight vector immediately gives the first equation below, and likewise, use of (2.171) with some straightforward manipulation of the resulting expression gives the second equation below:

$$\mu_A = \lambda^{-1} \cdot \left(\boldsymbol{\mu}'\mathbf{C}^{-1}\boldsymbol{\mu} - \mu_{\text{gmv}}\mathbf{1}'\mathbf{C}^{-1}\boldsymbol{\mu}\right) \tag{2.186}$$

$$\sigma_A^2 = \lambda^{-2} \cdot \left(\boldsymbol{\mu}'\mathbf{C}^{-1}\boldsymbol{\mu} - \mu_{\text{gmv}}\mathbf{1}'\mathbf{C}^{-1}\boldsymbol{\mu}\right). \tag{2.187}$$

It follows that the information ratio $\text{IR} = \mu_A/\sigma_A$ of the optimal active portfolio is given by

$$\text{IR} = \sqrt{\boldsymbol{\mu}'\mathbf{C}^{-1}\boldsymbol{\mu} - \mu_{\text{gmv}}\mathbf{1}'\mathbf{C}^{-1}\boldsymbol{\mu}} \tag{2.188}$$

which is independent of the risk aversion parameter. A plot of μ_A versus σ_A is called the *active efficient frontier*, and it follows that the active efficient frontier is a straight line through the origin with slope given by the above IR expression. Of course, as we shall see shortly, a plot of the portfolio mean return μ_P versus volatility σ_P will not be a straight line.

Recalling the formula (2.47) for the volatility of a GMV portfolio, and the formula (2.112) for the special constant d, the IR of the optimal active portfolio can also be written in the form

$$\begin{aligned}\mathrm{IR} &= \frac{1}{\sqrt{\mathbf{1}'\mathbf{C}^{-1}\mathbf{1}}}\sqrt{(\boldsymbol{\mu}'\mathbf{C}^{-1}\boldsymbol{\mu})(\mathbf{1}'\mathbf{C}^{-1}\mathbf{1}) - (\mathbf{1}'\mathbf{C}^{-1}\boldsymbol{\mu})^2} \\ &= \sigma_{\mathrm{gmv}}\sqrt{d}. \end{aligned} \tag{2.189}$$

It is interesting to note that the right-hand side just above also appears in the MaxQU efficient frontier formula(2.129), where the MaxQU mean μ_{qu} in creases with d, but increases with σ_{gmv} only for a fixed increment $\sigma_{\mathrm{qu}}^2 - \sigma_{\mathrm{gmv}}^2$. Here the IR increases with σ_{gmv} without any such condition.

The MaxQU active weight vector formula (2.184) depends on the non-intuitive Lagrange multiplier variable $\lambda > 0$, and portfolio managers need a version of the formula that depends on the tracking error σ_A. This is easily obtained by taking the square root of both sides of the active variance expression (2.187), solving for λ^{-1}, making use of (2.188), and substituting the result into (2.184), which gives:

$$\mathbf{w}_A = \frac{\sigma_A}{\mathrm{IR}}\mathbf{C}^{-1}\left(\boldsymbol{\mu} - \mu_{\mathrm{gmv}}\mathbf{1}\right) \tag{2.190}$$

$$= \frac{\mu_A}{\mathrm{IR}^2}\mathbf{C}^{-1}\left(\boldsymbol{\mu} - \mu_{\mathrm{gmv}}\mathbf{1}\right). \tag{2.191}$$

Note that the equality of the two right-hand sides of the above equation imply that

$$\mu_A = \mathrm{IR} \cdot \sigma_A \tag{2.192}$$

which states the important result that *the active efficient frontier is linear with slope equal to the information ratio*.

Example 2.19. Active Efficient Frontier Portfolio of 3 Stocks

The 3 stocks are the same as in Figure (2.4) and in Figure (2.16) of Example (2.14). The returns of the three stocks are uncorrelated and their means and variances are $\boldsymbol{\mu}_3 = (0.10, 0.04, 0.02)$ and $\boldsymbol{\sigma}_3^2 = (0.20, 0.15, 0.10)$. The information ratio of the MaxQU active portfolio information ratio, computed using the formula (2.189) and turns out to be $\mathrm{IR} = 0.36$, and the active efficient frontier is displayed in Figure (2.26).

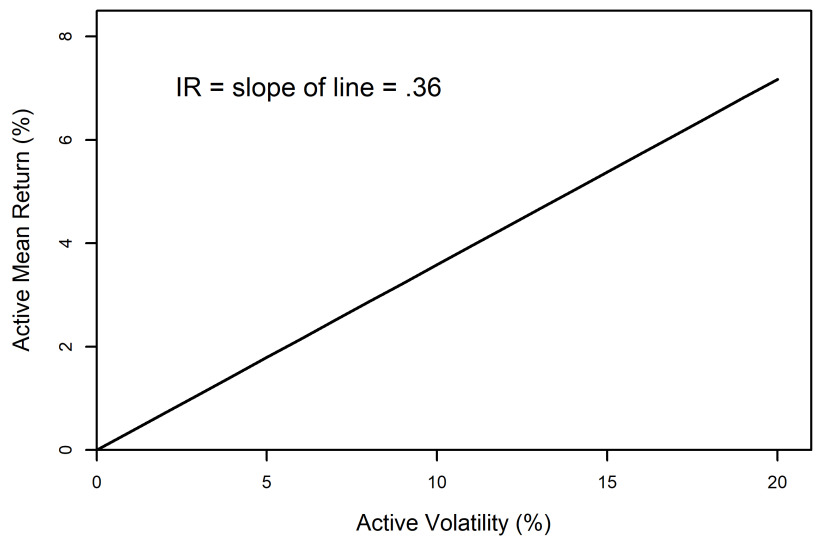


Fig. 2.26 Active Efficient Frontier for Three Stocks

Table 2.5 displays the active weights, along with their sum rounded to 3 digits, along the active efficient frontier for tracking error (TE) values 2%, 5% and 10%.

Table 2.5 Weights Along the Active Efficient Frontier

	wtA1	wtA2	wtA3	wtAsum
TE 2%	0.088	0.007	-0.095	0
TE 5%	0.219	0.018	-0.238	0
TE 10%	0.439	0.037	-0.476	0

A Serious Deficiency of MaxQU Active Portfolios

While the main ideas and theoretical foundation MaxQU active portfolios are quite attractive, they none-the-less suffer from a lack of control over portfolio total volatility, which we also refer to as absolute volatility. In fact MaxQU active portfolios can be quite inefficient relative to a MVO efficient frontier, as is revealed by deriving an expression for the total variance σ_P^2 of an active managers portfolio as a function of the total mean return μ_P. This is easily done using the basic formula $\sigma_P^2 = \mathbf{w}_P'\mathbf{C}\mathbf{w}_P$ with $\mathbf{w}_P = \mathbf{w}_B + \mathbf{w}_A$:

$$
\begin{aligned}
\sigma_P^2 &= (\mathbf{w}_B + \mathbf{w}_A)'\mathbf{C}(\mathbf{w}_B + \mathbf{w}_A) \\
&= \sigma_B^2 + \sigma_A^2 + 2\mathbf{w}_B'\mathbf{C}\mathbf{w}_A \\
&= \sigma_B^2 + \sigma_A^2 + 2\frac{\mu_A}{\mathrm{IR}^2}(\mu_B - \mu_{\mathrm{gmv}}) \\
&= \sigma_B^2 + \sigma_A^2 + 2\frac{\mu_P - \mu_B}{\mathrm{IR}^2}(\mu_B - \mu_{\mathrm{gmv}}) \qquad (2.193)
\end{aligned}
$$

where we have substituted (2.191) for $\mathbf{w}_A$ in going from the second line to the third line above, and used the full-investment assumption for the benchmark portfolio.

In using the above expression, one can assume that the mean return μ_B of the benchmark portfolio is greater than the mean return μ_{gmv} of the GMV portfolio. Otherwise, one would be better off to use the GMV portfolio as a benchmark.

Example 2.20. We use the formula (2.193) to compute the absolute mean return and variance frontier of a MaxQU active portfolio based on the 3 stocks of Example (2.19), using a benchmark portfolio with mean $\mu_B = 0.045$ and volatility $\sigma_B = 0.12$. The results are shown in Figure 2.27, along with the efficient frontier of a fully-invested portfolio that was previously computed and displayed in Figure 2.16. In Example (2.3) we computed the GMV mean return and volatility for these three stocks to be $\mu_{\mathrm{gmv}} = 0.037$ and $\sigma_{\mathrm{gmv}} = 0.077$, respectively, and those values are consistent with those values for the GMV portfolio in the figure.

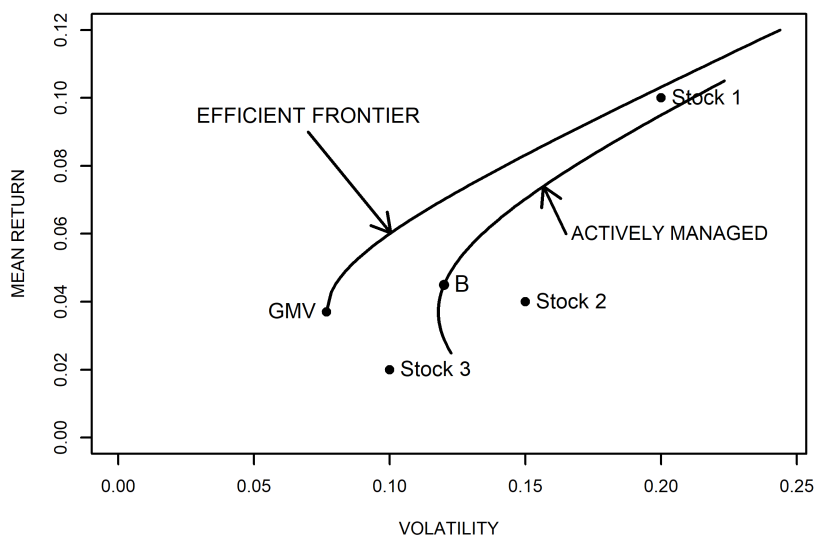


Fig. 2.27 Efficient Frontier for 3 Stocks and Inefficient Frontier of MaxQU Active Portfolio

Improving on Naive TEV Constrained Active Portfolio Management

Figure 2.27 reveals how an active portfolio manager who only uses a TE constraint can considerably under perform relative to the standard total mean returns versus volatility efficient frontier. Note that at a given position on the actively managed frontier, a larger mean portfolio return is sacrificed and a considerable increase in volatility occurs, relative to the efficient frontier. In recognition of this, Jorion (2003) showed that adding a total portfolio variance constraint

$$(\mathbf{w}_B + \mathbf{w}_A)' \, \mathbf{C} \, (\mathbf{w}_B + \mathbf{w}_A) = \sigma_P^2 \tag{2.194}$$

to the active portfolio optimization problem (2.179) - (2.168) with σ_P^2 specified by the portfolio manager, can result in considerably improved absolute returns performance.

The Appendices of Jorion (2003) contain algebraically involved formulas for computing a volatility-constrained MVO frontier, which turns out to be an ellipse in μ_P versus σ_P^2 space, and a distorted ellipse in μ_P versus σ_P space, whose size and position depends on the tracking error (TE) the manager wishes to use. The left-hand and right-hand plots in Figure 2.28 display Figures 4 and 5 of Jorion (2003), where the "Absolute Risk" horizontal axis is σ_P. The distorted ellipse in the left-hand plot is for tracking error volatility TEV = 4% and the one in the right-hand plot is for TE = 2%. In fact, the TEV = 11.5% and TEV = 23% dotted lines are also distorted ellipses, but with centers are far to the right off the plot region. The solid dot benchmark of course corresponds to TE = 0, and as TEV increases from 0 the distorted ellipse gets larger and larger until it touches the efficient frontier when TEV = 11.5%, and then continues to get larger until TEV = 23%, when it touches the MaxQU frontier without the total volatility constraint (and then moves further to the right, where it is of no interest due to be sub-optimal relative to the MaxQU frontier without the total risk constraint.

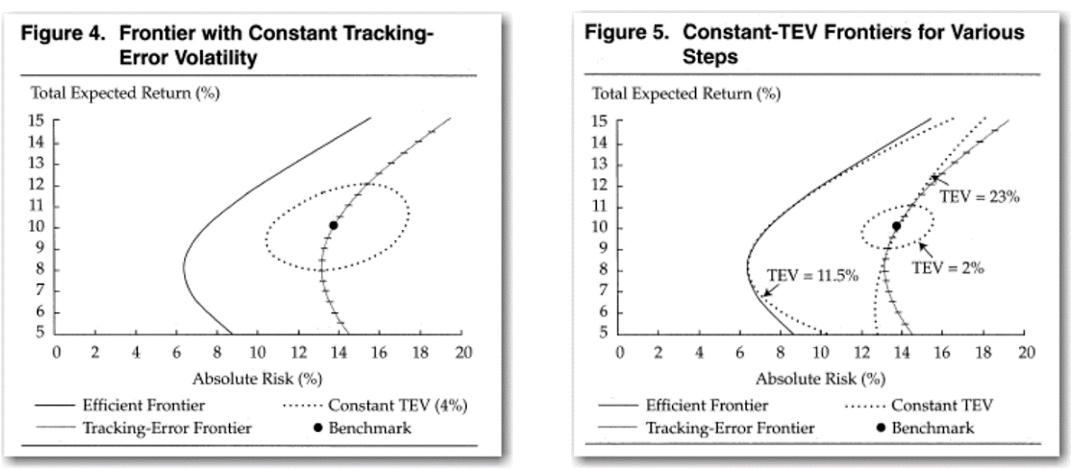


Fig. 2.28 The Jorion (2003) Figure 4 (left) and Figure 5 (right)

Figure 2.28 reveals that their clearly exist MaxQu active portfolios with constrained total variance σ_P^2 that out-perform portfolios on the MaxQU active portfolio frontier without such constraints. This is in

particular true for the practically relevant range of TE values between 2% and 10%. In his concluding comments Jorion states:

> "When myopically focusing on excess returns, the active manager ignores the total risk of the portfolio. As a result, optimization of excess returns that includes the benchmark assets will always increase total portfolio risk relative to the benchmark.
>
> This outcome is reinforced by the widespread use of information ratios as performance measures. Because information ratios consider only tracking error risk, a focus on information ratios ignores total portfolio risk."

2.4.5 Expected Utility Maximization and MVO

According to the theory of Von Neumann and Morgenstern (1947) (VNM) many decades ago, an investor will construct her investment portfolio based on the principle of expected utility maximization (EUM) using of a utility function $U(\mathrm{v})$ that has certain specified properties. That is, with a beginning of period investment wealth value v_0, the investor will construct a portfolio that maximizes the expected utility $\mathrm{E}[U(\mathrm{V})]$ where V is the random end of investment period wealth. The EUM theory is based on the assumption that an investor's behaviors satisfies a certain set of axioms. These axioms are fairly abstract, much more so than the axioms of coherent risk measures discussed in Chapter 8, and it is generally difficult to verify that investors behaviors actually satisfy all the axioms.[24] Fortunately, the abstract axioms lead to two simple properties that a utility function $U(\mathrm{v})$ must have for a EUM risk-averse investor who, at any level of end-of–period wealth v always wants more wealth. A risk-averse investor is one who always prefers a certain wealth v to a random wealth that has expected value v. The two utility function properties are that the function $U(\mathrm{v})$ is *strictly increasing*, and concave. A utility function that has first derivative $U'(\mathrm{v})$ is strictly increasing if $U'(\mathrm{v}) > 0$ or all v, and a utility function that has a second derivative is concave if $U''(\mathrm{v}) < 0$ for all v. The second property above implies that the first derivative is decreasing, and in context of a good dinner meal this behavior can be described as "the last bites are never as satisfying as the first bites". We refer to any such utility function as a *VNM Utility Function*.

The left-hand graph in Figure 2.29 illustrates the shape of a concave utility function. The straight-line in the graph illustrates the fact that a concave utility function is one for which a straight-line connecting two distinct points on the utility function curve lies below the curve.[25] While a risk averse EUM investor is one whose differentiable utility function $U(\mathrm{v})$ is concave, an expected utility maximizing investor who is *risk seeking* uses a utility function that is *convex*, such as the cup-shaped curve in the right-hand plot of Figure 2.29. An investor with a utility function that has a constant first derivative, and hence a zero second derivative, as shown by the straight line in that figure, is said to be a *risk neutral* investor.

[24] None-the-less, Markowitz in his 1959 Cowles Foundation monograph Markowitz (1959), devoted almost 40 pages to "The Expected Utility Maxim".

[25] Academic treatments of concavity sometimes use the term "weakly concave" to describe a utility function that has one or more straight line segments, and correspondingly the straight-line segment connecting any two points on the curve lies below or on the curve. However, this case is not commonly encountered in EUM discussions.

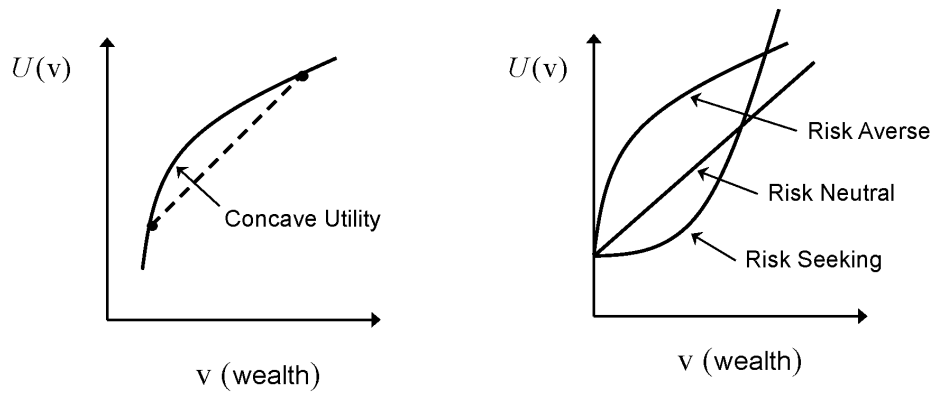


Fig. 2.29 A Concave Utility Function and Three Utility Shapes for Three Risk Preference Types

The most common family of increasing concave utility functions occurring in the academic literature is the family of *power utility functions* given by

$$U(\mathrm{v};\gamma) = \frac{\mathrm{v}^{\gamma} - 1}{\gamma}, \quad \mathrm{v} > 0, \quad \gamma < 1, \quad \gamma \neq 0. \tag{2.195}$$

The *log utility function*

$$U_{\log}(\mathrm{v}) = \ln(\mathrm{v}), \quad \mathrm{v} > 0 \tag{2.196}$$

may be shown (see Exercise 2.27) to be the limiting case of power utility functions as $\gamma \to 0$. The left-hand part of Figure 2.30 displays the log utility function and power utility functions for $\gamma = -1$ and $\gamma = 1/2$.

A small detail concerning the above utility functions is that their argument is end-of-period value (wealth) and portfolio optimization is carried with respect to end-of-period return r. For purposes of comparing the performance of different portfolios in terms of expected utility, it will suffice take beginning-of-period wealth to be 1 unit of local currency, say \$1 in the U.S. Then in the case of arithmetic returns one would let $\mathrm{v} = 1 + r$, which would result in a simple translation of the utility function by one unit in the positive direction, so that $r = -1$ gives $U(0)$ and $U(r) \to U(\infty)$ as $r \to +\infty$.[26]

[26] In the case of log return $\tilde{r} = \ln(\mathrm{v}) - \ln(1) = \ln(\mathrm{v})$, we have $\mathrm{v} = \exp(\tilde{r})$, and the utility function has the annoying form $U(\exp(\tilde{r}))$ which goes to $U(0)$ as $\tilde{r} \to -\infty$ and goes to $U(\infty)$ as $\tilde{r} \to +\infty$.

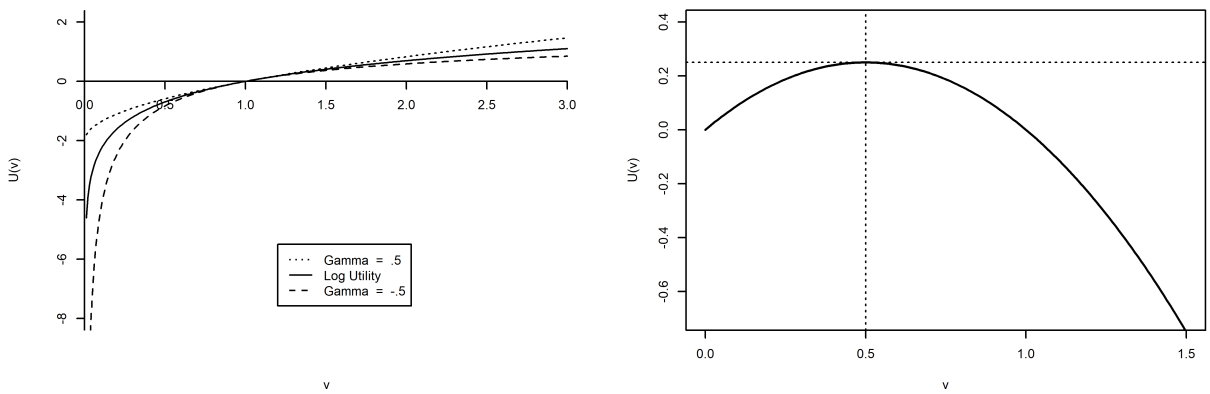


Fig. 2.30 Power Utility Functions and Quadratic Utility Function

Since maximum quadratic utility (MaxQU) is at the heart of MVO portfolio theory, and it is natural to think of MVO as arising from a *quadratic utility function* of the form

$$U(\mathrm{v}) = \mathrm{v} - b\mathrm{v}^2, > 0, \mathrm{v} > 0 \tag{2.197}$$

where the end-of-period value v can be converted to a return as described above. A quadratic utility function with $b = 1$ is shown in the right-hand graph in Figure 2.30. It is easy to check (see Exercise 2.29) that a quadratic utility function is increasing for $\mathrm{v} < (2b)^{-1}$ and decreasing for $\mathrm{v} > (2b)^{-1}$. Thus quadratic utility fails to satisfy the basic EUM investor utility function requirement of being a strictly increasing function.

Under the unrealistic assumption that returns are normally distributed, it can be proved that EUM with a VNM utility function leads uniquely to an MVO portfolio, see for example Chapter 4 of Ingersoll (1987). The same is true under the assumption that returns are log-normally distributed, as has been shown by Bodnar et al. (2020). One often finds arguments that quadratic utility MVO provides a good approximation to an EUM portfolio based on a VNM utility function in the realistic case that returns are non-normal with fat tails and skewness. Most frequently a Taylor series argument is used, but is not adequately justified with respect to dropping terms that involve significant third and fourth moment representations of skewness and fat tails. A frequently quoted reference that uses empirical analysis to justify MVO is the article by Levy and Markowitz (1979). But the market data they used is not at all representative of the wide range of current portfolio assets in use, whose returns often exhibit quite non-negligible skewness and fat tails.

Our view is that attempts to justify quadratic utility as a universally good approximation to proper EUM utility functions is quite fruitless, and the MVO portfolio construction principle is a sound one that is supported by considerable theory and successful use in practice. On the other hand the log utility function has an important place in portfolio theory because it is unique in providing a multi-period portfolio optimization result, which MVO is not. Consequently, we discuss log-optimal portfolio

theory and results in a later chapter. Furthermore, the principle of mean-versus-downside risk based portfolio optimization, which is discussed briefly in the next section, has the potential to provide better risk adjusted performance than MVO for some asset types in some market conditions.

Risk Aversion Functions

Academics in financial econometrics have introduced two closely related utility functions based risk aversion formulas, which are good to be aware of, namely *absolute risk aversion* (ARA) and *relative risk aversion* (RRA) formulas. The ARA and RRA risk measures were introduced by Pratt (1964) and Arrow (1971) (see also Section 1.2 of Pennacchi (2008)), and their formulas are:

$$\mathrm{ARA}(\mathrm{v}) = -\frac{U''(\mathrm{v})}{U'(\mathrm{v})} \tag{2.198}$$

$$\mathrm{RRA}(\mathrm{v}) = -\mathrm{v}\frac{U''(\mathrm{v})}{U'(\mathrm{v})}. \tag{2.199}$$

The first derivative of utility appearing in the denominators above represents the *marginal utility of wealth*, and the negative of the second derivative appearing in the numerators represents the *degree of riskiness*. As such, it is natural that they appear as they do in a ratio, for if the latter is large, it may be offset by a large marginal utility of wealth. It is clear from the definition of an ARA and RRA functions, that if two utility functions are equivalent by virtue of an affine transformation $U_2 = a + bU_1$, $b > 0$, they have the same ARA and RRA functions.

The derivation of the ARA function in Appendix C shows that it represents the *amount* of initial wealth that an investor is willing to pay to avoid the risk associated with the random outcome of an investment. On the other hand, the derivation of the RRA function in Appendix C shows that it represents the *fraction* of initial wealth an investor is willing to pay to avoid the risk associated with the random outcome of an investment.

It is commonly accepted that investors usually have decreasing absolute risk aversion and constant relative risk aversion (CRRA). It is straightforward to show (see Exercise 2.26) that EUM investors who use power utility or log utility utility have CRRA and decreasing ARA.

Certainty Equivalent

Sometimes portfolio managers want to compare portfolios that have different expected utilities in terms of monetary units rather the dimensionless units of utility functions. One way to do so, is to define a dollar amount $\mathrm{v_c}$, called the *certainty equivalent value*, that satisfies the equation:

$$U(\mathrm{v_c}) = \mathrm{E}[U(\mathrm{V})] \tag{2.200}$$

where V is random end of period wealth. It follows that for an EUM investor's strictly increasing and concave utility function, one has $v_c < E(V)$. A justification of this claim is provided in Appendix C.

Appendix C contains mathematical details concerning utility functions and EUM. Further mathematical depth on the topic may be found for example in Chapter 1 of Pennacchi (2008), or in online documents such as Levin (2006). "Choice under uncertainty" at `https://web.stanford.edu/~jdlevin/Econ%20202/Uncertainty.pdf`.

2.5 CAPM, Zero Beta CAPM, and the Beta Factor

2.5.1 The Capital Asset Pricing Model

The *Capital Asset Pricing Model* (CAPM), which was introduced in Treynor (1961), Treynor (1962), and Sharpe (1964), and extended in Lintner (1965)) and Mossin (1966), is a simple model of capital market equilibrium that holds under the following highly idealized assumptions:

1. All investors are mean–variance optimal investors who have the same single period investment horizon, and use the same mean vector and covariance matrix of asset returns, though their levels of risk aversion are not necessarily identical. Furthermore, a risk-free asset (cash) exists and all investors agree that it is risk-free.
2. There are no restrictions on borrowing or lending at a common risk free rate r_f.
3. Securities markets are perfect in the sense that assets are infinitely divisible, short sales and leverage are allowed without limit, and there are no taxes or transaction costs.
4. The market is in equilibrium.

Under the first three assumptions, all investors hold the same tangency portfolio T of risky assets in combination with some amount of the risk–free asset, and that amount varies from investor to investor depending upon their individual level of risk aversion.

Under the final assumption, the tangency portfolio T must be the market portfolio M. The reasoning is as follows. If all participants hold identical views on the expected returns, variances and covariances of every asset, their portfolios must hold each asset in the same proportion, and must therefore have the same *capital allocation line* (CAL). In equilibrium, supply equals demand, and, as the proportional holding of each asset is the same for all investors, the sum of their holdings in each asset must equal its market capitalization, i.e., the tangency portfolio T must be the market portfolio M, and correspondingly the CAL becomes the *capital market line* (CML). The algebraic version of this rationale is provided in Appendix D.2.

Let the market single period return r_M have mean $\mu_M = \mathrm{E}[r_M]$ and variance $\sigma_M^2 = \mathrm{var}(r_M)$, and let the i–th security, among a group of N securities, have a single period return r_i with mean $\mu_i = \mathrm{E}[r_i]$. Under the above assumptions, the CAPM asserts that

$$\mu_i - r_f = \frac{\mathrm{cov}(r_i, r_M)}{\sigma_M^2} \cdot (\mu_M - r_f),\ i = 1, 2, \cdots, N \tag{2.201}$$

where the CAPM *beta* β_i of the i–th security is:

$$\beta_i = \frac{\mathrm{cov}(r_i, r_M)}{\sigma_M^2},\ i = 1, 2, \cdots, N. \tag{2.202}$$

Substituting r_M, the return of the market, for r_i on the right–hand side of (2.201) shows that β_M, the beta of the market, is equal to 1. A proof of the CAPM is provided in Appendix D.3.

The CAPM beta parameter β_i reflects the degree to which an asset's expected excess return is greater than or less than that of the market, according to the degree to which its beta is greater than or less than one. The quantity $\mu_{M,e} = \mu_M - r_f$ is the *market risk premium* for investing in the market, i.e., it is the expected additional amount an investor is rewarded for investing the risky market portfolio, relative to investing in a risk–free asset. The quantity $\mu_{i,e} = \mu_i - r_f = \beta_i \cdot \mu_{M,e}$ is the *security risk premium*, i.e., the expected additional amount the investor is rewarded for investing in a risky security, relative to investing in a risk–free asset.

As we will see in Chapter 11, there is an intuitive explanation for why expected returns should be an increasing function of beta: a security that does well in bad times (i.e. a low beta security that does not lose much value when the market declines) has insurance–like properties, and will be valued highly by investors on account of them. In equilibrium, its price will be bid up on account of the demand for it, and it will have a correspondingly low expected return. Likewise, a security that does well in good times (i.e. a high beta security whose value increases when the market rises but which declines sharply when markets fall) has the exact opposite properties, and will be valued lowly by investors on account of them. In equilibrium, its price will be bid down on account of the lack of demand for it, and it will have a correspondingly high expected return.

It is important to keep in mind that for a fixed market variance σ_M^2, β_i is proportional to $\text{cov}(r_i, r_M)$, and the latter represents the contribution of the i–th security to the risk of the market portfolio (recall (2.33) in Section 2.2.2). Larger values of β_i result both in larger security contributions to market risk, and in a correspondingly larger security risk premium. We will shortly see that with an observational model for security returns and market returns in hand, larger values of beta for a security, or portfolio of securities, also result in a larger variance for the security or the portfolio of securities.

Example 2.21. A security has a $\beta_i = 1.2$,the expected return of the market is $\mu_M = 6$, and the risk-free rate is $r_f = 3\%$. Then the market risk-premium is $6\% - 3\% = 3\%$,and risk premium of the security is $1.2 \times 3\% = 3.6\%$.

The *security market line* (SML) is defined by writing the CAPM equation (2.201) as

$$\mu_i = r_f + (\mu_M - r_f)\beta_i, \tag{2.203}$$

with μ_M and r_f fixed, β_i the independent variable, and μ_i the dependent variable. According to the CAPM, all securities should lie on the SML. Unfortunately, as we discuss subsequently, this frequently fails to be the case in practice.

Portfolio Beta

Suppose P is fully-invested portfolio with portfolio returns $r_P = \sum_{i=1}^{N} w_i r_i$ and portfolio mean returns $\mu_P = \sum_{i=1}^{N} w_i \mu_i$. Substituting (2.203) for the μ_i, and using the full-investment constraint $\sum_{i=1}^{N} w_i = 1$, results in:

$$\mu_P = r_f + \frac{\sum_{i=1}^{N} w_i \operatorname{cov}(r_i, r_M)}{\sigma_M^2} \cdot (\mu_M - r_f).$$

Since $\sum_{i=1}^{N} w_i \operatorname{cov}(r_i, r_M) = \operatorname{cov}\left(\sum_{i=1}^{N} w_i r_i, r_M\right) = \operatorname{cov}(r_P, r_M)$, we have the portfolio level CAPM:

$$\mu_P = r_f + \beta_P \cdot (\mu_M - r_f), \qquad \beta_P = \frac{cov(r_P, r_M)}{\sigma_M^2}. \tag{2.204}$$

CAPM as a Pricing Model

The presence of the word "Pricing" in the CAPM may not yet seem meaningful, and so we now show that it is indeed a pricing model. Let $p_{i,0}$ be the known beginning–of–period price of an asset and let $E(p_{i,1})$ be the expected value of its random end–of–period price $p_{i,1}$. Then the expected end–of–period return μ_i can be expressed as

$$\mu_i = \mathrm{E}\left[\frac{p_{i,1} - p_{i,0}}{p_{i,0}}\right]. \tag{2.205}$$

Substituting this expression into the left–hand side of the CAPM relationship 2.201 gives

$$\frac{\mathrm{E}\left[p_{i,1}\right]}{p_{i,0}} - 1 = r_f + \beta_i \cdot (\mu_M - r_f). \tag{2.206}$$

and rearranging it we have the CAPM pricing formula:

$$p_{i,0} = \frac{\mathrm{E}\left[p_{i,1}\right]}{1 + r_f + \beta_i \cdot (\mu_M - r_f)}. \tag{2.207}$$

According to the CAPM, the current price of an asset increases with increasing expected end of period price $\mathrm{E}\left[p_{i,1}\right]$, decreases with increasing β_i, and decreases with increasing excess market mean return. The CAPM formula (2.201) states that the expected excess return of a security increases with increasing beta, but increased expected return is not obtained as a free lunch. Instead expected excess return is a "risk premium" that is earned by taking on risk, with increased risk resulting increased excess return. However, the larger the risk, the lower the present price should be for taking on that risk. The CAPM basically tells the investors that market risk is "priced", and the above formula gives the price of an asset explicitly.

A CAPM Time Series Model

The CAPM beta is typically estimated using a time series linear regression model of the form

$$r_{e,it} = \alpha_i + \beta_i r_{Me,t} + \varepsilon_{it},\ t = 1, 2, \cdots, T,\ i = 1, 2, \cdots, N \tag{2.208}$$

where $r_{e,it} = r_{it} - r_{f,t}$ is the return of an individual security in excess of the risk-free rate $r_{f,t}$ at time t, $r_{Me,t} = r_{M,t} - r_{f,t}$ is the return of the market in excess of $r_{f,t}$.The above model is a special case of a general time series *single factor model* (SFM), where the market excess return $r_{Me,t}$ is replaced by a benchmark factor return $r_{B,t}$. We discuss SFM's extensively with regard to their properties and model estimation methods in Chapter 6, so here we just briefly mention the behavior if expected values and variances of the terms of the above CAPM single factor model.

Taking expectations on both sides of (2.208) under the assumptions that the $r_{e,it}$ and $r_{Me,t}$ have constant means μ_i and μ_M across time, and the ε_{it} all have zero means, gives:

$$\mu_i - r_f = \alpha + \beta_i \cdot (\mu_M - r_f) \tag{2.209}$$

In general the intercept α is non-zero, in which case the above equation is not identical to the CAPM equation (2.208) – *it coincides with the CAPM only if* $\alpha_i = 0$. Under additional standard SFM assumptions discussed in Chapter 4, including the assumption of a constant risk-free rate r_f, equating variances on both sides of (2.208) results in:[27]

$$\sigma_i^2 = \beta_i^2 \sigma_M^2 + \sigma_{\varepsilon_i}^2,\ i = 1, 2, \cdots, N. \tag{2.210}$$

The above expression shows that the security variance σ_i^2 can be decomposed into two parts. The first part, $\beta_i^2 \sigma_M^2$, increases with the absolute value of β_i, and is a *systematic risk* that depends on the common *market risk* σ_M^2 of all securities in a portfolio. The second part, $\sigma_{\varepsilon_i}^2$, is known as *idiosyncratic risk* or *residual risk*.

Strengths of the CAPM

The greatest strength of the CAPM is that it is the simplest possible model of capital market equilibrium. In addition, it yields the following insights and benefits:

- Only non-diversifiable risk should be priced (prior to the development of the CAPM, investors thought that total risk should be priced)
- The CAPM identifies the market as a primary source of risk for all securities

[27] The variance of excess returns relative to a fixed risk-free rate r_f is the same as the variance of returns, i.e., $\text{var}(r_{e,it}) = \text{var}(r_{it}) = \sigma_i^2$ and $\text{var}(r_{Me,t}) = \text{var}(r_{M,t}) = \sigma_M^2$, and similarly $\text{cov}(r_{Me,t}, \varepsilon_{it}) = \text{cov}(\varepsilon_{it}, r_{M,t}) = 0$.

- Beta is useful in predicting security in the context of the variance model (2.210). For this purpose investors will likely use an index such as the S&P 500 or the Russell 1000 as a proxy for the market portfolio
- A side benefit is that the mean return and variance mathematics of the time series CAPM model are usable in the context of active portfolio management, as discussed in this chapter and in Chapter 5.

Weaknesses of the CAPM

> "One of the greatest tragedies of life is the murder of a beautiful theory by a gang of brutal facts"
> Benjamin Franklin

The greatest weakness of the CAPM is that its assumptions are unrealistic, particularly the assumption that all investors are efficient mean-variance investors who use the same returns mean vector and covariance matrix. The assumption that all investors can freely borrow and lend at a constant risk–free rate is also quite untenable. Finally, the CAPM predicts an affine relationship between the *expected* return of the market and the *expected* return of individual securities, but expected returns are unobservable. Tests of the CAPM therefore are necessarily based on *realized* returns, which can differ from expected returns, and sometimes quite dramatically.

2.5.2 The Zero Beta CAPM and the Beta Factor

Motivated by the CAPM unrealistic assumption of unlimited borrowing of a risk-free asset, Black (1972) studied the effects of restrictions on borrowing in a seminal paper, and derived the *Zero Beta CAPM*, in which a so-called *minimum variance zero beta* portfolio plays the role of the riskless asset in the standard CAPM. Given that beta is proportional to a covariance, a zero beta portfolio is also called a *zero-covariance* portfolio. Details concerning zero-covariance portfolios are provided in Section D.6 of Appendix D. Of the many extensions of the CAPM that have been derived over the years, only the Zero Beta CAPM appears to be of significant practical interest.

In a contemporaneous paper, Black et al. (1972)[28] measured the returns of ten equally weighted portfolios (the "beta portfolios") formed using all NYSE stocks over the period 1926–1964. The portfolios were reconstituted annually using a decile–ranking of security betas measured over a trailing 5–year period, and their returns were computed in each year starting from the end of 1931 to the end of 1965. The performance of low beta portfolios was consistently higher than the CAPM would suggest, while the performance of high beta portfolios was found to be consistently lower than the CAPM would suggest.

The left hand plot in Figure 2.31 , which is drawn from an updated version of the BJS study in Black (1993), makes clear that the slope of the security market line that is actually observed from 1931–1965

[28] This paper is often referred to as the Black–Jensen–Scholes, or BJS, study.

is lower than the slope predicted by the CAPM, or, in other words, that the CAPM alpha's of low beta portfolios are positive, while those of high beta portfolios are negative.

Black's results for the period 1965–1991 (i.e. the period following the publication of the CAPM in Sharpe (1964)), are even more striking, as the right hand plot in Figure 2.32 clearly illustrates. Now the relationship between beta and return is essentially flat, in complete opposition to the basic prediction of the CAPM. Black (1972) argues that this flattening of the Security Market Line is driven in large part by borrowing restrictions, which prevent investors from levering low β portfolios, and thus increasing demand for, and lowering the expected return of, high β portfolios.

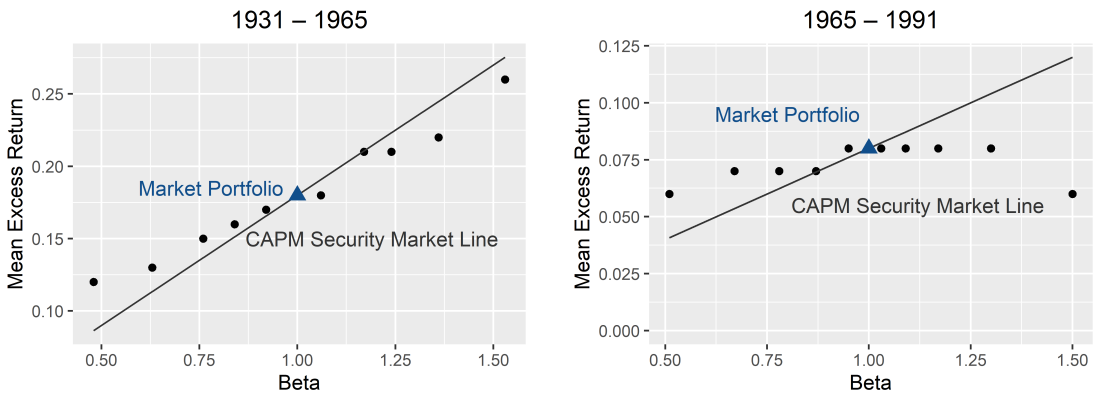


Fig. 2.31 Mean Return vs Beta of 10 Beta Portfolios: 1931–1965 and 1965-1991. Data Source: Fischer Black. 1993. Beta and Return., *Journal of Portfolio Management* 20 (1): 8–18. Exhibits 3 and 4, DOI: https://doi.org/10.3905/jpm.1993.409462

On the other hand, the empirical relationship between standard deviation and beta displayed in Figure 2.32 is modeled remarkably well by a straight line, indicating that beta is a good predictor of portfolio risk. The slope of the line is related, but is not identical, to the standard deviation of the market portfolio[29], and drops by about a third from the first period to the second, reflecting a drop in market volatility.

[29] The lack of perfect equivalence is driven by the fact that the slope is estimated from a set of portfolios whose security weights differ significantly from those of the market portfolio, and which are not optimized to minimize their variance.

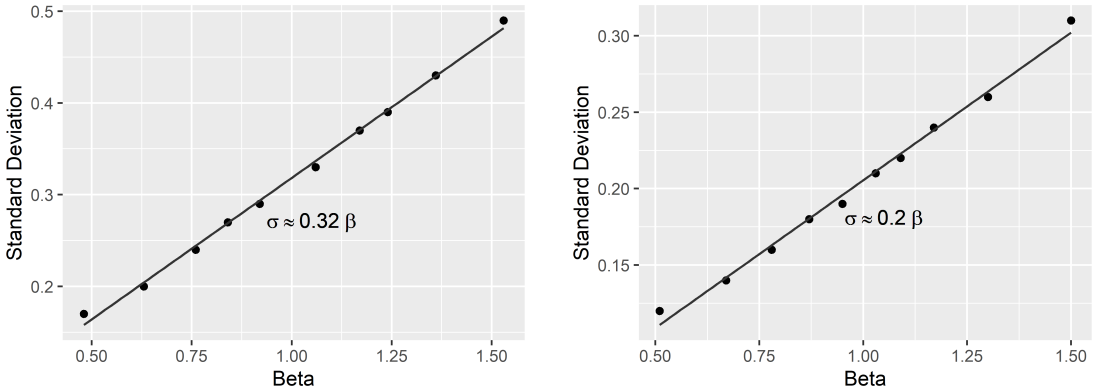


Fig. 2.32 Standard Deviation vs Beta of 10 Beta Portfolios: 1931–1965 and 1965-1991. Data Source: Fischer Black. 1993. Beta and Return., *Journal of Portfolio Management* 20 (1): 8–18. Exhibits 3 and 4, DOI: https://doi.org/10.3905/jpm.1993.409462

Empirically, Black et al. (1972) showed that security returns are modeled well by an equation of the form

$$\mu_i - r_z = \alpha + \beta_i \cdot (\mu_M - r_z), \tag{2.211}$$

where r_z is a mean return that is not equal to the risk-free rate r_f, and is empirically typically larger than r_f. The derivation of the Zero Beta CAPM in Black (1972) makes clear that the absence of a riskless asset forces r_z to equal the expected return of a *minimum variance zero–beta portfolio* relative to the market portfolio M, which we denote by $\mu_{\mathrm{zc}(M)}$. A derivation of the above Zero Beta CAPM formula is provided in Appendix D.4.

In the Zero Beta CAPM the mean $\mu_{\mathrm{zc}(M)}$ of the zero covariance portfolio $\mathrm{zc}(M)$ substitutes for a risk-free asset. But of course when there is a risk-free asset with rate r_f, it replaces $\mu_{\mathrm{zc}(M)}$.

The Beta Factor and Beyond

Black (1993) studied the validity of the standard CAPM by using a *beta factor* that is constructed from the decile beta portfolios in Figure 2.31 by applying weight $1 - \beta_i$ to the i–th decile portfolio, whereby high beta portfolios have negative weights and low beta portfolios have positive weights[30]. If the CAPM holds, then the excess returns of the beta portfolio should be reliably negative (convince yourself why). Black tested this hypothesis for six contiguous 10–year intervals starting with the 1930's and ending with the 1980's, and found that μ_e, the mean excess return of the beta factor portfolio, was negative only for the decade of the 1930's, thereby making the CAPM quite implausible. His results are displayed in Table 2.6.

[30] The beta factor first appeared in published form simultaneously in Black (1972) and Black et al. (1972).

Table 2.6 History of the β factor. All NYSE Stocks

Period	μ_e	σ_e	$t(\mu)$
1/31–12/39	-0.05	0.17	-0.94
1/40–12/49	0.03	0.10	1.06
1/50–12/59	0.08	0.06	4.25
1/60–12/69	0.03	0.07	1.32
1/70–12/79	0.01	0.10	0.18
1/80–12/91	0.09	0.08	3.90

Data Source:
Data contained within Exhibit 6 of Black, F. 1993. Beta and Return.
The Journal of Portfolio Management 20 (1): 8-18.
DOI: https://doi.org/10.3905/jpm.1993.409462

Black (1993) is exceptionally instructive, both for its clear linkage of theory and data, and for the simplicity of its econometric methodology. Black points out that the high levels of noise in financial markets makes it unlikely that complex econometric models will prove of great value in resolving debates, and stresses the importance of having a theoretical foundation for any hypothesis that one tests. He encapsulates his wisdom in a single sentence that should guide all researchers in finance: *"Lack of theory is a tipoff: watch out for data mining!*". We shall return to this when we study the estimation of expected returns in Chapter 11.

In the intervening years since Black (1993) was published, many studies have cast doubt on the validity of the CAPM. For example, there is considerable evidence that betas estimated using the Fama-MacBeth Fama and MacBeth (1973) cross–sectional regression method indicate either a negative relationship, or no relationship at all, between beta and the cross-section of returns (see Section 8.6 of Bali et al. (2016)). Many explanations have been given for the failure of the CAPM to predict returns, and to this day there is no clear explanation for its failure that improves upon Black's addition of borrowing restrictions to the basic model.

Over the years, the use of the CAPM single-factor model (2.208) has been supplanted by the use of multi-factor models using many firm characteristic factors such as firm Size (log of market capitalization in $M), book-to-market (BM), earnings-to-price (EP), etc., and other factors such as momentum, the Fama–French factors (SMB, HML, and others), often alongside Beta, as discussed in Chapters 6 and 11. Even so, the Zero Beta CAPM stands out for its economic intuition and the simplicity of the method used to test it. A particularly interesting paper that is in no small part motivated by it is the *Betting Against Beta* paper by Frazzini and Pedersen (2014). We discuss this paper, as well as estimators of expected return and the cost of equity capital in Chapter 11.

2.5.3 The Arbitrage Pricing Theory

In Section 2.5.1, the use of mean-variance optimization, with a number of additional assumptions that include the existence of market equilibrium and the existence of a risk-free asset that can be borrowed and sold without frictions, led us to the CAPM equation (2.201). That latter is a very special single factor model that predicts a linear relationship between systematic (or non-diversifiable) risk and expected return. The CAPM's prediction that expected returns are influenced only by systematic risk is striking, as is its prediction that the market portfolio is the sole source of systematic risk.

If, instead, we assume that security returns are described by a linear multi-factor model, make certain assumptions about the linear model and diversification of asset specific risk, and assume the absence of arbitrage possibilities in well functioning capital markets, we are led to an alternative model of expected return referred to as the *Arbitrage Pricing Theory* (abbreviated APT). It is assumed that asset returns r_i are given by the linear factor model

$$r_i = a_i + \sum_{k=1}^{K} b_{ik} f_k + \epsilon_i \; i = 1, 2, \cdots, n, \; K < n \tag{2.212}$$

where K is the number of risk factors, where f_k is the random value of the k-th risk factor, b_{ik} is the *sensitivity*, also called the *exposure*, of the i-th return to the k-th risk factor, the a_i are intercepts, and ϵ_i is a random risk contribution that is specific to the i-th return r_i. The model is conveniently written in the vector and matrix form

$$\mathbf{r} = \mathbf{a} + \mathbf{B}\mathbf{f} + \boldsymbol{\epsilon} \tag{2.213}$$

where $\mathbf{b}$, $\mathbf{a}$ and $\boldsymbol{\epsilon} = (\varepsilon_1, \varepsilon_2, \cdots, \varepsilon_n)'$ are n–dimensional vectors, $\mathbf{B}$ is an $n \times K$ matrix, and $\mathbf{f}$ is a K–dimensional vector.

The above linear factor model generalizes the CAPM time series model (2.208) at a fixed time and ignoring a risk-free rate, in a natural and intuitive way. It allows for multiple sources of systematic risk, with each source of risk contributing to a security's expected return in proportion to its exposure to it. As we point out below, the APT captures the impact of the covariances between assets and a set of exogenous risk factors. The CAPM, in contrast, focuses solely on the covariance between assets and the market portfolio, and can be viewed as a special case of the APT in which all systematic risk is channeled through a single factor – the market.

Linear Factor Model Assumptions.

A proof of the APT is based on the following mostly standard assumptions for a linear regression model with random coefficients. It is assumed that the random vectors $\mathbf{f}$ and $\boldsymbol{\epsilon}$ have zero mean values, $\mathrm{E}[\mathbf{f}] = \mathrm{E}[\boldsymbol{\epsilon}] = 0$, from which it follows that $\mathbf{a}$ is the mean of $\mathbf{r}$. [31] Furthermore, it is assumed that the f_k and ε_i are uncorrelated, i.e., $\mathrm{E}(f_k \varepsilon_i) = 0, \forall\, i, k$, since if these are not uncorrelated a given ε_i can effect all the r_i, and as such each ε_i would not represent risk specific to just the i-th asset. For convenience, and without

[31] Note that if the original f_k have non-zero mean values μ_k, then the model is unchanged by replacing $\alpha_i + \sum_{k=1}^{K} b_{ik} f_k$ by $\tilde{\alpha}_i + \sum_{k=1}^{K} b_{ik} \tilde{f}_k$ where $\tilde{f}_k = f_k - \mu_k$ and $\tilde{\alpha}_i = \alpha_i + \sum_{k=1}^{K} b_{ik} \mu_k$.

loss of generality, it is assumed that the random vector $\mathbf{f}$ has a K dimensional identity covariance matrix $\mathbf{I}_K$, which means that the f_k have variances 1, and f_k and f_j are uncorrelated for $k \neq j$.[32] It follows that the covariance between the return r_i and f_j is $\text{cov}(f_j, r_i) = \text{cov}(f_j, \sum_{k=1}^{K} b_{ik} f_k + \epsilon_i) = b_{ij}$. Finally, instead of the frequent assumption that the errors ε_i are uncorrelated, it is only assumed that ϵ has a positive definite covariance matrix $\mathbf{C}$.

Asymptotic Arbitrage

A portfolio *asymptotic arbitrage* exists if a portfolio can be acquired at zero cost, and if it returns a non-zero certain return asymptotically as the number of assets n tends to infinity. An asymptotic non-zero certain portfolio return is obtained by a portfolio whose asymptotic mean return is non-zero and whose asymptotic variance is zero. We will show that when asymptotic arbitrage is ruled out, the intercepts α_i in (2.212) asymptotically depend upon the specific asset i only through the exposures b_{ik}. The precise statement of the APT is that

$$a_i = \lambda_0 + \sum_{k=1}^{K} b_{ik}\lambda_k + u_i, \; i = 1, 2, \cdots, n \tag{2.214}$$

where λ_k is the *factor price of risk*, also called the *risk premium*, associated with the k–th factor, and the following *APT condition* is satisfied:

$$\lim_{n\to\infty} \frac{1}{n} \sum_{i=1}^{n} u_i^2 = 0. \tag{2.215}$$

This means that the a_i of the r_i in (2.212) are given for large n by the approximation

$$a_i \approx \lambda_0 + \sum_{k=1}^{K} b_{ik}\lambda_k \tag{2.216}$$

where the mean-squared-error between the left-hand and right-hand sides above go to zero as $n \to \infty$. Note that for the linear model (2.212) the a_i in general depend on asset i in a completely unknown manner relative to market variables. A great strength of the APT is that the a_i in the APT approximation (2.216) depend on asset i only through the known factor exposures b_{ik}. All the assets have the same common expected value component λ_0, and the summation term in (2.216) contributes different expected value components for different assets through the factor exposures b_{ik} and the risk premia $\lambda_1, \lambda_2, \cdots, \lambda_K$.

The proof of the above result was originally due to Ross (1976). Subsequently a simpler proof based on the clever use of ordinary least squares regression was provided by Huberman (1982) and used in Chapter 3.3 of Pennacchi (2008). In Appendix D.6 we provide a compact version of that proof using the well-known least squares (LS) regression vector projection and orthogonality conditions.

The absence of arbitrage is more fundamental than equilibrium (which lies at the heart of the CAPM), because it takes relatively few rational investors to bid away opportunities for arbitrage in liquid capital

[32] In case $\mathbf{f}$ has a covariance matrix $\mathbf{C}$ that is assumed to be positive definite, it can be factored into the product $\mathbf{C} = \mathbf{C}^{1/2}\mathbf{C}^{1/2}$ of its positive definite square-root matrices (see Appendix A) whose inverse exist, and then we have $\mathbf{Bf} = \mathbf{BC}^{1/2}\mathbf{C}^{-1/2}\mathbf{f} = \check{\mathbf{B}}\check{\mathbf{f}}$, where $\text{cov}(\check{\mathbf{f}}) = \mathbf{I}_K$.

markets. Unfortunately, the theory does not specify what the factors in the APT should be, and the factors identified in Chen et al. (1986) do not seem to have maintained their explanatory power over time. Even so, the powerful economic intuition that underlies the APT has stood the test of time: linear factor models have become a mainstay of risk models in all asset classes, while factor investing has exploded in popularity.

2.6 Minimum Downside Risk Portfolios

In his Cowles Foundation Monograph, Markowitz (1959) was aware that the symmetric character of variance made it a less than satisfactory risk measure. This led him to devote an entire Chapter of his monograph (Chapter 9) to a discussion of semi–variance (SV), which intuitively captures the downside nature of risk, and provides a viable alternative to variance (V) for portfolio construction. There he states:

> "Analyses based on S tend to produce better portfolios than those based on V".

However, by the time his book Markowitz (1987) had been published and become well known, and he had received the 1990 Nobel Prize for his pioneering work on minimum variance (MVO) portfolio construction, the variance-based method had become firmly established in the curricula of academic graduate programs in finance and economics, including MBA programs, and in leading commercial portfolio management software products. Consequently, there was little incentive to develop the needed computational methods for constructing and evaluating the performance of minimum SV portfolios. The success of variance-based optimization over the subsequent decades was driven largely by two factors:

1. Using the the variances and covariances of a portfolios constituent securities, the variance of a portfolio can be computed as a simple covariance matrix based quadratic form in the portfolio weights. Furthermore, MVO optimal portfolios can be computed using Quadratic Programming, with linear equality and inequality constraints readily incorporated into the optimization.
2. The use of parsimonious factor models, using a variety of intuitively meaningful factors, made mean-variance optimization and factor risk analysis practical for portfolios with very large numbers of stocks, e.g., 1000 to 3000.

Interest in minimum downside risk optimal portfolios increased substantially after the turn of the century with the publication of the seminal Rockafellar and Uryasev (2000) paper in which it is shown that the construction of a minimum *conditional Value–at–Risk* (cVaR) portfolio is a convex optimization problem. The following is a simple description of the Rockafellar and Uryasev method, in which we use the equivalent *expected shortfall* (ES) downside risk measure in place of cVaR.

Recall the expected shortfall (ES) estimator expression (2.79), with loss taken as a positive quantity. Assuming for convenience $v_0 = 1$ and γT is an integer[33], (2.79) reduces to

$$\widehat{\mathrm{ES}}_\gamma = -\frac{1}{\gamma T} \sum_{r_t \leqslant r_{(\gamma T)}} r_t. \tag{2.217}$$

where $r_{(\gamma T)}$ is an estimate $\hat{q}_\gamma$ of the unknown quantile $q_\gamma(F)$ of the unknown distribution function F of the r_t. For the case where r_t are portfolio returns $r_t = \mathbf{w}'\mathbf{r}_t$, the above expression becomes

$$\widehat{\mathrm{ES}}_\gamma(\mathbf{w}) = -\frac{1}{\gamma T} \sum_{\mathbf{w}'\mathbf{r} \leqslant q_\gamma(\mathbf{w}'\mathbf{r})} \mathbf{w}'\mathbf{r}. \tag{2.218}$$

A *minimum expected shortfall* (MES) portfolio is one whose weight vector $\mathbf{w}$ minimizes $\widehat{\mathrm{ES}}_\gamma(\mathbf{w})$. In view of the fact that portfolio weight dependent quantity $q_\gamma(\mathbf{w}'\mathbf{r})$ is a nonlinear function of the portfolio weight vector $\mathbf{w}$,computation of an MES portfolio would appear to be a daunting task . However, Rockafellar and Uryasev (2000) showed that the equation (2.218) can be represented in a clever way, such that the opimization problem can be solved using linear programming. The interested reader is referred to the detailed discussion of minimum downside risk portfolios in Chapter 9.

The following example illustrates the viability of obtaining MES portfolios that outperform MV portfolios in terms of their mean return for a range of levels of ES. By definition of the MES optimality, the MES frontier is an efficient frontier by virtue of having the minimum possible ES risk for each mean return value. At each such mean return value, an MVO portfolio has an ES risk that is at least as large, and typically larger, than that of the corresponding MES portfolio. Consequently the MVO frontier using ES risk will lie below and to the right of the MES efficient frontier, as is illustrated by the following example.

Example 2.22. MES Efficient Frontier Versus MVO Frontier for a Portfolio Based on Weekly Returns of 10 Smallcap Stocks.

Figure 2.33 displays the weekly returns of 10 small-cap stocks for the two-year period from the beginning of 2004 to the end of 2005.

[33] In the general case where γT is not an integer, it is replaced by the ceiling function $\lceil \gamma T \rceil$

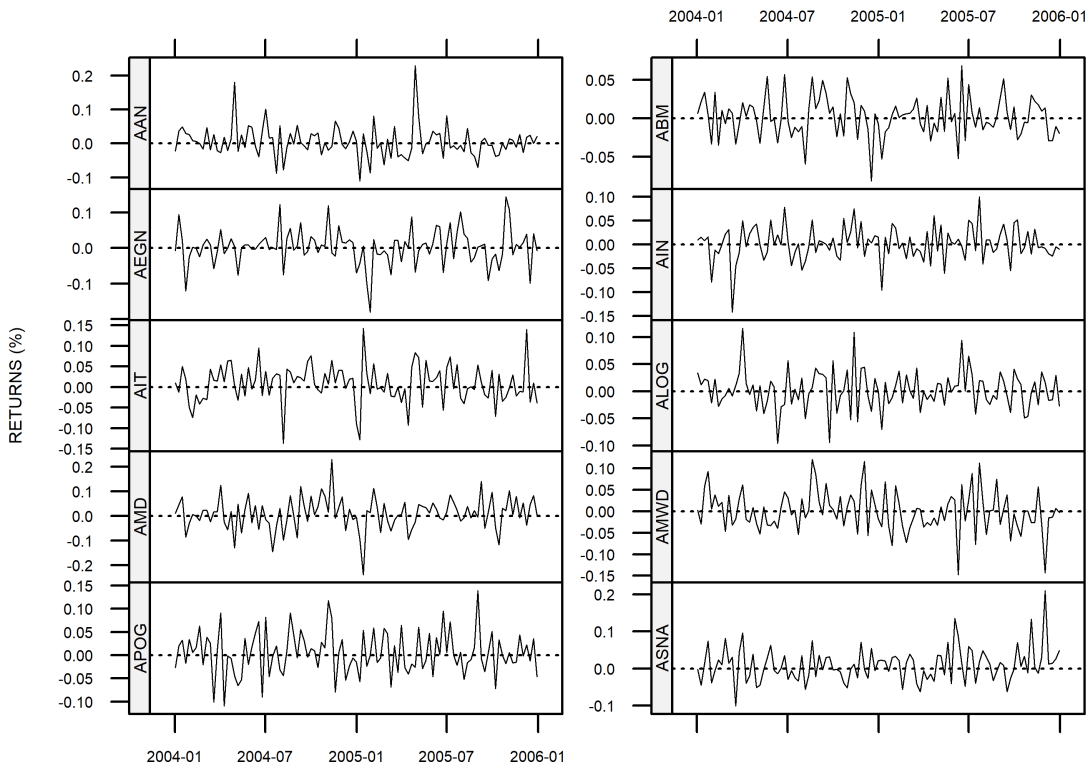


Fig. 2.33 Weekly returns of 10 smallcap stocks for 2004 and 2005.

Figure 2.34 displays the 5% ES efficient frontier (the solid line), and the MVO frontier (the dashed line). As the legend indicates, at the 0.35% weekly mean return of the global minimum ES (GMES) portfolio, the MVO portfolio has $100 \cdot (0.0426/0.0374 - 1) = 14\%$ higher risk than the GMES portfolio, and at the 0.00426 risk of the that MVO portfolio, the MES portfolio has a return that is almost twice as high. All in all, the MES portfolio is preferable to the MVO portfolio when assessed in terms of 5% ES risk.

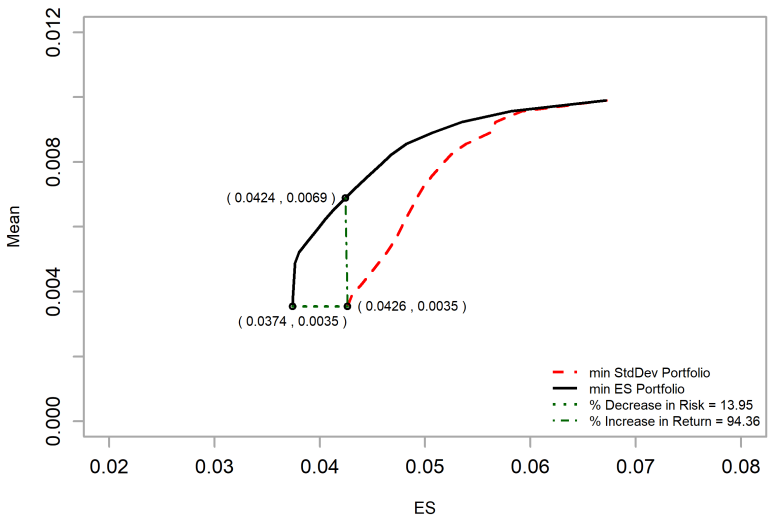


Fig. 2.34 MES efficient frontier and MVO frontier versus 5% ES risk, for weekly returns of 10 smallcap stocks from January 2005 to December 2006.

That said, one may quite reasonably wonder how the MVO and MES portfolios compare in terms of their relative return versus standard deviation risk. This comparison is provided in Figure 2.35, where we see that the MVO portfolio percent reduction in standard deviation risk is 12% as compared with the percent reduction in MES risk in Figure 2.34 of 14%, and the MVO portfolio percent increase in return in Figure 2.35 is only 67%, as compared with the MES increase in return of 94% in Figure 2.34. Overall, the investor who considers MES to be the risk measure of choice will be pleased with its performance relative to MVO as a risk measure, and the investor who considers MVO to be the risk measure of choice will be pleased with it, but somewhat less so than the MES investor.

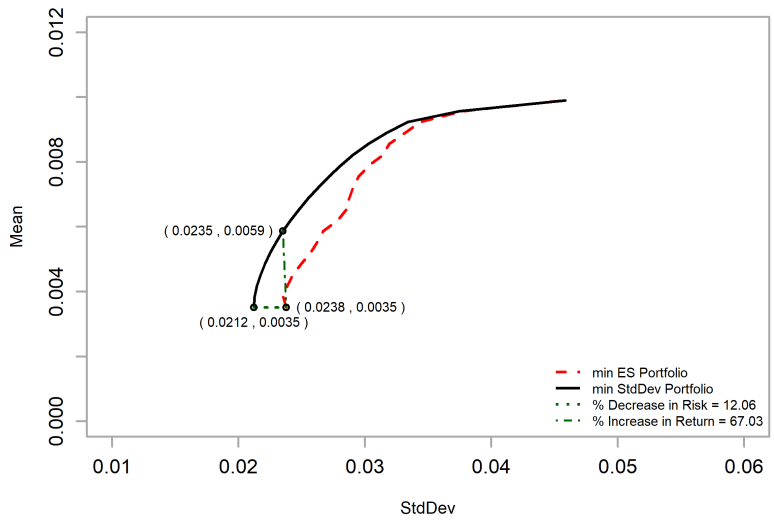


Fig. 2.35 MVO efficient frontier and ES frontier versus standard deviation risk, for weekly returns of 10 smallcap stocks from January 2005 to December 2006.

The use of MES based portfolio optimization has not emerged as a common practice for the following two reasons. First, there does not exist a rich tractable mathematical theory for ES portfolio optimization comparable to that of mean variance analysis as reflected in Section 2.4. Second, there currently exists a broad range of commercial and open source MVO portfolio software products and educational materials that have been developed at great expense over a long period of time. Correspondingly, there is little incentive to develop new MES based portfolio products without clear evidence that such products will deliver better risk adjusted returns. However, the Basel III standard requires the use of ES in place of VaR for portfolio risk analysis, and has resulted in an increasing awareness of ES as a risk measure;[34] it might well prove to be the stimulus required to hasten the adoption of MES based portfolio optimization.

With the above in mind, Chapter 9 discusses several downside portfolio risk minimization methods in detail, along with some promising applications results. The discussion includes not only the MES method, but also the Markowitz *minimum semi-variance* (MSV) method, and a *minimum coherent second moment* (MCSN) portfolio method that is closer in spirit to the Markowitz minimum semi-variance (SV) method than is MES.

2.7 Robust Estimators

Portfolio construction and analysis has traditionally relied on classical parameter estimation and model fitting methods, including the use of sample covariance matrices and least squares (LS) model fitting.

[34] See for example Bank-International-Settlements (2019)

The use of classical estimators is often justified by their virtue of being maximum likelihood estimators under the idealized and often unrealistic assumption that returns are normally distributed. Unfortunately, all classical normal distribution maximum-likelihood estimators (MLE's) are subject to the adverse influence of outliers that commonly occur in asset returns and related factor model variables. While least squares is further justified by virtue of it being a *best linear unbiased estimator* (BLUE), the linearity of LS exposes it to the influence of outliers, and even a single outlier can have an unbounded influence on an LS estimate.

Fortunately, the theory and computation of robust estimators have evolved extensively since the late 1970's following the seminal articles by Tukey (1960), Huber (1964), Huber (1973), and Hampel (1974), and the books by Huber (1981), Hampel et al. (1986), Rousseeuw and Leroy (1987), and Huber and Ronchetti (2009). We note that Rousseeuw and Leroy (1987) covers the detection of outliersand includes computational algorithms for this pupose.

The more recent Maronna et al. (2019) book, which is our primary reference, focuses both on the theoretical basis of bias robust estimators and their practical implemantation using robust computational algorithms, many of which are implemented in the open source R package `RobStatTM` (`https://cran.r-project.org/web/packages/RobStatTM/index.html`).

In data oriented terms, a robust parameter estimator or model fit is one that:

- Is not much influenced by outliers
- Provides a good fit to the bulk of the data
- Provides reliable detection of outliers
- Results in standard errors, confidence intervals and t-statistics that are not much influenced by outliersThere currently exist no books on portfolio construction and risk analysis that contain substantial coverage of robust methods for dealing with outliers in asset returns and factor exposure. This book fills that gap by discussing the main types of robust estimators that are useful in the context of portfolio optimization and risk analysis, and demonstrating their usefulness in such applications.

In this section we introduce robust estimators of the mean and standard deviation in the context of:

1. A returns model known as a signal-plus-noise model to engineers, and a location and scale distribution model to statisticians, and
2. A returns noise mixture distribution model that behaves like a normal distribution in the central region, but whose fat tails generate returns outliers.

With regard to the second item, we focus in this section on the returns mixture distribution model

$$F(r) = (1-\gamma)N(r;\mu,\sigma^2) + \gamma H(r), \;\; 0 \leqslant \gamma \leqslant 0.5 \tag{2.219}$$

where $N(r;\mu,\sigma^2)$ is a normal distribution function with mean μ and standard deviation σ, $H(r)$ is an unknown and completely unrestricted distribution that generates outliers, and γ is the fraction of

outliers than can be generated by $H(r)$[35]. The latter can result in bias and increased variance of sample mean and sample standard deviation estimates. In portfolio management practice, the fraction γ of outliers is typically in the range of 0.01 to 0.20, depending on the asset class and frequency of observations. Lower capitalization stocks and higher observation frequencies (e.g., weekly and daily) tend to have higher fractions γ of outliers, as well as more severe outlier generating distributions $H(r)$, while larger capitalization stocks and lower frequency observations (e.g., monthly) tend to have fewer outliers and less severe outlier generating distributions $H(r)$. For vivid illustrative examples, see the normal distribution qqplots of microcap, smallcap, midcap and largecap stocks, for monthly, weekly and daily returns respectively in Figures 3.11, 3.12 and 3.13.

The primary goal of robust estimators of the mean μ and standard deviation σ for the family of distributions (2.219) is to minimize estimator bias and variance inflation due to outliers induced by $H(r)$. Since our focus is on estimating μ and σ, we will often refer to these robust estimators as robust *mean estimators* and robust *standard deviation* estimators, even though these estimators are known more generally in the robust statistics and theory literature as robust *location and dispersion* estimators respectively. We follow that literature in introducing robust mean and standard deviation *M-estimators* that apply intuitively appealing nonlinear weights that mitigate the influence of outliers to observations. These robust estimators can be conveniently computed using a simple iterative method, and their in controlling for the influence of outliers is demonstrated with simple applications.

2.7.1 Robust Mean Estimators

The returns model most commonly used for robust mean and standard deviation estimators is the *signal plus noise* form

$$r_t = \mu + s \cdot \epsilon_t, \qquad t = 1, 2, \cdots, T, \tag{2.220}$$

where the ϵ_t are independent and identically distributed (i.i.d.), with $s > 0$, $\mathrm{E}(\epsilon_t) = 0$ and $\mathrm{var}(\epsilon_t) = \mathrm{E}(\epsilon_t^2) = 1$. It follows that the r_t have mean $\mathrm{E}(r_t) = \mu$, and standard deviation $s = \sigma = \mathrm{E}^{1/2}(r_t - \mu)^2$. This model is known to statisticians as a *location and scale model* with location parameter μ and scale parameter s.[36]

The scale parameter s is included in the above model for two reasons. The first is that it is needed for the robust estimators of location we describe shortly, as well as for robust estimation of regression model parameters in later chapters. The other reason is the mathematical one that for fat-tailed distributions of ϵ_t with infinite variance such as a t-distribution with 2 or fewer degrees of freedom, the scale parameter

[35] Multivariate generalized forms of the normal mixture model (2.219) are also used in later chapters for factors, and for joint returns and factors.

[36] The reason for this terminology is that for any errors distribution function $F_\epsilon(x)$ for the ϵ_t, the distribution $F_r(x)$ of r_t is $F_r(x) = F_{\epsilon_t}\left(\frac{x-\mu}{s}\right)$, where the parameter μ translates F_{ϵ_t} to different locations and s scales F_{ϵ_t} to different widths. The corresponding probability density of r_t has the form $f_r(x) = \frac{1}{s} f_{\epsilon_t}\left(\frac{x-\mu}{s}\right)$, where $f_{\epsilon_t}(x)$ is the probability density of ϵ_t.

in (2.220) is a meaningful way in which to describe its "width". Furthermore, for fat-tailed distributions whose mean does not exist, e.g., the Cauchy distribution, which is a special case of a t-distribution with 1 degree of freedom, μ is still a meaningful parameter that usefully describes its location.

When robustly estimating μ in the location and scale model (2.220), the parameter s is a *nuisance parameter* that needs to be robustly estimated in order to allow the estimation of the desired robust mean estimate $\hat{\mu}$. To estimate s, we follow the common practice of using a robust median absolute deviation about the median (MADM) estimate $\hat{s}_{\text{MADM}} = \hat{s}_{\text{MADM}}(\mathbf{r})$ described in Section 2.7.2 equations (2.229) and (2.230).

In his seminal paper Huber (1964), Peter Huber introduced a family of robust "maximum likelihood type" *location M-estimators* $\hat{\mu}$ of μ in (2.220) for the special case where $s = 1$.[37]. A Huber M-estimator $\hat{\mu}$ for the model (2.220), with s replaced by its estimate $\hat{s}_{\text{MADM}}$, is a solution of the minimization problem

$$\hat{\mu} = \underset{\mu}{argmin} \sum_{t=1}^{T} \rho\left(\frac{r_t - \mu}{\hat{s}_{\text{MADM}}}\right) \tag{2.221}$$

where $\rho(x)$ is a robustifying *rho* function that is non-negative, symmetric, bounded, and non–decreasing for $x > 0$, with $\rho(0) = 0$. Differentiating 2.221 with respect to μ and evaluating the derivative at $\hat{\mu}$ results in the *estimating equation*

$$\sum_{t=1}^{T} \psi\left(\frac{r_t - \hat{\mu}}{\hat{s}_{\text{MADM}}}\right) = 0 \tag{2.222}$$

where the *psi* function is $\psi(x) = \rho'(x) = d\rho(x)/dx$ is odd since $\rho(x)$ is symmetric. In standard least squares estimation, $\rho(x) = {}^{x^2}/_2$ and $\psi(x) = x$

There are many possible choices for the rho and psi function pairs for the above robust mean estimators, as well as for regression model extensions discussed in later chapters.[38] However, only two such rho and psi function pairs have a strong theoretical foundation, namely the functions $\rho_{\text{Huber}}(x)$ and $\psi_{\text{Huber}}(x)$ discovered by Huber (1964), and the functions $\rho_{\text{mOpt}}(x)$ and $\psi_{\text{mOpt}}(x)$ discovered by Yohai and Zamar (1997). The theoretical foundations and mathematical formulas for these two types of M-estimators are discussed the Chapter 16.

Here we focus on the empirical features of the Huber and mOpt rho and psi functions, whose shapes are displayed in Figure 2.36, with regard to minimizing the influence of outliers.[39]

---

[37] The family of M-estimators contains maximum-likelihood estimators as special cases, e.g., for ϵ_t with a normal distribution or t-distribution distribution. But the family also contains estimators that are not MLE's for any distribution function, such is the case for the specific location M-estimator recommended in this Section.

[38] For example, the SAS software product offers 10 such pairs in the regression context. See `documentation.sas.com/doc/en/statug/15.2/statug_rreg_details01.htm`.

[39] These estimators each have their own tuning constants to control the tradeoff between robustness toward outliers and having a variance not much larger than that of the least squares estimator when the data has a normal distribution. The Huber and mOpt rho functions in Figure 2.36 are set so that their variance is only about 5.3% larger than that of least squares for normally distributed returns. For further details see the Section on Location M-Estimator Asymptotic Efficieny in Chapter 16.

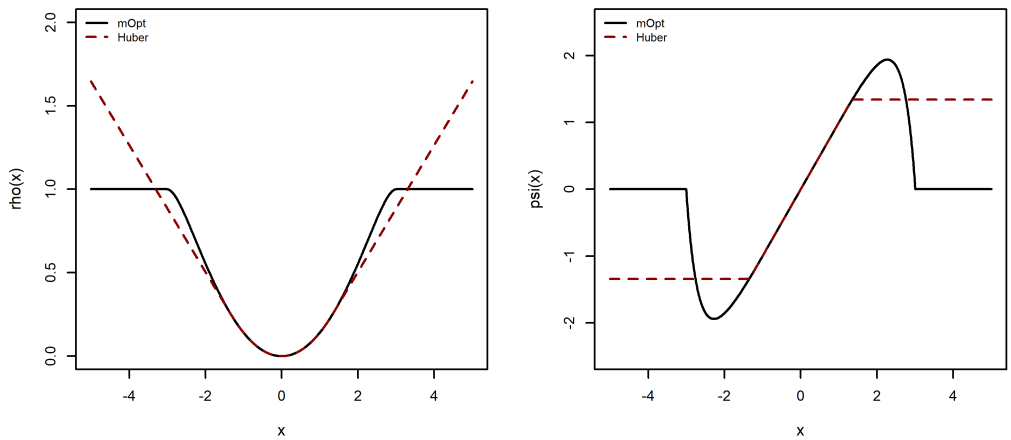


Fig. 2.36 Rho and psi functions for mOpt (solid) and Huber (dashed) estimators.

While the Huber estimator has been well-known to statisticians for many decades, which is not the case for the mOpt estiamtor, the latter is the preferered estimator for the following data oriented reasons. The arguments of the rho and psi functions in (2.221) and (2.222) are the robustly scaled residuals $(r_t - \hat{\mu}_{\text{mOpt}})/\hat{s}_{\text{MADM}}$. With regard to the rho functions: the mOpt rho function is a *bounded* function of the r_t, thereby limiting the influence of outliers on the estimator, but the Huber rho function is unbounded, allowing outliers to have an *unbounded* influence on the estimator. With regard to the psi functions: for the mOpt psi function, all sufficiently large outliers result in zero values for the summands of (2.222), but for the Huber psi function, all sufficiently large outliers result in constant positive or negative values for those summands. Strong theoretical reasons are provided in for preferring the mOpt estimator to the Huber estimator for the robust time series and cross-section regressions in Chapter 6 and Chapter 7.

The nonlinear estimating quation (2.222) can be written in a convenient weighted mean form that considerably facilitates computing its solution. To see this, we first multiply and divide the left-hand side of that equation by $(r_t - \hat{\mu}_{\text{mOpt}})/\hat{s}_{\text{MADM}}$so that

$$\sum_{t=1}^{T} \frac{\psi\left(\frac{r_t - \hat{\mu}}{\hat{s}_{\text{MADM}}}\right)}{\frac{r_t - \hat{\mu}}{\hat{s}_{\text{MADM}}}} \frac{r_t - \hat{\mu}}{\hat{s}_{\text{MADM}}} = 0\,. \tag{2.223}$$

We now define the *weight* function in terms of the psi function as

$$\text{w}(x) = \frac{\psi(x)}{x} \tag{2.224}$$

so that we have time-varying weights

$$\mathrm{w}_t(\hat{\mu}) = \mathrm{w}\left(\frac{r_t - \hat{\mu}}{\hat{s}_{\mathrm{MADM}}}\right) \tag{2.225}$$

and the estimating equation (2.222) can be rewritten as

$$\sum_{t=1}^{T} \mathrm{w}_t(\hat{\mu}) \cdot (r_t - \hat{\mu}) = 0\,. \tag{2.226}$$

The latter is equivalent to the nonlinear weighted mean estimator:

$$\hat{\mu} = \frac{\sum_{t=1}^{T} \mathrm{w}_t(\hat{\mu}) \cdot r_t}{\sum_{t=1}^{T} \mathrm{w}_t(\hat{\mu})}. \tag{2.227}$$

Figure 2.37 displays the shapes of the weight functions $w(x)$ for the Huber and mOpt estimators, which are obtained by dividing $\psi_{\mathrm{Huber}}(x)$ and $\psi_{\mathrm{mOpt,c}}(x)$, respectively by x.[40].

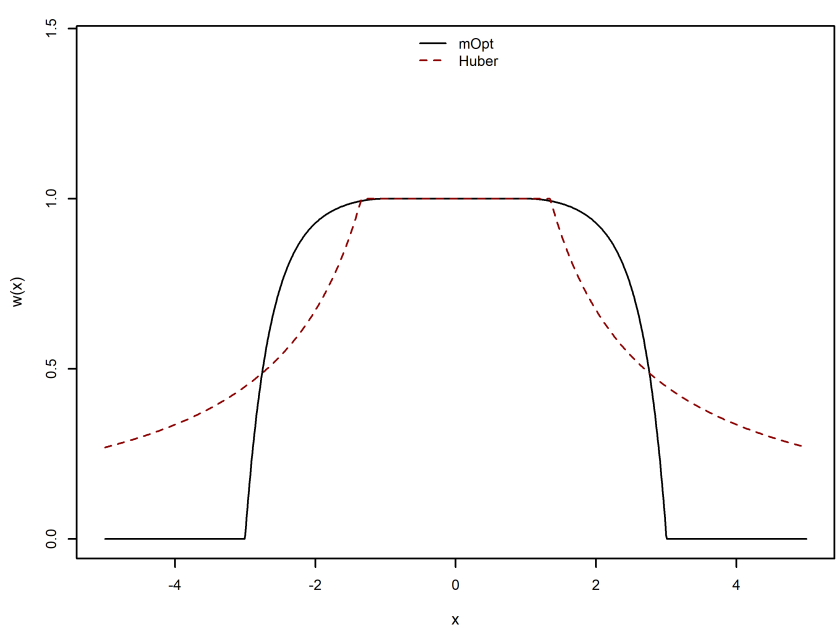


Fig. 2.37 Weight functions for mOpt and Huber estimators, tuned for 95% normal distribution efficiency.

The above Figure shows that the time t mOpt weight function $\mathrm{w}_{\mathrm{mOpt},t}(r_t) = \mathrm{w}_{\mathrm{mOpt}}\left(\frac{r_t - \hat{\mu}_{\mathrm{mOpt}}}{\hat{s}_{\mathrm{MADM}}}\right)$ assigns a weight of 1 to all sufficiently small scaled residuals $(r_t - \hat{\mu}_{\mathrm{mOpt}})/\hat{s}_{\mathrm{MADM}}$, and smoothly transitions to zero weight for all scaled residuals whose absolute value is larger than than 3.0. Returns whose scaled residuals exceed 3 are said to be *outliers* and are *rejected*. As such, the mOpt estimator is a robust version of the non-robust classic 3-sigma edit rule.[41]

[40] The formulas for these psi functions, and the corresponding weight functions will be provided in Chapter 16

[41] See, for example, Pukelsheim (1994) and Andrushkiw et al. (2006).

On the other hand, the Huber weight function assign a smaller weight than the mOpt weight function for some modest values of the robustly scaled residuals less than 3.0, and never assigns zero weight to observations r_t, no matter how large. As such, the Huber estimator never fully rejects outliers, but only gives them decreasing weight. These properties make the mOpt estimator preferable to the Huber estimator. Additional theoretical considerations discussed in Chapter 16 further justify our preference for the mOpt estimator.

The nonlinear weighted mean equation (2.227) is easily solved using the *iterated weighted mean* (IWM) algorithm

$$\hat{\mu}^{k+1} = \frac{\sum_{t=1}^{T} \mathrm{w}_t(\hat{\mu}^k) \cdot r_t}{\sum_{t=1}^{T} \mathrm{w}_t(\hat{\mu}^k)}, \qquad k = 0, 1, 2, \cdots \tag{2.228}$$

starting with the initial estimator $\hat{\mu}^0 = \mathrm{MED}(\mathbf{r})$. For examples in this book, the IWM algorithm is computed using the `RobStatTM` function `locScaleM`, for the which IWM algorithm terminates when the relative change condition $(\hat{\mu}^k - \mu^{k-1})/\mu^{k-1} < 10^{-4}$ is reached. A relatively simple formula for the standard error (SE) of the IWM estimator is provided in Chapter 16.

Example 2.23. Robust mean versus sample mean for the returns of a fixed income arbitrage (FIA) hedge funds style index.

The fixed income arbitrage (FIA) hedge fund style index displayed in Figure 2.38 is one of the twelve EDHEC hedge fund style indexes shown in Figure 2.40. The horizontal dashed line in Figure 2.38 is located at the sample mean, and the horizontal solid line corresponds is located at the robust mean estimate. The three substantial negative returns outliers in August, September and October 1998 are due to the Russian financial crisis that started in August 1998. The values of the sample mean and robust mean estimates are shown in the first column of Table 2.7. The small sample mean value of 0.13 is due to the distorting influence of those three substantial negative outliers, as a consequence of which the sample mean is a bad fit to the bulk of the returns.

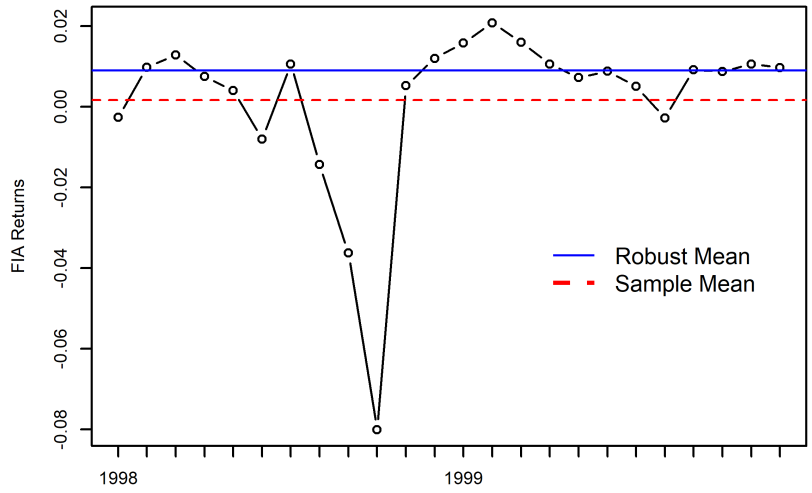


Fig. 2.38 Sample mean and robust estimate of mean for FIA hedge funds style index returns.

	Estimate (%)	Std. Error (%)	t-Stat	Sharpe Ratio
Sample Mean	0.17	0.43	0.39	0.08
Robust Mean	0.90	0.13	7.01	1.43

Table 2.7 Sample mean and robust mean estimate values with standard errors for FIA hedge fund returns

On the other hand, the robust mean estimator (2.228) computed with the `RobStatTM` package function `locScaleM` has the much larger value 0.90, which fits the bulk of the FIA index returns quite well. A key aspect of the way in which the robust mean estimate works is that all sufficiently large outliers are *rejected*, which means they are in effect deleted from the data set and thus have no influence on the estimator's value. The four smallest returns in Figure 2.38 are rejected as outliers, and the last three of these occur during the Russian financial crisis.

The second column of Table 2.7 contains the standard errors of the estimates of the sample mean and the robust estimate, respectively. The sample mean's large standard error of 0.43% reflects the considerable inflating influence of the outliers on the sample standard deviation used to compute the standard error.[42] By way of contrast, the robust mean's standard error, also computed with `locScaleM`, has the much smaller value of 0.13%, which is not surprising in view of the fact that the robust mean estimate is based on the FIA returns with the four smallest returns rejected.

[42] Recall that the standard error of the sample mean is $\mathrm{SE}_{\bar{r}_n} = \hat{\sigma}/\sqrt{n}$, where $\hat{\sigma}$ is the square root of the sample variance estimator $\hat{\sigma}^2 = n^{-1}\sum_{t=1}^{n}(r_t - \bar{r}_n)^2$. The large outlier returns in August, September and October 1998 result in very large values of $(r_t - \bar{r}_n)^2$ for these three months, which considerably inflates the standard error.

In view of the values of the sample mean and the robust mean, and their standard errors, it is not surprising that sample mean based t–statistic of 0.39 leads one to accept the null hypothesis that the FIA returns mean is equal to zero, but the positive robust t–statistic of 7.01 leads to strong rejection of that hypothesis.

The Robust Mean Versus the Sample Median

The sample median $\text{MED}(\mathbf{r})$ initial estimator for the IWM algorithm is a much simpler estimator than the robust mean estimator itself, so why do we bother with the latter? There are two main reasons in favor of the robust mean:

1. For normally distributed returns, the sample mean is the maximum-likelihood estimator, which has the minimum possible estimator variance. By way of comparison, it can be shown that the variance of the sample median is about 57% higher that that of the sample mean, while the variance of the robust mean estimator is only about 5.3% higher than that of sample mean. The robust mean is therefore preferred;
2. Computing the standard error (SE) of the sample median is complicated, while computing the SE of $\hat{\mu}$ is relatively simple, as is shown in Section 18.2.5 of Chapter 16;[43]
3. As discussed in Chapters 6, 7, and 16, the location M-estimator defined by (2.221) and (2.222) generalize to robust regression M-estimators.

On the other hand, for very large sample sizes the median may be preferred to the robust mean. The reason for this is that the variances of the robust mean and sample median go to zero at an asymptotic rate $1/T$ as the sample size $T \longrightarrow \infty$. But the bias of these estimators due to distributions $H(r)$ that are asymmetric around μ does *not* go to zero, and the sample median is known to minimize such bias.[44]

2.7.2 Robust Standard Deviation Estimators

There are two main types of robust sample standard deviation estimators, a simple medians based estimator, and a *dispersion parameter* M-estimator. The next two sub-sections discuss each of these two types of robust standard deviation estimators in turn.

Medians Based Robust Standard Deviation Estimators

The robust standard deviation estimate $\hat{s}_{\text{MADM}}(\mathbf{r})$ used in equations (2.225) is the calibrated sample *median absolute deviation about the median* (MADM)

[43] Computation of the SE of the sample median requires estimating the probability density of the returns evaluated at the sample median.

[44] See the "Estimator Asymptotic Bias" section in Section 18.1.2 of Chapter 16.

$$\hat{s}_{\text{MADM}}(\mathbf{r}) = 1.4826 \cdot \text{MADM}(\boldsymbol{r}) \tag{2.229}$$

where

$$\text{MADM}(\boldsymbol{r}) = \text{MED}\left(|\mathbf{r} - \text{MED}(\boldsymbol{r})|\right) \tag{2.230}$$

and $\mathbf{r} - \text{MED}(\boldsymbol{r})$ is the set of *centered returns* obtained by subtracting the value of *sample median* $\text{MED}(\boldsymbol{r})$ from each of the returns r_t.[45] Assuming no tied values in the data sample $\mathbf{r} = (r_1, r_2, \cdots, r_T)$, the sample median $\text{MED}(\boldsymbol{r})$ is defined in terms of ordered data $r_{(1)} < r_{(2)} < \cdots < r_{(T)}$ as

$$\text{MED}(\boldsymbol{r}) = \begin{cases} r_{(T+1)/2}\,, & T \text{ odd} \\ \dfrac{r_{(T/2)} + r_{(T/2+1)}}{2}\,]\,. & T \text{ even} \end{cases} \tag{2.231}$$

In words with regard to the ordered data: the sample media is the "middle" data value for T odd, and the "average of the two middle data points" for T even.

The reason for the multiplier 1.4826 in (2.229) is that as $T \longrightarrow \infty$, the $\text{MADM}(\boldsymbol{r})$ estimator for a $N(\mu, \sigma^2)$ returns distribution converges to $0.6745\,\sigma$. Thus, to make $\text{MADM}(\boldsymbol{r})$ an asymptotically unbiased estimator of σ, $\text{MADM}(\boldsymbol{r})$ needs to be divided by 0.6745, which is equivalent to multiplying it by 1.4826.

Robust Standard Deviation M-Estimators

We next introduce a robust standard deviation estimator of returns r_t generated by the location and scale model (2.220).[46] When the errors ϵ_t for this model have a standard normal distribution, the *scale parameter* s is the standard deviation of the errors ϵ_t. However, since the returns r_t differs from $s\epsilon_t$ only by the additive constant mean μ, s is also the standard deviation of the returns r_t. As such, the *errors scale parameter* s also serves as an r_t *dispersion parameter* that measures the size of the scatter of the r_t about μ.

A highly robust way to estimate s is to first compute the *centered returns*

$$\mathring{r}_t = r_t - \hat{\mu}_{\text{MED}},\ t = 1, 2, \cdots, T \tag{2.232}$$

and then compute a *scale M-estimator* (Mscale) $\hat{s}$ of the residuals by solving the equation

[45] In some of the robust statistics literature MADM is abbreviated to MAD. But the latter has the unfortunate consequence that it is also sometimes used as the acronym for the *mean absolute deviation* estimator. So we use MADM in this book.

[46] Such estimators were treated in Huber (1964) and Huber (1981), and the treatment in Section 2.5 of Maronna et al. (2019) starts with maximum-likelihood estimators (MLE's) of scale, and generalizes them to Mscale estimators.

$$\frac{1}{T}\sum_{i=1}^{T}\rho\left(\frac{\mathring{r}_t}{\hat{s}}\right)=0.5 \tag{2.233}$$

where the "rho" function $\rho(x)$ has the same properties as for the robust location estimator minimization (2.221), i.e., $\rho(x)$ is non-negative, symmetric, bounded, and non–decreasing for $x>0$, with $\rho(0)=0$. It is important to be aware that boundedness of the rho function is necessary for robustness of the estimator $\hat{s}$. For example use of the unbounded quadratice rho function $\rho(x)=x^2/2$ results in a sample standard deviation estimator that is quite non-robust toward outliers.

Multiplying and dividing the left-hand side of equation by $(\mathring{r}_t/\hat{s})^2$ and re-arranging the equation, results in

$$\hat{s}^2=\frac{2}{T}\sum_{t=1}^{T}\left[\rho\left(\frac{\mathring{r}_t}{\hat{s}}\right)\left(\frac{\hat{s}}{\mathring{r}_t}\right)^2\right]\mathring{r}_t^2. \tag{2.234}$$

A robust *Mscale weight function* is defined as

$$\mathrm{w}_{\mathrm{Mscale}}(x)=\frac{\rho(x)}{x^2} \tag{2.235}$$

and with this weight function[47], the estimating equation (2.234) may be written in the *weighted mean square* form:

$$\hat{s}^2=\frac{2}{T}\sum_{t=1}^{T}\mathrm{w}_{\mathrm{Mscale}}\left(\frac{\mathring{r}_t}{\hat{s}}\right)\mathring{r}_t^2\,. \tag{2.236}$$

Note that since the above Mscale weight function (2.235) decreases at a quadratic rate, the summands in (2.236) are bounded, which is needed for robustness toward outliers.

The above equation lends itself to the iterative weighted mean square (IWMS) form

$$\left(\hat{s}^{(k+1)}\right)^2=\frac{2}{T}\sum_{t=1}^{T}\mathrm{w}_{\mathrm{Mscale}}\left(\frac{\mathring{r}_t}{\hat{s}^{(k)}}\right)\mathring{r}_t^2\,,\quad k=0,1,2,\cdots \tag{2.237}$$

with $\hat{s}^{(0)}=\hat{s}_{\mathrm{MADM}}(\mathbf{r})$ given by (2.229).

We recommend an mOpt based robust scale estimator weight function that is defined as

$$\mathrm{w}_{\mathrm{Mscale,\,mOpt}}(x)=\frac{\rho_{\mathrm{mOpt}}(x)}{x^2} \tag{2.238}$$

[47] Although one seldom encounters $r_t=0$, most robustifying $\rho(x)$ of interest are locally quadratic with $\rho(0)=\rho'(0)=0$ and $\rho''(0)$ positive, so that one can define $\mathrm{w}_{\mathrm{Mscale}}(0)=\rho''(0)$.

where the shape of $\rho_{\text{mOpt}}(x)$ is displayed in the left-hand plot of Figure 2.36.[48] Figure 2.39 displays the shape of $\text{w}_{\text{Mscale, mOpt}}(x)$. Observe that the weight function for the scale estimator is defined in terms of ρ_{mOpt}, while the weight function for the location estimator is defined in terms of ψ_{mOpt}.

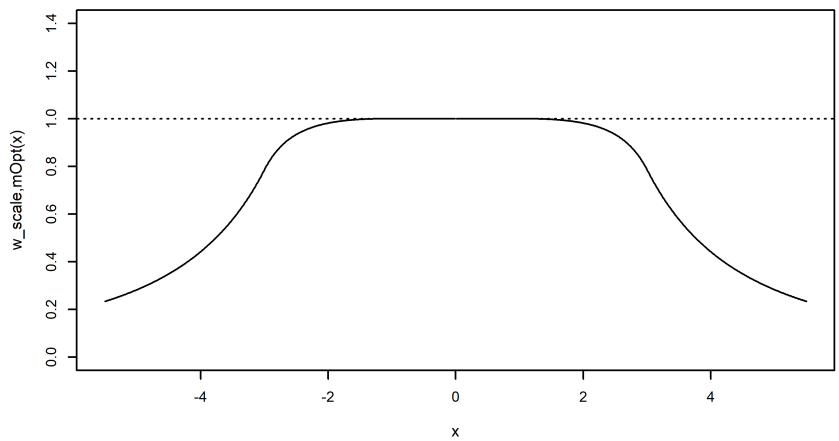


Fig. 2.39 The $\text{w}_{s,\text{mOpt}}(x)$ weight function.

Example 2.24. Standard deviation and robust standard deviation estimators for the EDHEC hedge funds style indexes returns.

Figure 2.40 shows the time series of the monthly returns of 12 (out of 13) `edhec` Hedge Fund Indexes for the period 1998 - 1999. The `edhec` data set is included in the `PerformanceAnalytics` package, where it is currently available from January 1997 to May 2021.[49]

[48] The value of$\text{w}_{\text{Mscale, mOpt}}(0)$ is defined as the limit of the right-hand side of (2.238) as $x \downarrow 0$.

[49] For details concerning the edhec data, which is from the EDHEC Risk and Asset Management Research Center in France, see the `PerformanceAnalytics` help for the `edhec` data set. In addition to the 12 hedge fund style indexes in Figure 2.40, the edhec data also includes a fund-of-hedge funds (FOF) index.

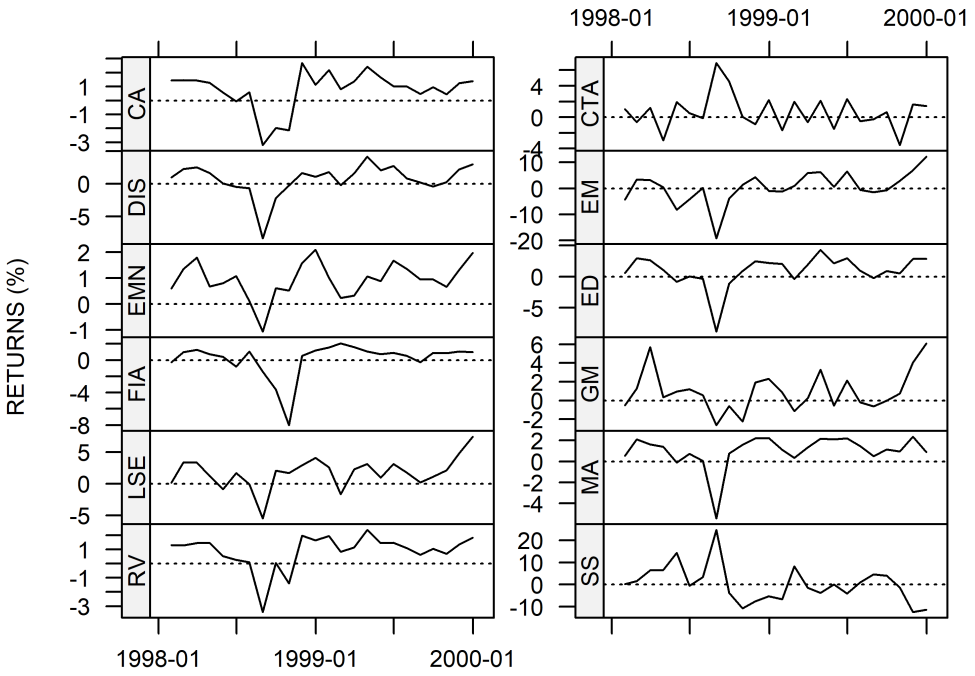


Fig. 2.40 EDHEC hedge fund indexes monthly returns for 1998 and 1999.

The dominant returns outliers for most of the hedge fund style indexes returns are the negative outliers associated with the financial crisis associated with the Russian default in the summer of 1998. One expects that sample standard deviation estimates will be inflated by these outliers, while robust standard deviation estimates will not be very much inflated. The results in Table 2.8 for the StdDev, MADM, and RobSD robust standard deviation estimates confirm this expectation. Interestingly, the MADM and RobSD estimates tend to be similar.

Table 2.8 Classic standard deviation StdDev, and MADM, and RobSD robust standard deviation estimates for 12 EDHEC hedge fund indexes for 1998 and 1999.

HFindex	StdDev	MADM	RobSD
CA	1.39	0.64	0.69
CTA	2.25	1.91	1.85
DIS	2.40	1.79	1.64
EM	6.11	4.11	4.61
EMN	0.68	0.58	0.59
ED	2.52	1.96	1.85
FIA	2.10	0.53	0.57
GM	2.17	1.82	1.80
LSE	2.47	1.81	2.24
MA	1.56	1.07	0.99
RV	1.21	0.70	0.80
SS	8.33	6.77	7.03

2.7.3 Simple Robust Outliers Detection and Cleaning

Sometimes it is desirable to "clean" data by shrinking outliers prior to model fitting, especially so in the case of cross-section factor models in which a stock's factor exposures, such the log of market capitalization (Size) and earnings-to-price (EP), among others, tend to be plagued with outliers to different (factor-dependent) extents. Outliers in a set $\mathbf{f} = (f_1, f_2, \cdots, f_N)$ of factor exposures at a given time are easily detected in Step 1 below, using any robust mean estimate $\hat{\mu}_{\text{Rob}}(\mathbf{f})$ and robust standard deviation estimate $\hat{\sigma}_{\text{Rob}}(\mathbf{f})$. Then Step 2 below can be used to "clean the outliers" by shrink shrinking them toward $\hat{\mu}_{\text{Rob}}(\mathbf{f})$:

1. Define the i-th factor value f_i to be an outlier if it falls outside the interval $(\hat{\mu}_{\text{Rob}}(\mathbf{f}) - k \cdot \hat{\sigma}_{\text{Rob}}(\mathbf{f}), \hat{\mu}_{\text{Rob}}(\mathbf{f}) + k \cdot \hat{\sigma}_{\text{Rob}}(\mathbf{f}))$, and
2. Shrink all outliers f_i to their nearest interval endpoint.

While the robust mean and robust standard deviation estimators $\hat{\mu}_{\text{Rob}}(\mathbf{f})$ and $\hat{\sigma}_{\text{Rob}}(\mathbf{f})$ of the previous section can be computed using the `RobStatTM` package function `locScaleM` and used for data outlier-cleaning, it will often suffice for routine data cleaning purposes to use the median based robust location and scale estimators $\text{MED}(\boldsymbol{r})$ and $\hat{s}_{\text{MADM}}(\mathbf{r})$ given by (2.231) and (2.229) respectively. Although published references are not available, it appears to be common knowledge in the quantitative finance industry that this simple medians based robust data cleaning method is used by some leading commercial portfolio management software solutions providers.

Often the value $k = 3.0$ is used, in which case the shrinkage method is a form of "robust three-sigma editing" (see Pukelsheim (1994) and Andrushkiw et al. (2006)). However, other values of k may be preferred, as, for example, in Section 3.3.2, where a case is made for the choice $k = 4$ in the context of

robust estimation of skewness and kurtosis, as well as in the section on robust cleaning of exposures outliers in Chapter 7.

2.8 Exercises

Exercise 2.1. (**Computing**) This exercise assumes that you have read the documents: (1) PCRA book "Chapter 2 Foundations" through at least Section 2.4, and (2) "PCRA Reproducibility Code.pdf.", both of which will have been provided to you by your Instructor. Use what you have learned in the latter document to access the R script `Ch 2 Foundations demo.R`. Run that script and confirm that it creates all the Figures and Tables in this Chapter, except for a few Figures that were not created by the code. Note also that the Tables produced by the code are in standard R format, whereas the Tables in the book chapter pdf files were produced with the useful `kableExtra` package. If you have any problems with this, please let your Instructor know, and the instructor get back to PCRA authors on the them.

Exercise 2.2. (**Computing**) Load `PCRA` and use `help(stocksCRSPmonthly)` to familiarize yourself with the various data components of the `stocksCRSPmonthly` `data.table` object. Next, familiarize yourself with the code that produces Figures 2.7–2.10, as follows. First, find the code chunk in the script `Ch 2 Foundations demo.R` that begins by loading the data.table package with `library(data.table)`. The next several code lines use `selectCRSPandSPGMI` function to select all 106 of the smallcap stocks in `stocksCRSPmonthly` during the time period January 1997 through December 2010, and save it as the `xts` time series object `stocksDat.`. The code then extracts ten smallcap stocks in columns 21 through 30 of `stocksDat`, creates the time series object `returns10Mkt` consisting of the returns of these ten tocks and the Market, and finally uses the PCRA function `tsPlotMP` to plot the time series in Figure 2.7. The code that follows computes Figures 2.8, 2.9 and 2.10, as indicated by the comment lines. The code then computes the rest of the Figures and Tables in Section 2.3. Your Exercise here is to run the code that produces all these figures and tables for each of the five groups of 20 stocks in columns 1-20, 21-40, 41-60, 61-80 and 81-100 of the `stocksDat` object, along with the `Market`. Then recommend which of the 5 groups of 20 stocks one should invest in for the best return on their investment, and state in words why you make the recommendation (there is no absolutely "right" answer). Also report on which group of 20 stocks results in the poorest overall performance, and state why.

Exercise 2.3. (**Computing**) Carry out the previous exercise for each of the following capitalization groups of `stocksCRSPmonthly` data: (a) microcap stocks, (b) midcap stocks, and (c) largecap stocks, and discuss any notable differences in the results across those groups.

Exercise 2.4. (**Computing**) *Stock Selection*. The above exercises beg the question of how one should select stocks for the best investment portfolio performance. It is generally infeasible to evaluate all subsets of a large number of stocks, such as the 106 smallcap stocks in `stocksCRSPmonthly`. One well-known method of choosing a fixed number N_{port} from a larger set of N stocks, is to rank all N of the

stocks in a preliminary step using some good ranking method, and choose the best N_{port} of them for the portfolio. Implement this method for the 106 smallcap stocks, using each of the risk estimators of Section 2.3.3, and write a small report of your analysis code and results. Which risk estimator gives the best performing portfolio, and how does this portfolio perform relative to your best portfolio in Exercise 19?

Exercise 2.5. (**Computing**) Download the annual returns of stocks, bonds and bills in the U.S. from Professor Aswath Damodaran's web page at `http://pages.stern.nyu.edu/~adamodar/New_Home_Page/datacurrent.html`, and compute the standard deviation of the S&P 500 over the periods 1931–1965 and 1965–1991. How do your results compare with the slopes of the lines in Figure 2.32? What is the percent drop in standard deviation from the first period to the second? How does it compare to the percent drop in the slope from the first period to the second?

Exercise 2.6. (**Computing**) (a) Read Sections 1 and 2 of `PCRA` package Vignette *Introduction to CRSP Stocks and SPGMI Factors in PCRA.pdf*, which will be provided by your instrutor; (b) Based on a careful reading of Section 2.3 of that Vignette, load into your R session the weekly returns and related data for the 294 stoccks in the `stocksCRSPmonthly` data set; (c) Use the R-code that created Table **??** to create a similar table based on the weekly returns; (d) Repeat (c) using daily returns of those 294 stocks.

Exercise 2.7. (**Computing**) Use the R acf function to plot autocorrelations with a maximum lag $k_{max} = 5$, and identify which hedge fund indexes have significant lag-3 autocorrelations, one of which has a significant lag-4 autocorrelation. Are there any qualitative features of the time series of those hedge fund indexes having significant lag-3 autocorrelations which might suggest such serial correlation?

Exercise 2.8. Use R to confirm that the covariance matrix (2.21) has as its inverse the covariance matrix (2.22), and to confirm that $\mathbf{C}^{-1}\mathbf{C} = \mathbf{C}\mathbf{C}^{-1} = \mathbf{I}$.

Exercise 2.9. Confirm that the sample covariance matrix form (2.26) has the pairwise sample covariances (2.24) as its elements.

Exercise 2.10. Show that the expression for σ^2_{gmv} in (2.46)is correct.

Exercise 2.11. Confirm the results in Example 2.3 for the GMV portfolio weight vector, mean return and volatility.

Exercise 2.12. Show that if a fully-invested portfolio has equal marginal contributions to risk, i.e., all equal MCTR_i, then it must be a global minimum variance portfolio.

Exercise 2.13. Derive the expression on the right-hand side of equation (2.81).

Exercise 2.14. (**Computing**). Use the entire cross-section of stocks during 1999-2000, compute the percentage of the stocks for which the incorrectly computed Sharpe Ratio, based on only the raw returns, exceeds each of the following percentages: 25%, 50%, 75%.

Exercise 2.15. In this exercise you will derive the following MVO portfolios formulas:

$$\lambda_1 = \frac{c\mu_P - a}{d}, \quad \lambda_2 = \frac{b - a\mu_P}{d}$$

where $a = \mathbf{1}'\mathbf{C}^{-1}\boldsymbol{\mu}$, $b = \boldsymbol{\mu}'\mathbf{C}^{-1}\boldsymbol{\mu}$, $c = \mathbf{1}'\mathbf{C}^{-1}\mathbf{1} > 0$, and $d = bc - a^2$. Substituting λ_1 and λ_2 into (2.109) to obtain the MVO weight vector formula $\mathbf{w} = \mathbf{w}_1 + \mu_P \mathbf{w}_2$ where

$$\mathbf{w}_1 = \frac{1}{d}\left(b\mathbf{C}^{-1}\mathbf{1} - a\mathbf{C}^{-1}\boldsymbol{\mu}\right)$$
$$\mathbf{w}_2 = \frac{1}{d}\left(c\mathbf{C}^{-1}\boldsymbol{\mu} - a\mathbf{C}^{-1}\mathbf{1}\right)$$

where

$$d = (\boldsymbol{\mu}'\mathbf{C}^{-1}\boldsymbol{\mu})(\mathbf{1}'\mathbf{C}^{-1}\mathbf{1}) - (\mathbf{1}'\mathbf{C}^{-1}\boldsymbol{\mu})^2.$$

It is shown in Appendix D that d is always non-negative, and is not equal to zero in practice.

Exercise 2.16. Derive the expression (2.111) for the covariance between two frontier portfolios. As convenient steps in doing so, you should show that using a and c from 2.15 one gets $c^{-1} = \sigma^2_{\text{gmv}}$ and $\frac{a}{c} = \mu_{\text{gmv}}$, where μ_{gmv} and σ^2_{gmv} are the mean and variance of the global minimum variance (GMV) portfolio.

Exercise 2.17. Show directly from the definition of the weight vector in (2.122) that the weights sum to zero.

Exercise 2.18. Show that a QU optimal portfolio with weight vector $\mathbf{w}_{\text{qu}}$ has a mean return given by the expression (2.124). *Hint*: For notational convenience, define the adjusted mean return vector as $\boldsymbol{\mu}_{\text{adj}} = \boldsymbol{\mu} - \mu_{\text{gmv}} \cdot \mathbf{1}$, and write $\mathbf{w}'\mathbf{C}\mathbf{w}_{\text{qu}} = \mathbf{w}'_{\text{gmv}}\mathbf{C}\mathbf{w}_{\text{gmv}} + 2\lambda^{-1}\mathbf{w}'_{\text{gmv}}\boldsymbol{\mu}_{\text{adj}} + \lambda^{-2}\boldsymbol{\mu}'_{\text{adj}}\mathbf{C}^{-1}\boldsymbol{\mu}_{\text{adj}}$. Then show that $\mathbf{w}'_{\text{gmv}}\boldsymbol{\mu}_{\text{adj}} = 0$, and $\boldsymbol{\mu}'_{\text{adj}}\mathbf{C}^{-1}\boldsymbol{\mu}_{\text{adj}} = d \cdot \sigma^2_{\text{gmv}}$.

Exercise 2.19. Show that a QU optimal portfolio with weight vector $\mathbf{w}_{\text{qu}}$ has a variance given by the expression(2.125).

Exercise 2.20. Using the positivity of d in (2.112), derive the inequalities in (2.136).

Exercise 2.21. Show that as the risk tolerance $\lambda^{-1}$of a QU–optimal portfolio goes to zero the portfolio weights become unbounded. Explain what happens to the cash position as the weights become unbounded.

Exercise 2.22. Derive the expression on the right–hand–side of (2.132).

Exercise 2.23. Derive the formula for $\boldsymbol{\gamma}$ in (2.163).

Exercise 2.24. Show that the Lagrange multiplier vector $\boldsymbol{\gamma}$ in (2.163) results in the weight vector $\mathbf{w}$ in (2.163).

Exercise 2.25. Show that the QU optimal weight vector in (2.163) is equal to the expression (2.121) when the only constraint is the full investment constraint defined by $\mathbf{A} = (1, 1, \cdots, 1)$ and $\mathbf{b} = 1$.

Exercise 2.26. Derive the risk aversion functions for the power utility functions (2.195) and the log utility function (2.196), and show that they both have decreasing ARA and constant RRA.

Exercise 2.27. Show that the log utility function is the limit of a power utility function as $\gamma \to 0$. Hint: Use L'Hospital's rule.

Exercise 2.28. Show that the quadratic utility function (2.197) is increasing for $\mathrm{v} < (2b)^{-1}$ and decreasing for $\mathrm{v} > (2b)^{-1}$.

Exercise 2.29. Derive the risk aversion functions for the quadratic utility function (2.197), and show that it has increasing ARA and RRA.

Exercise 2.30. Show that the exponential utility function $U_{\exp}(\mathrm{v}) = -\exp(-c \cdot \mathrm{v}), c > 0, \mathrm{v} \geqslant 0$ has constant ARA and increasing RRA.

Exercise 2.31. Show that if portfolio returns have a multivariate normal distribution, and an investor uses an exponential utility function, then the investor is an MVO investor.

Exercise 2.32. Show that if $\tilde{\mathbf{r}}$ has a $N(0,\sigma^2)$ distribution, then the asymptotic value of $\mathrm{MED}\,(|\tilde{\mathbf{r}}|)$ is 0.6745.

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